NFT

Neuroelectromagnetic Forward and Inverse Head Modeling Toolbox

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Swartz Center for Computational Neuroscience

Virtual EEGLAB workshop, June 2021

NFT code

https://github.com/sccn/NFT

Demo folders:

NFTplugin_demo_dipole
NFTplugin_demo_cortical

https://rdl-share.ucsd.edu/message/U7F0uMivgbGZJur64TXDac

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NFT

- A complete framework for accurate forward and inverse problem solution.
- Easy-to-use MATLAB environment with GUI and command-line functions.
- Ability to use available subject information
  - T1-weighted 3D MR images
  - Digitized sensor (electrode) locations

Comparison with Dipfit

- The realistic model in Dipfit is a three-layer MNI head model represented with 3000 vertices.
  - The forward matrices are pre-calculated, so there is no need for FP calculations. Faster solutions.

- NFT generates subject-specific models.
  - NFT does model generation and forward problem calculations.
  - More accurate.

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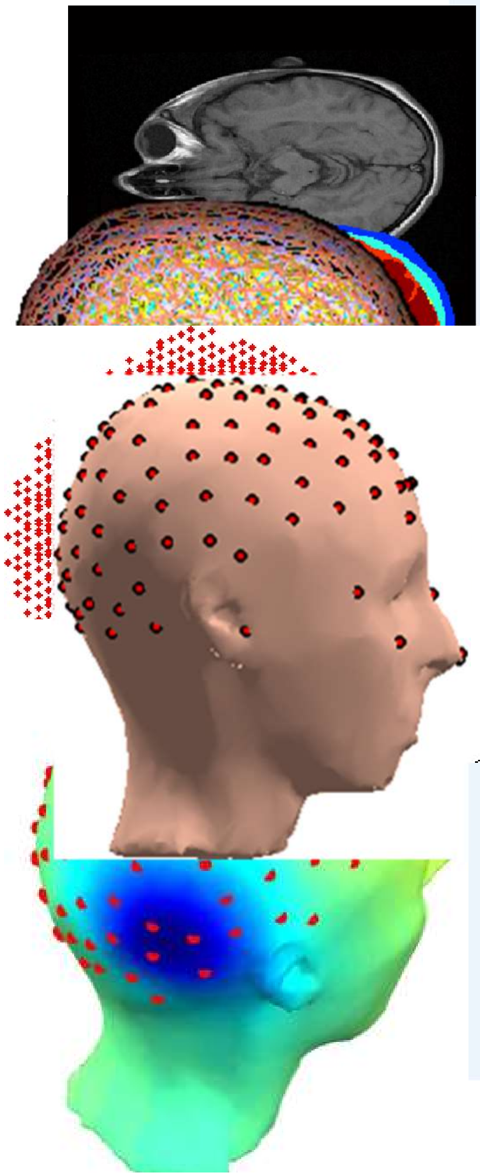
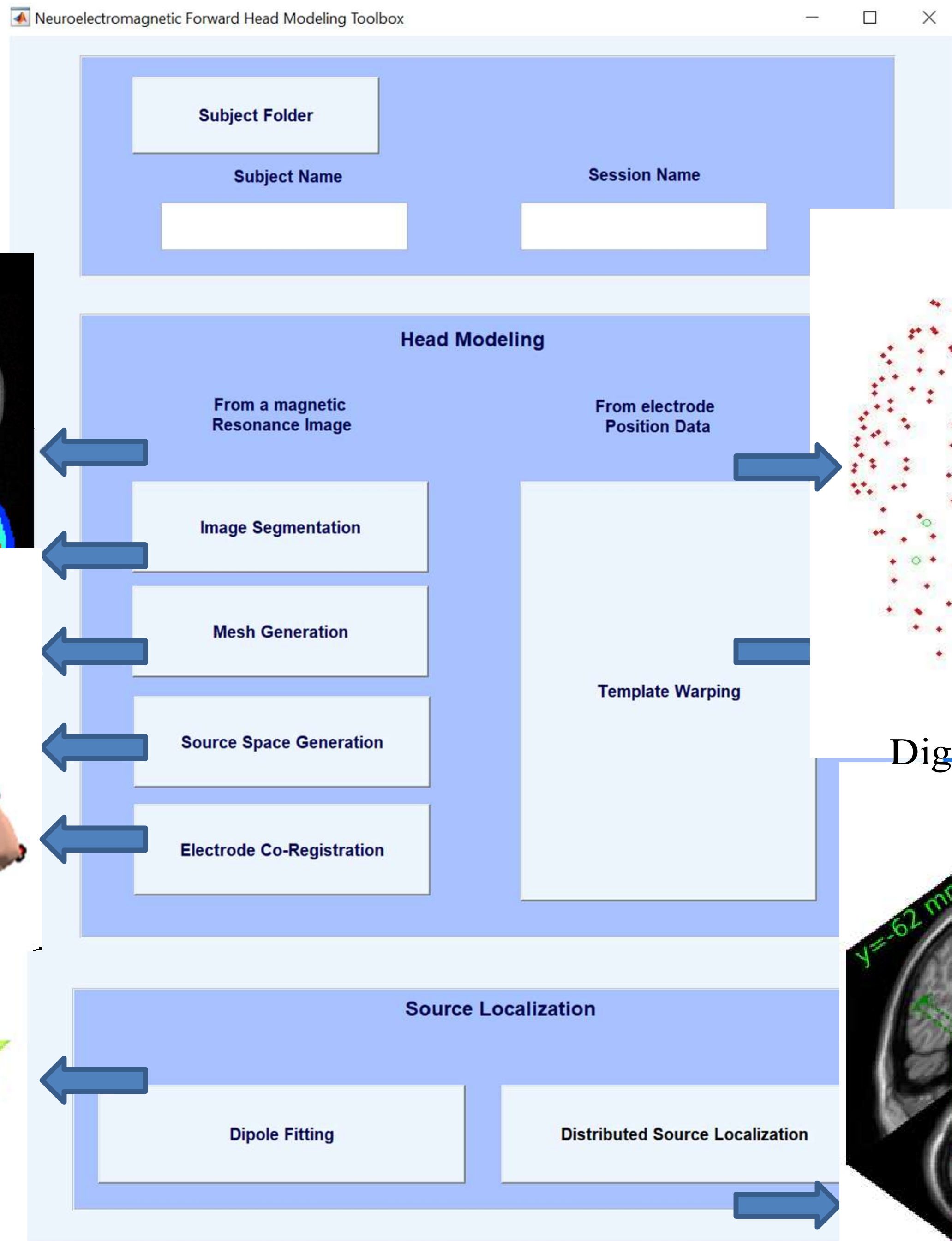
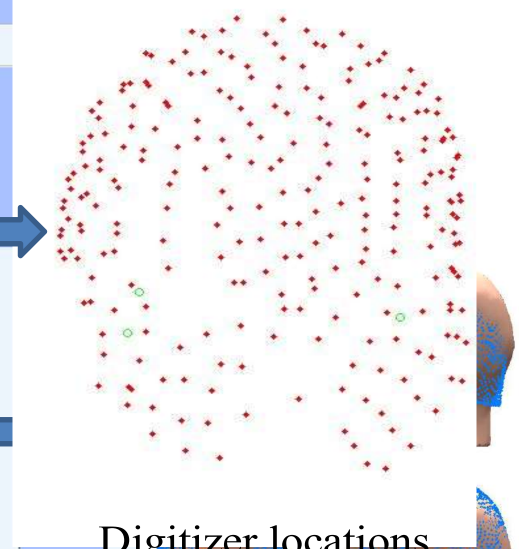
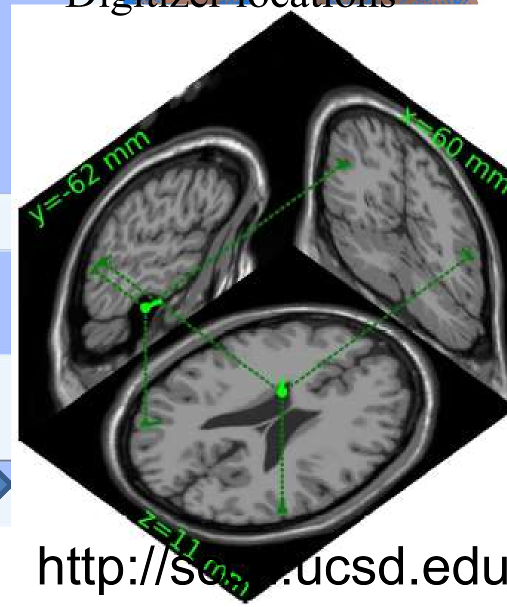
NFT





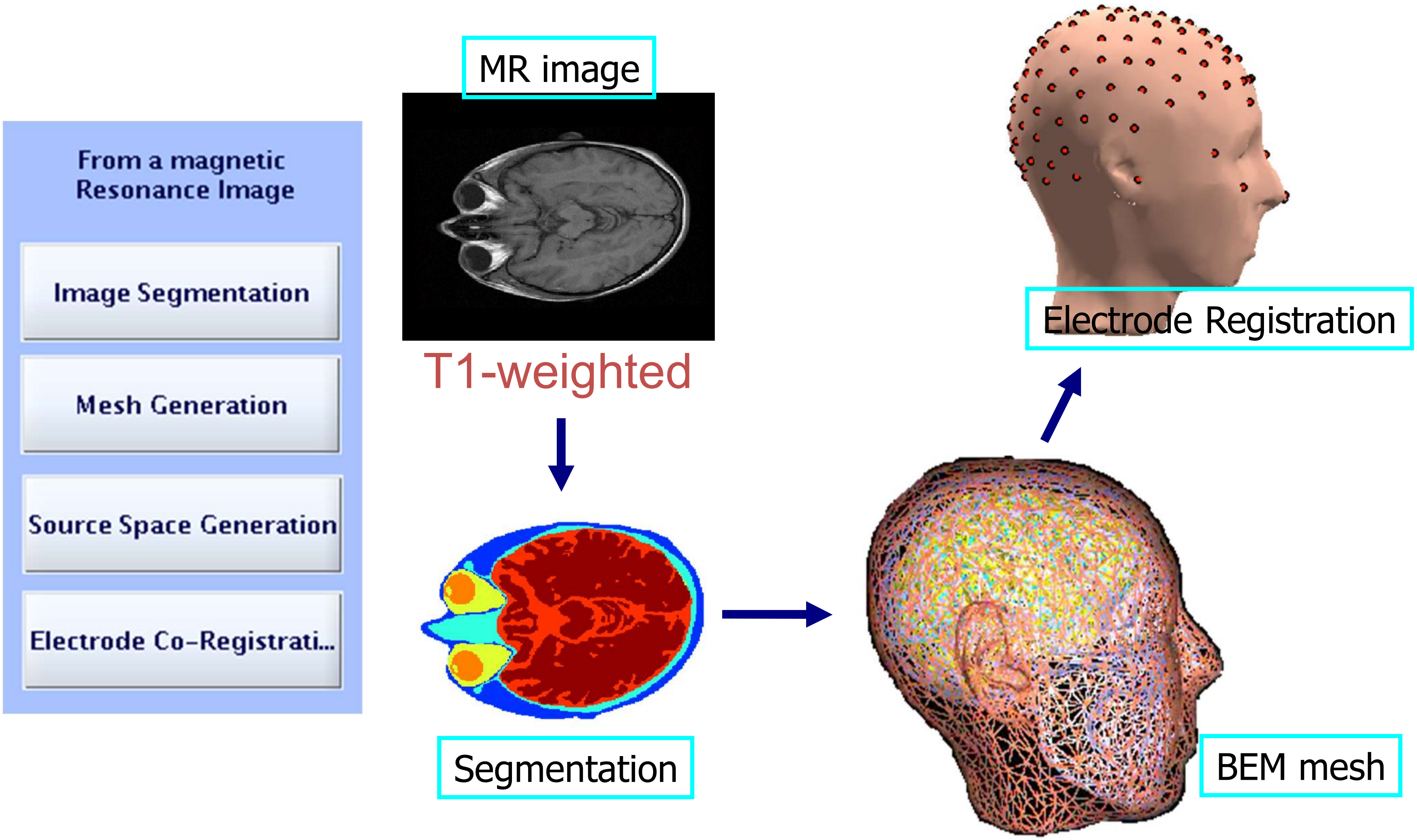


Digitizer locations



http://sc .ucsd.edu/nft

Head modeling from MR images

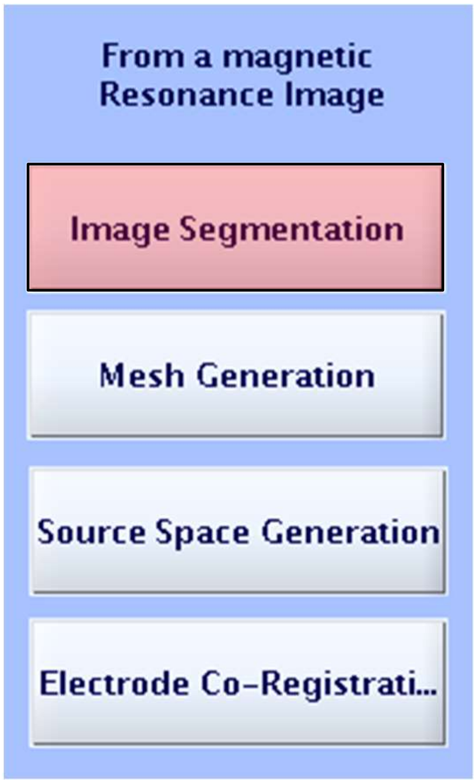


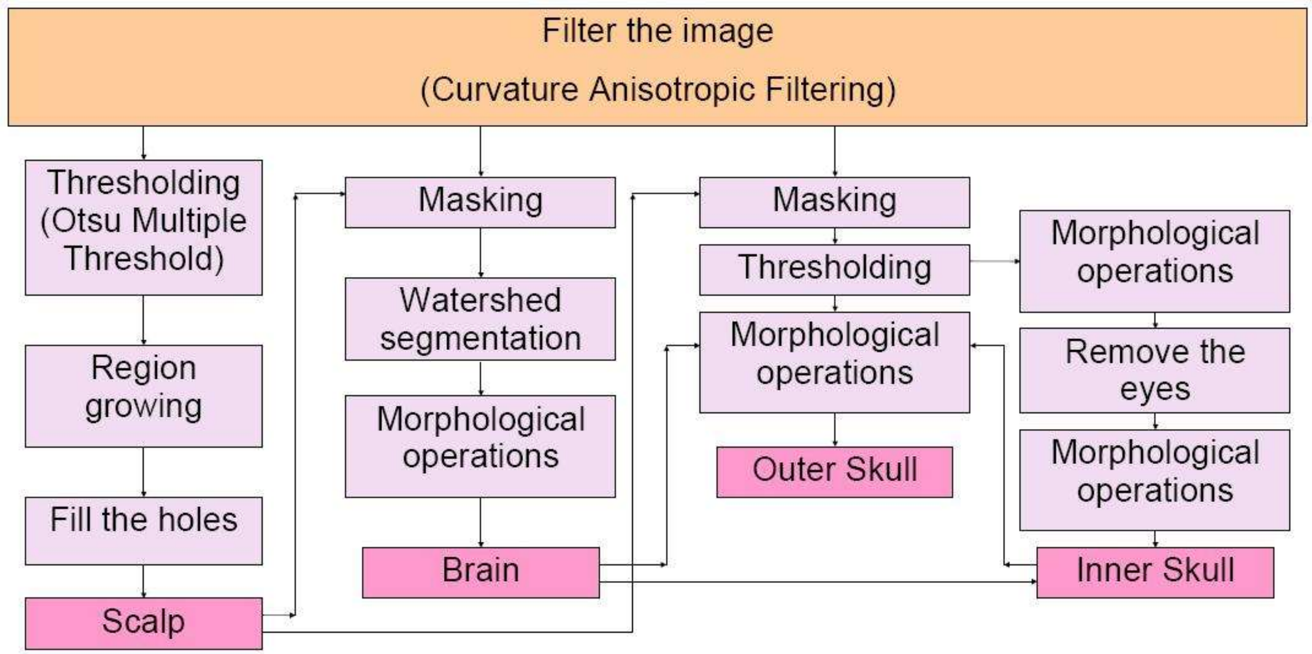
Preparing the MR Image

- Using FreeSurfer
  - Inhomogeneity correction
  - Convert to 1x1x1 volume
  - Arrange direction of the image
  - Save in analyze format

```
mri_convert -i brain.nii -ot analyze -o test1.img

mri_nu_correct.mni --i test1.img --o t1c.img --n 2

mri_convert -i t1c.img --conform_size 1 --
out_orientation PSR -ot analyze -o test2.img
```

Image Segmentation





Classifies four tissues from T1-weighted images
Scalp, Skull, CSF and Brain

Starting NFT

- To start from EEGLAB

  EEGLAB -> Tools -> NFT

- To start as a standalone toolbox

  addpath NFT directory

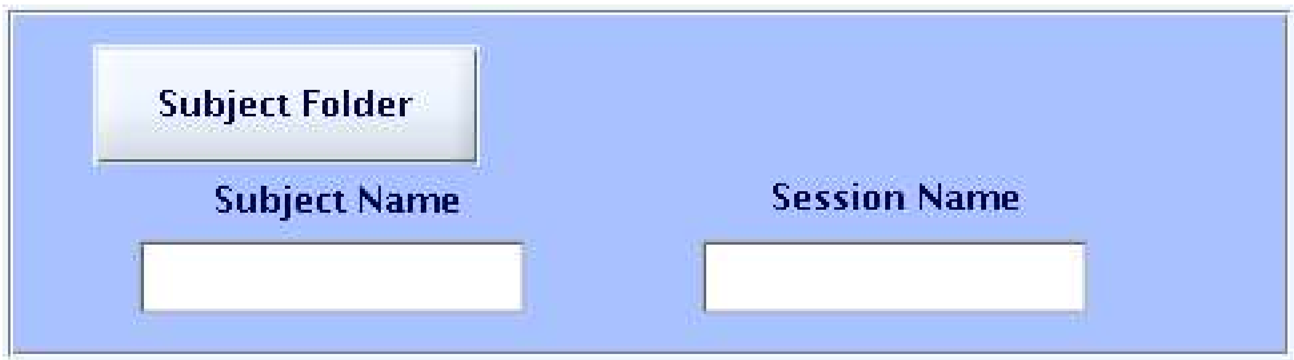
  Type ‘NFT’ in Matlab

Demo folders: NFTplugin_demo_dipole

NFTplugin_demo_cortical

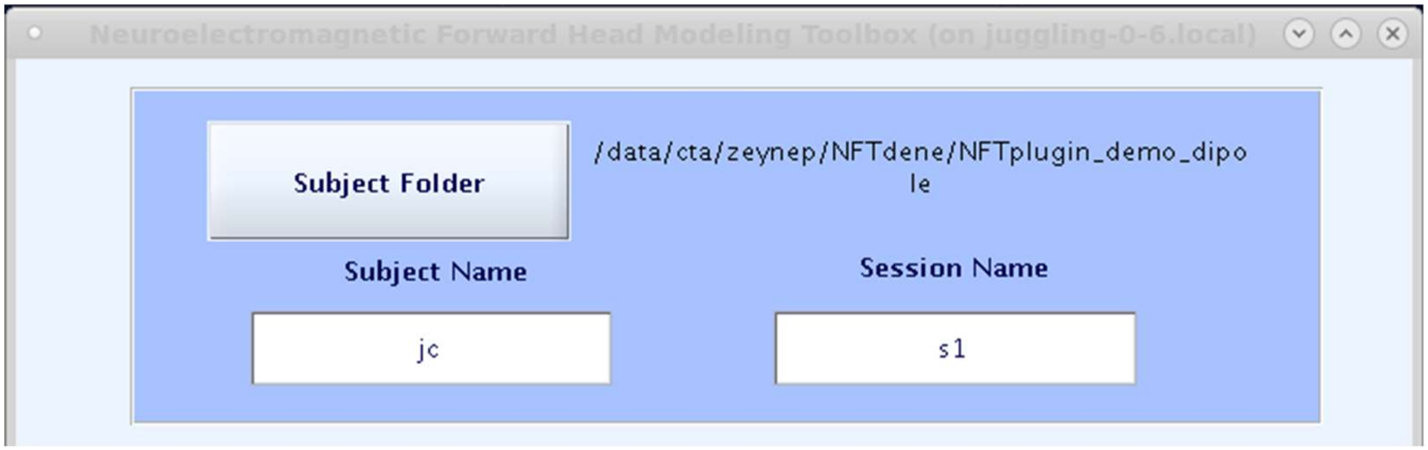
https://rdl-share.ucsd.edu/message/U7F0uMivgbGZJur64TXDac

Subject Selection



- Select subject folder
- Specify subject name
- Specify session name

Subject Selection



Select current folder as subject folder
Enter “jc” as subject name
Enter “s1” as the session name

Image Segmentation

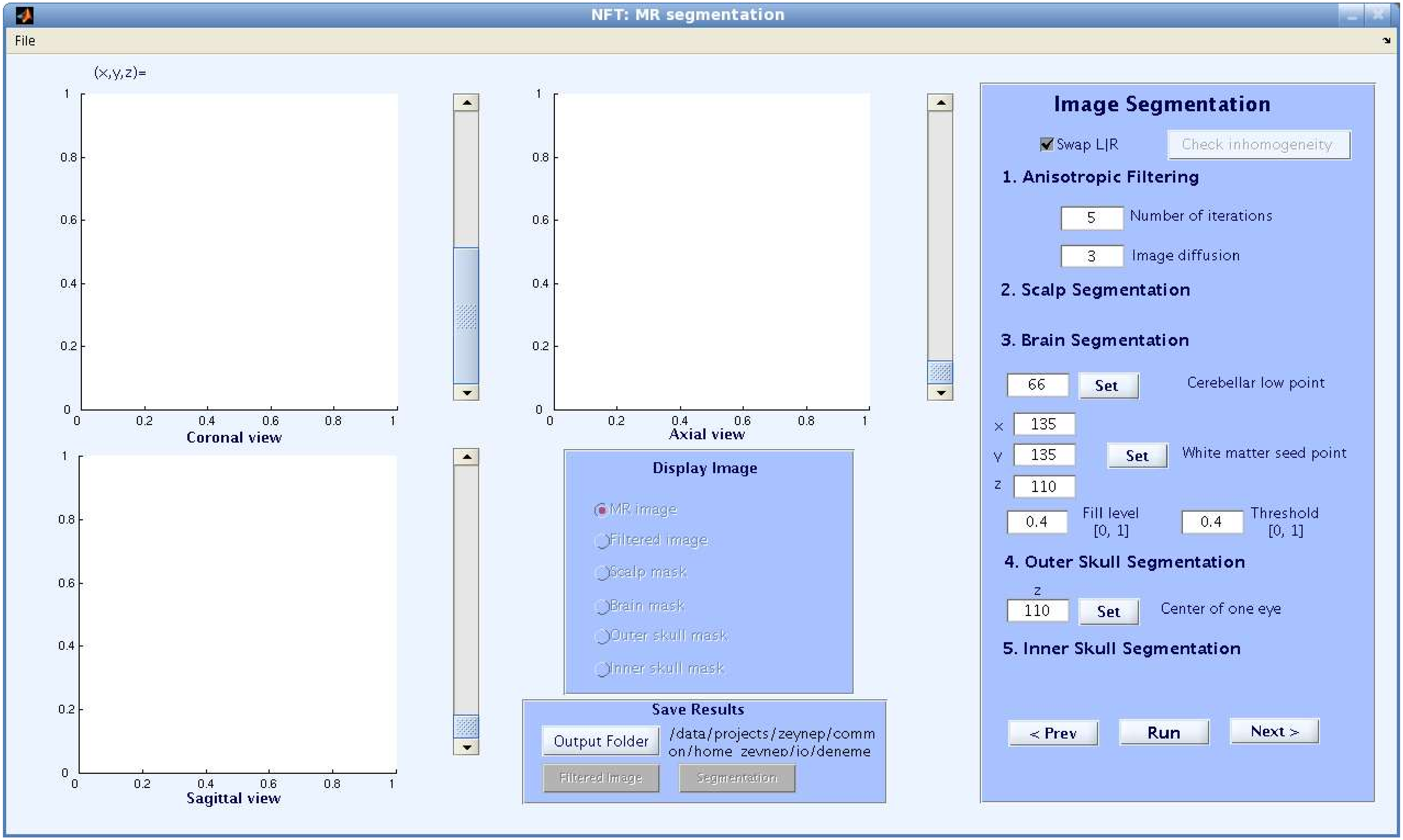
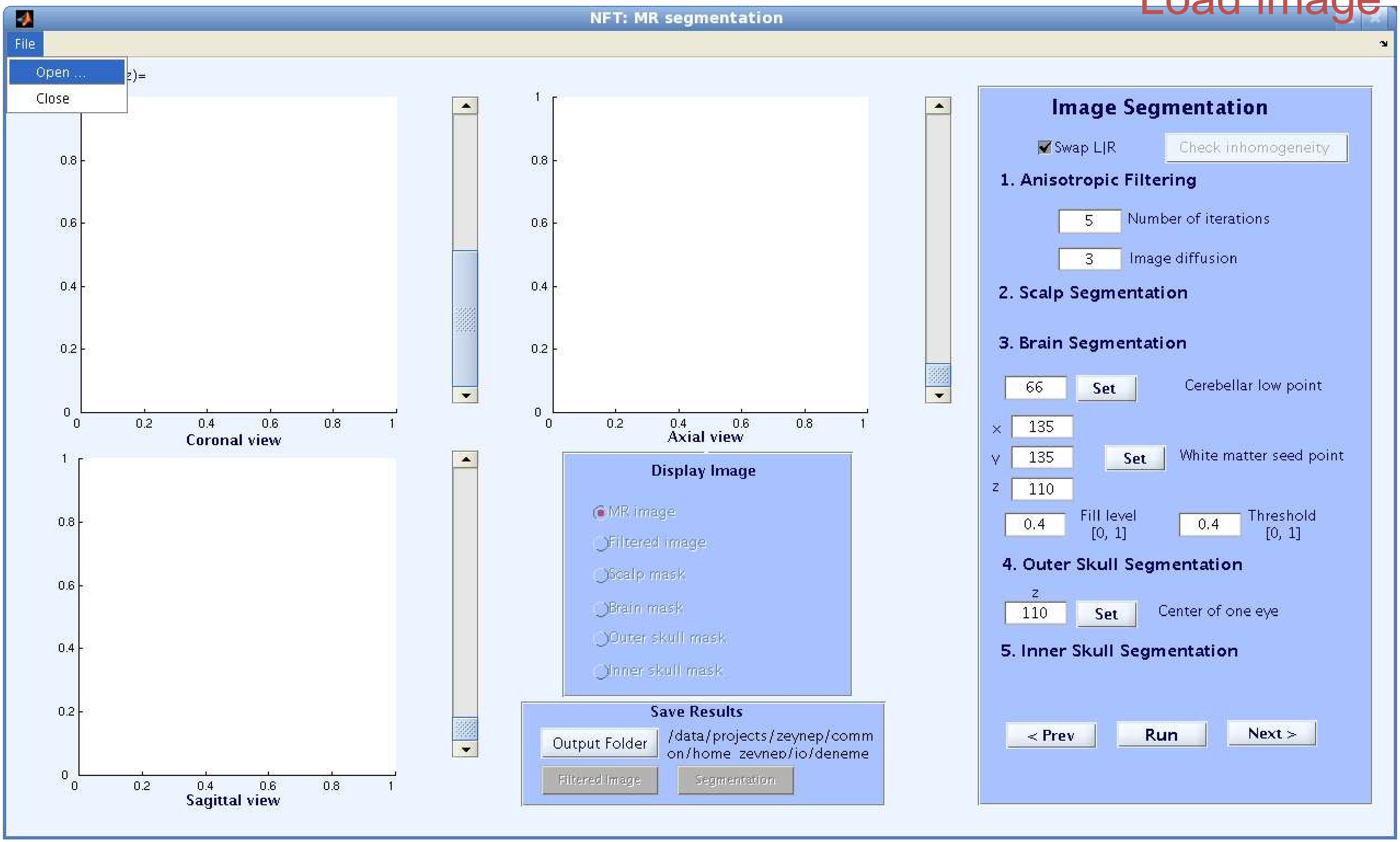


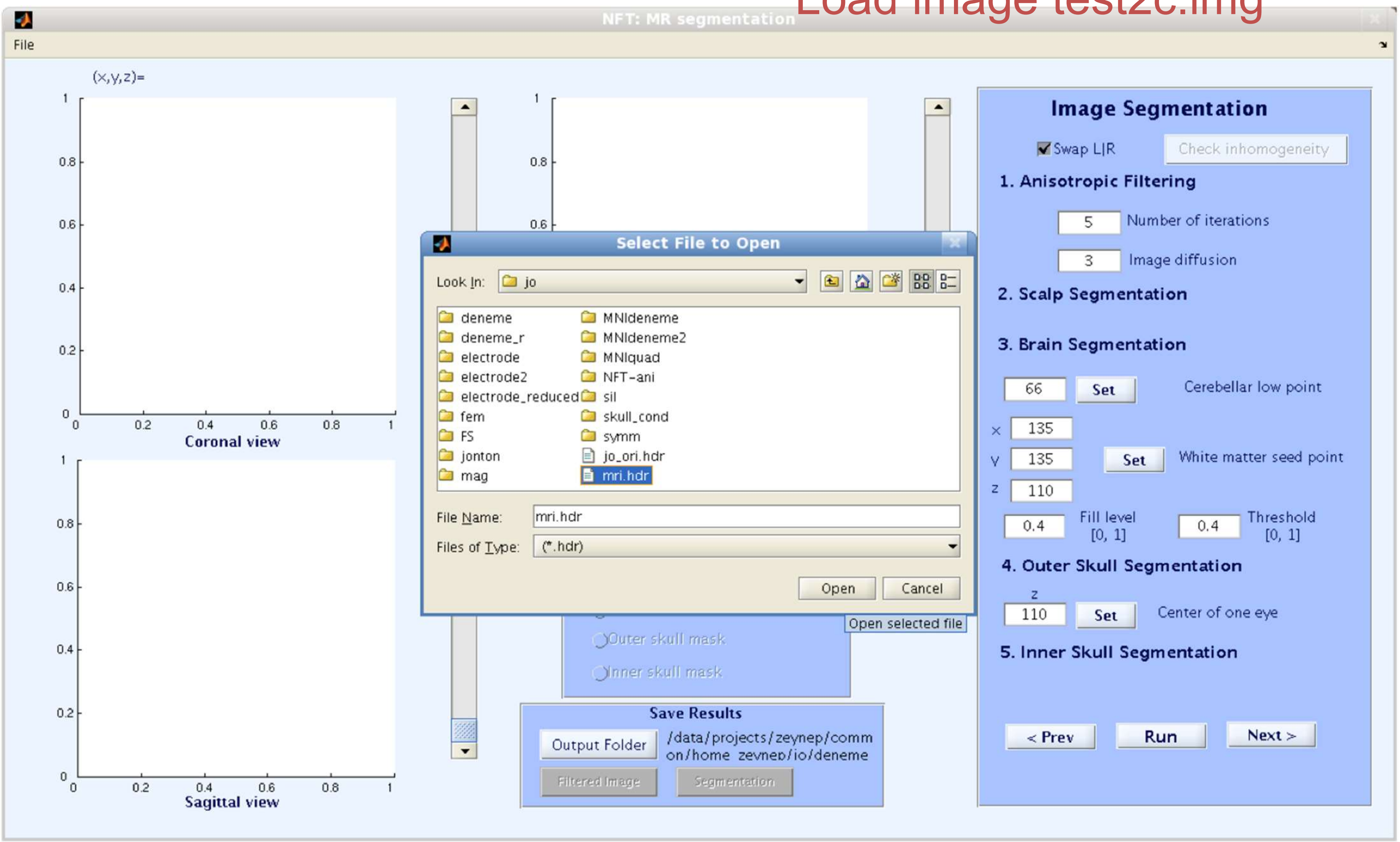
Image Segmentation

Load image



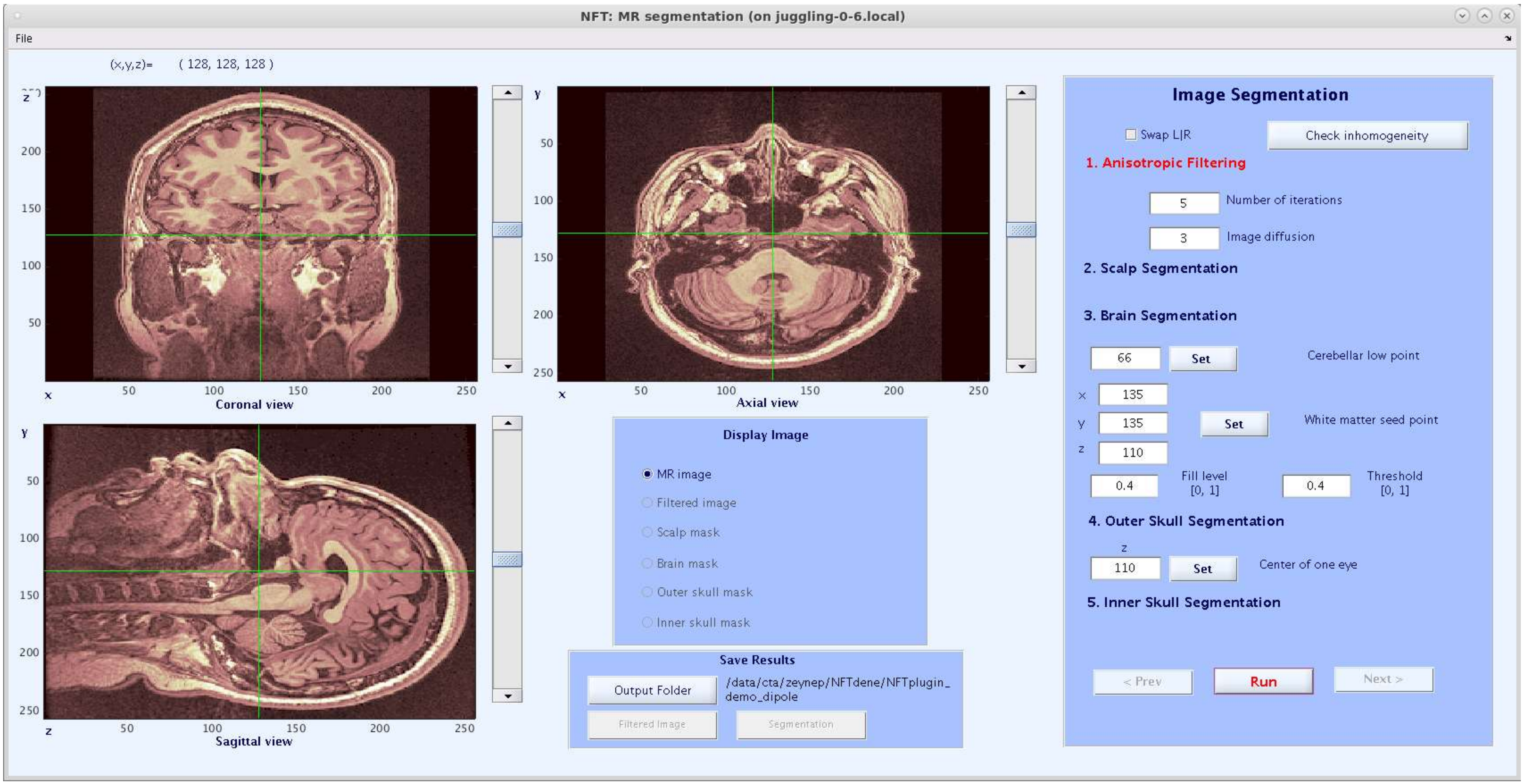
Segmentation

Load image test2c.img

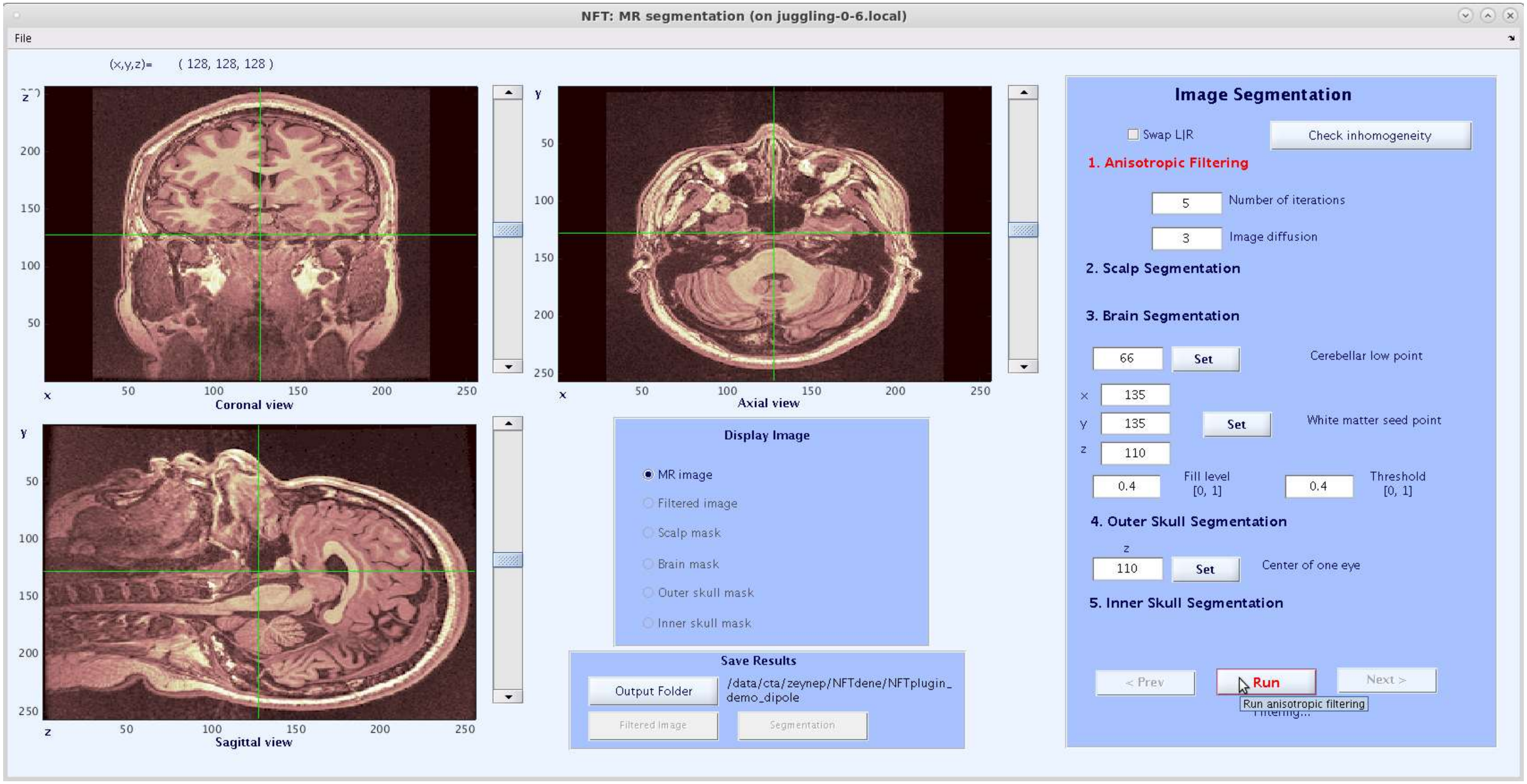


Segmentation

Run filtering

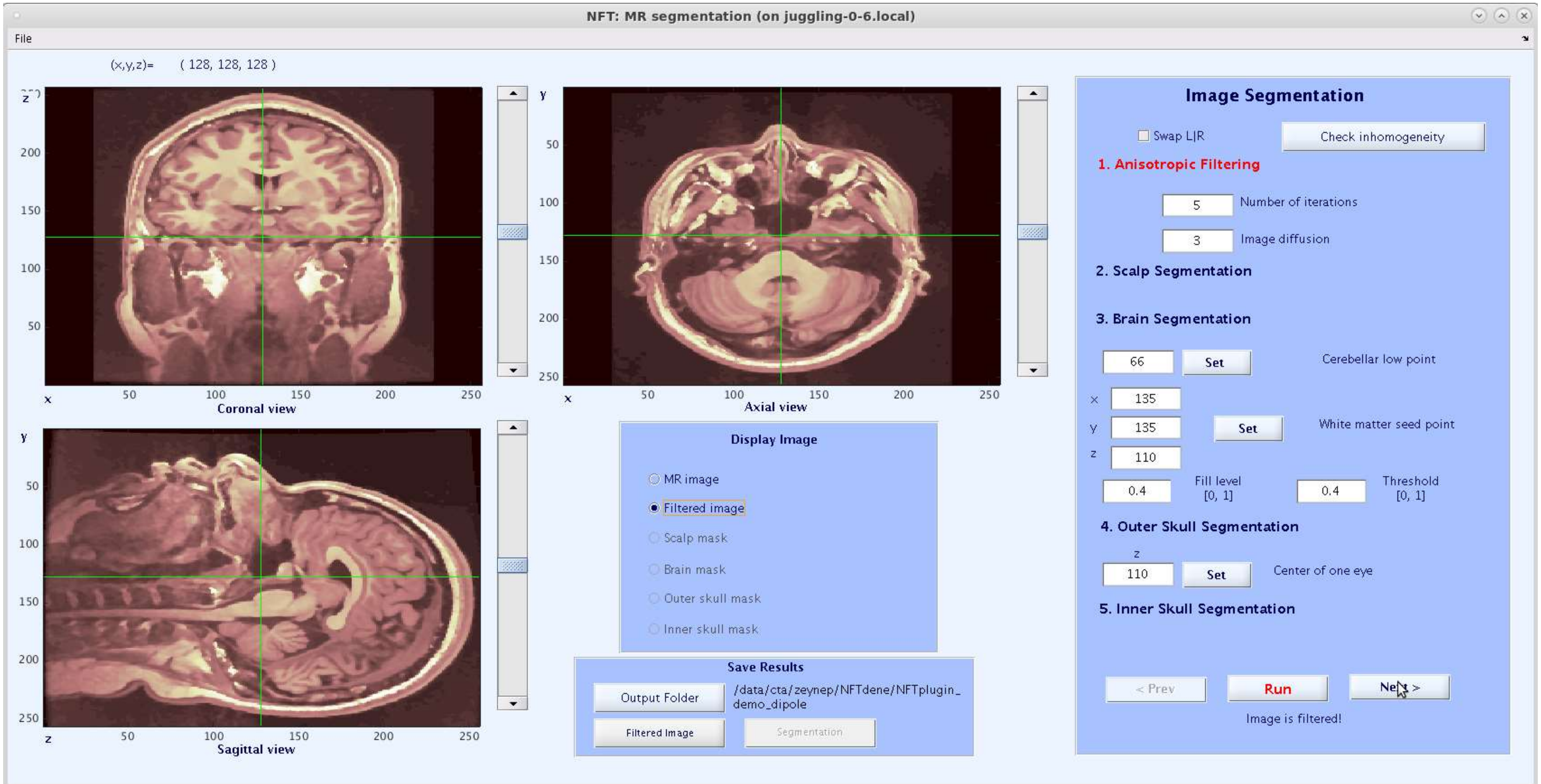


Segmentation



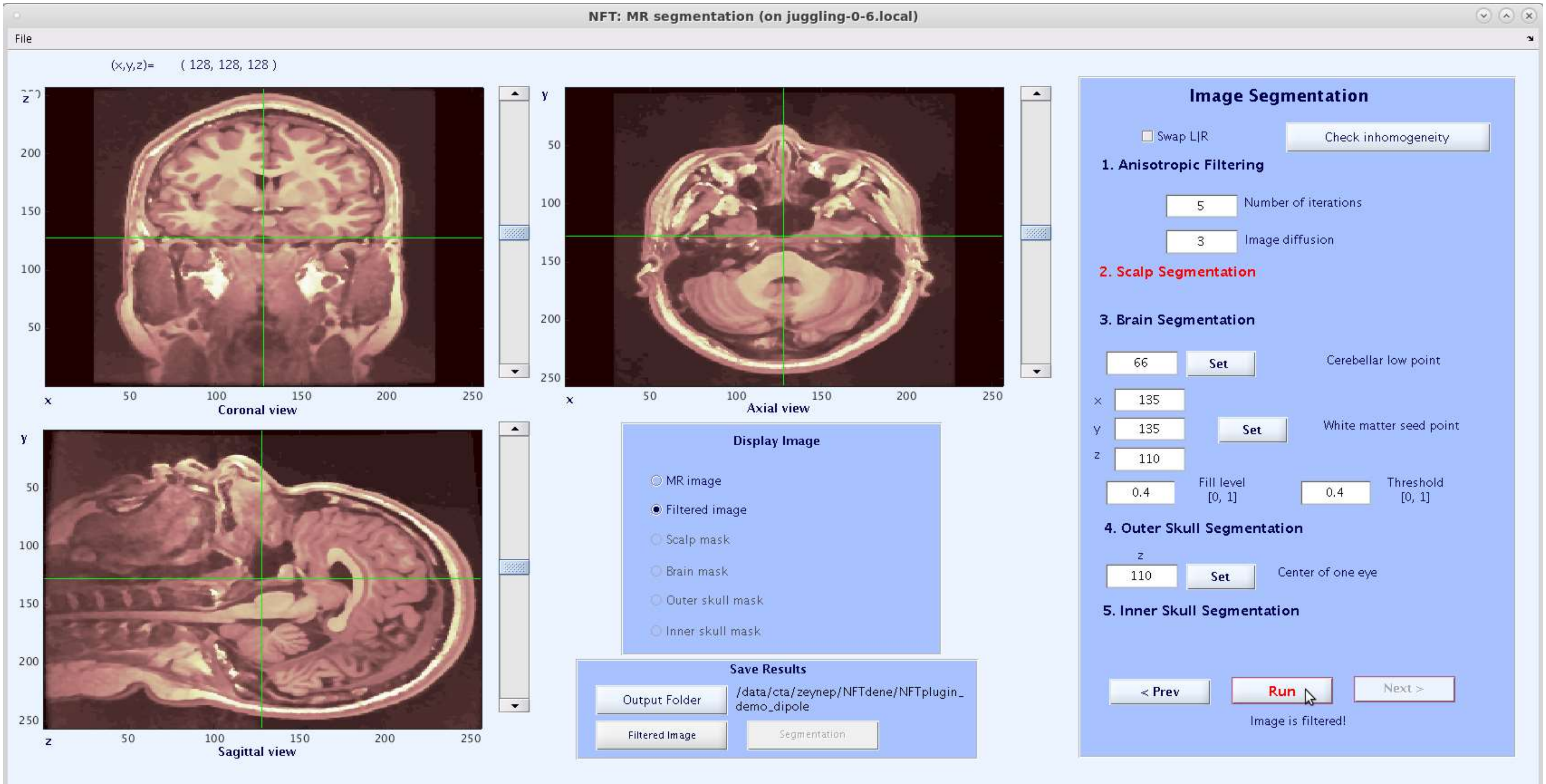
Segmentation

View filtered image



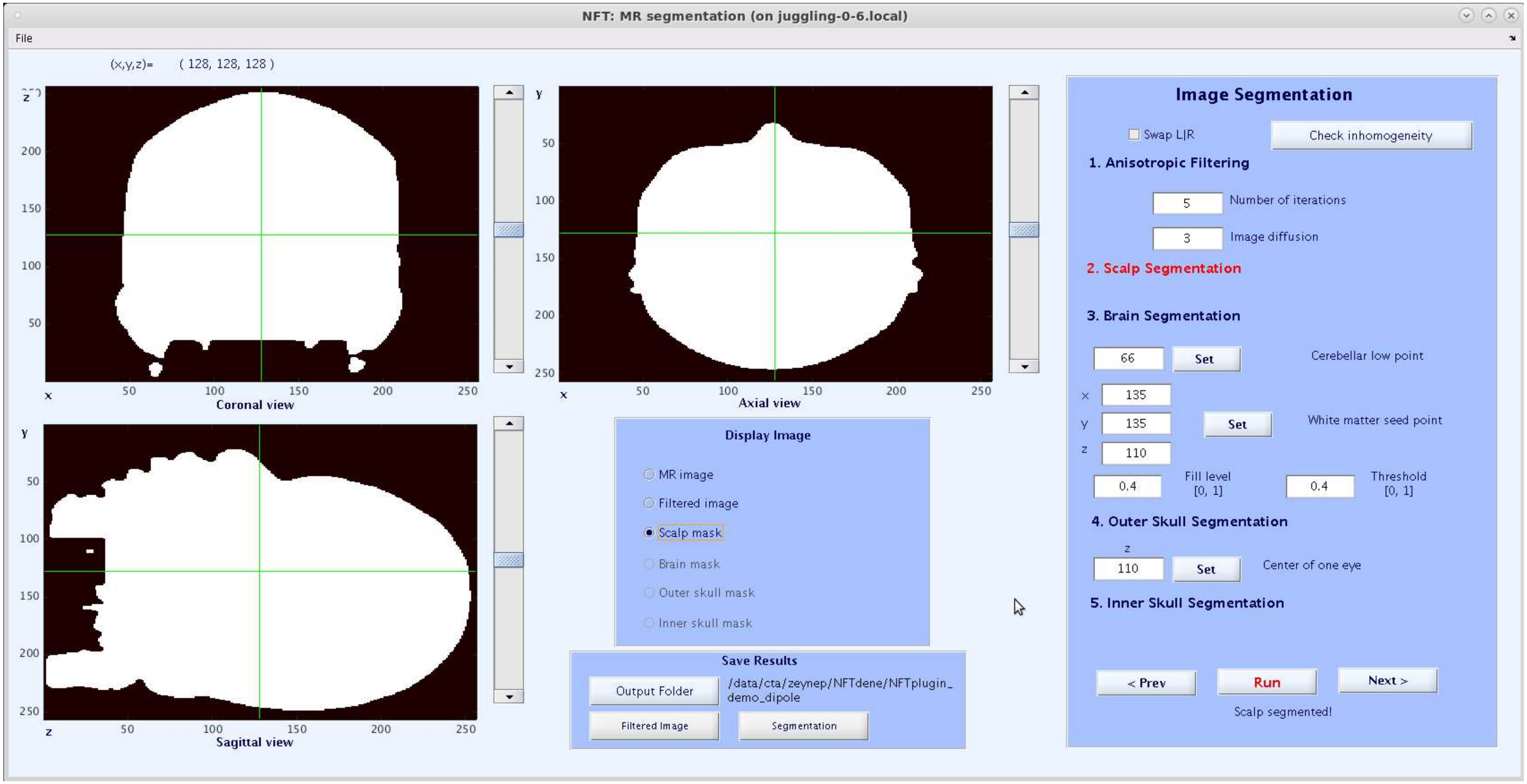
Segmentation

Click ‘Next’ for scalp segmentation and run scalp segmentation



Segmentation

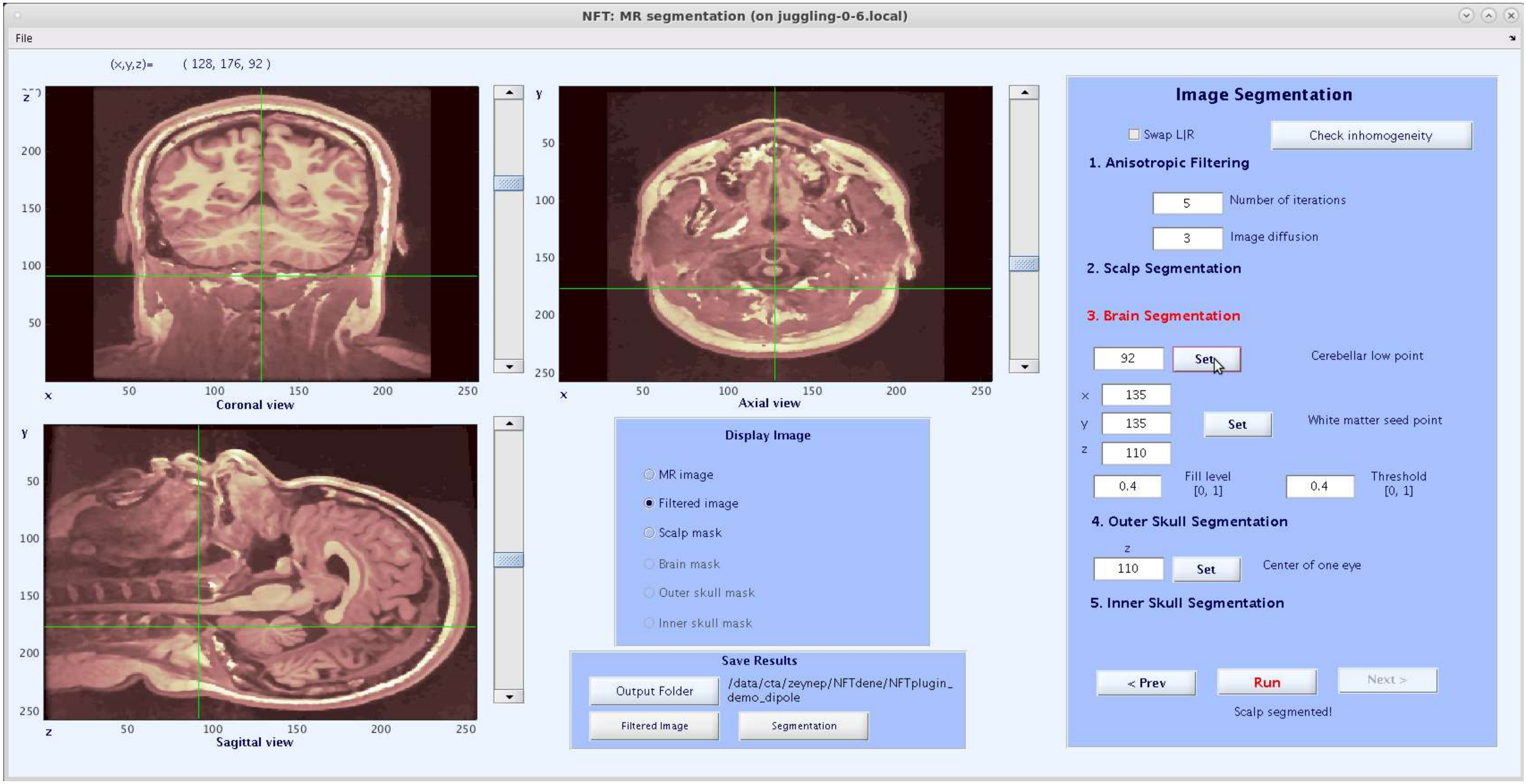
View scalp mask



Segmentation

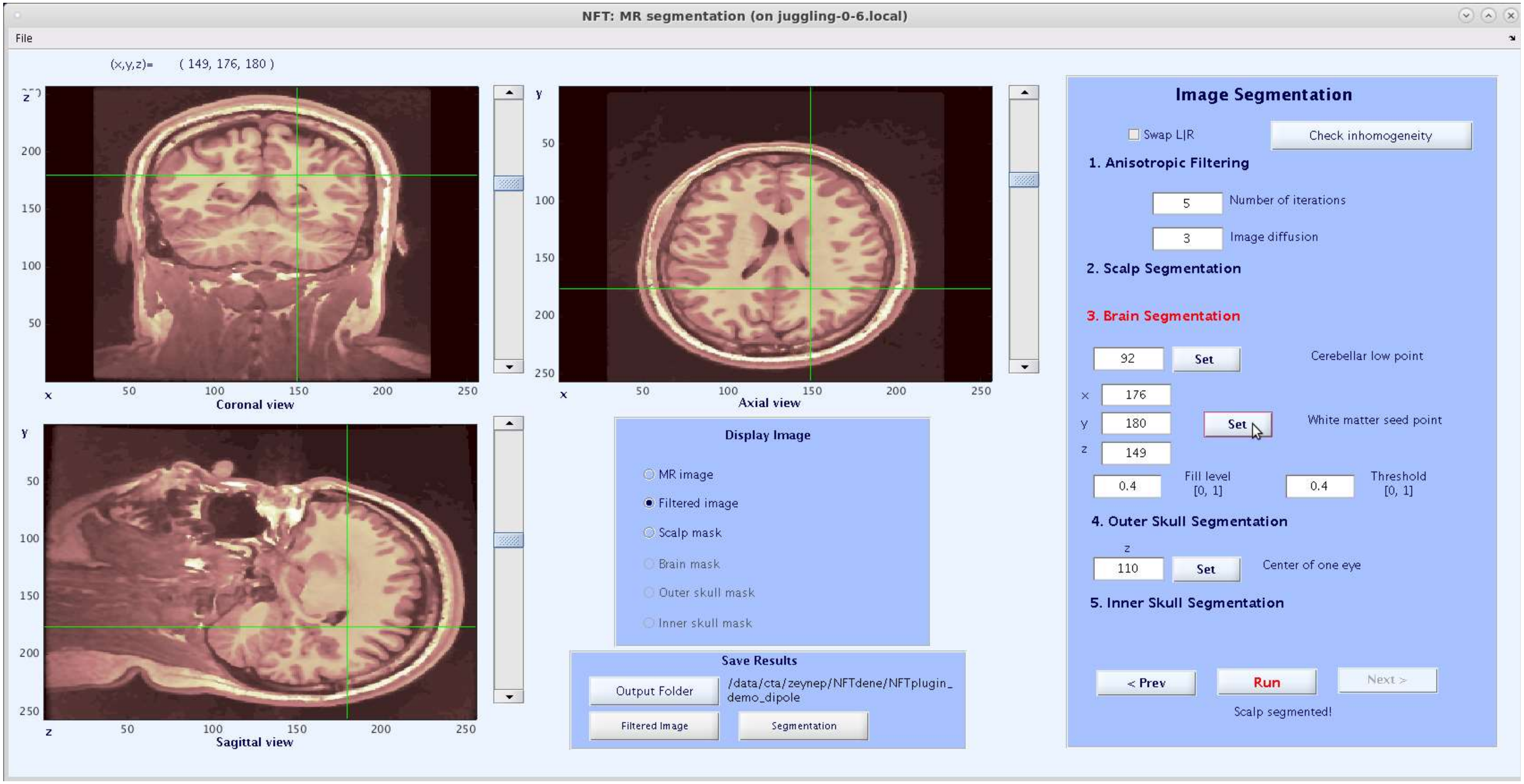
Click ‘Next’ for brain segmentation

Selection of cerebellar low point



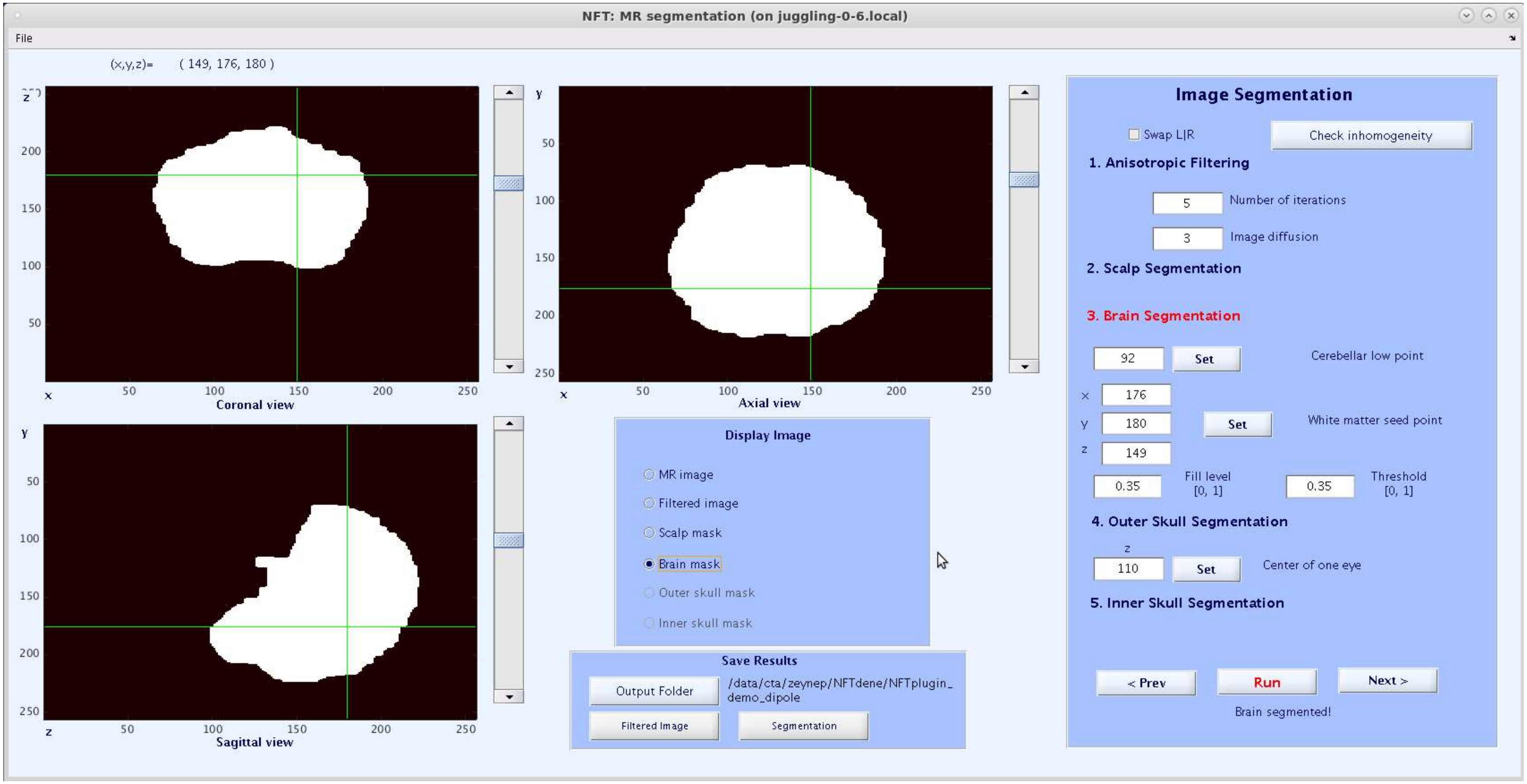
Segmentation

Selection of a white matter point



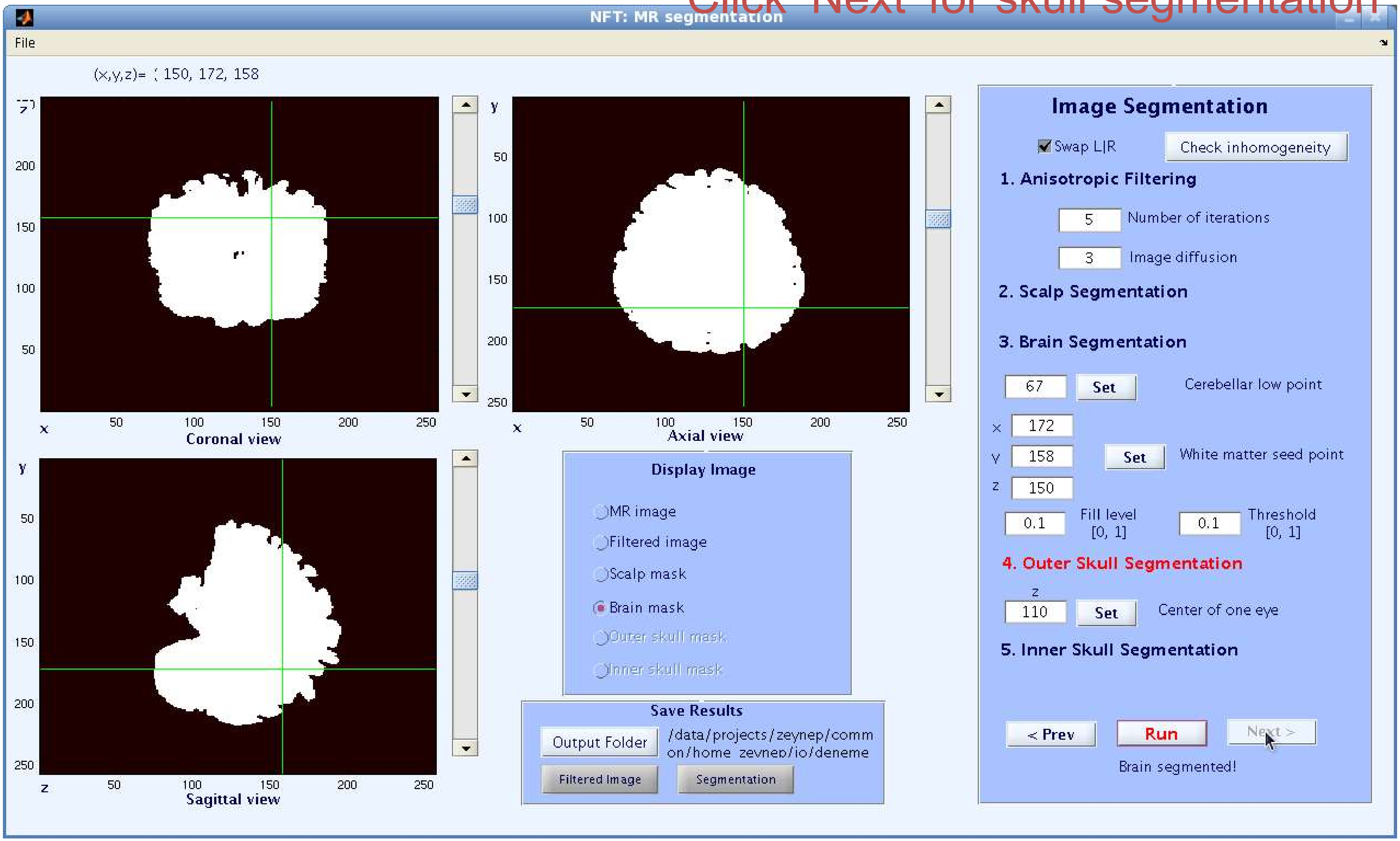
Segmentation

View brain mask



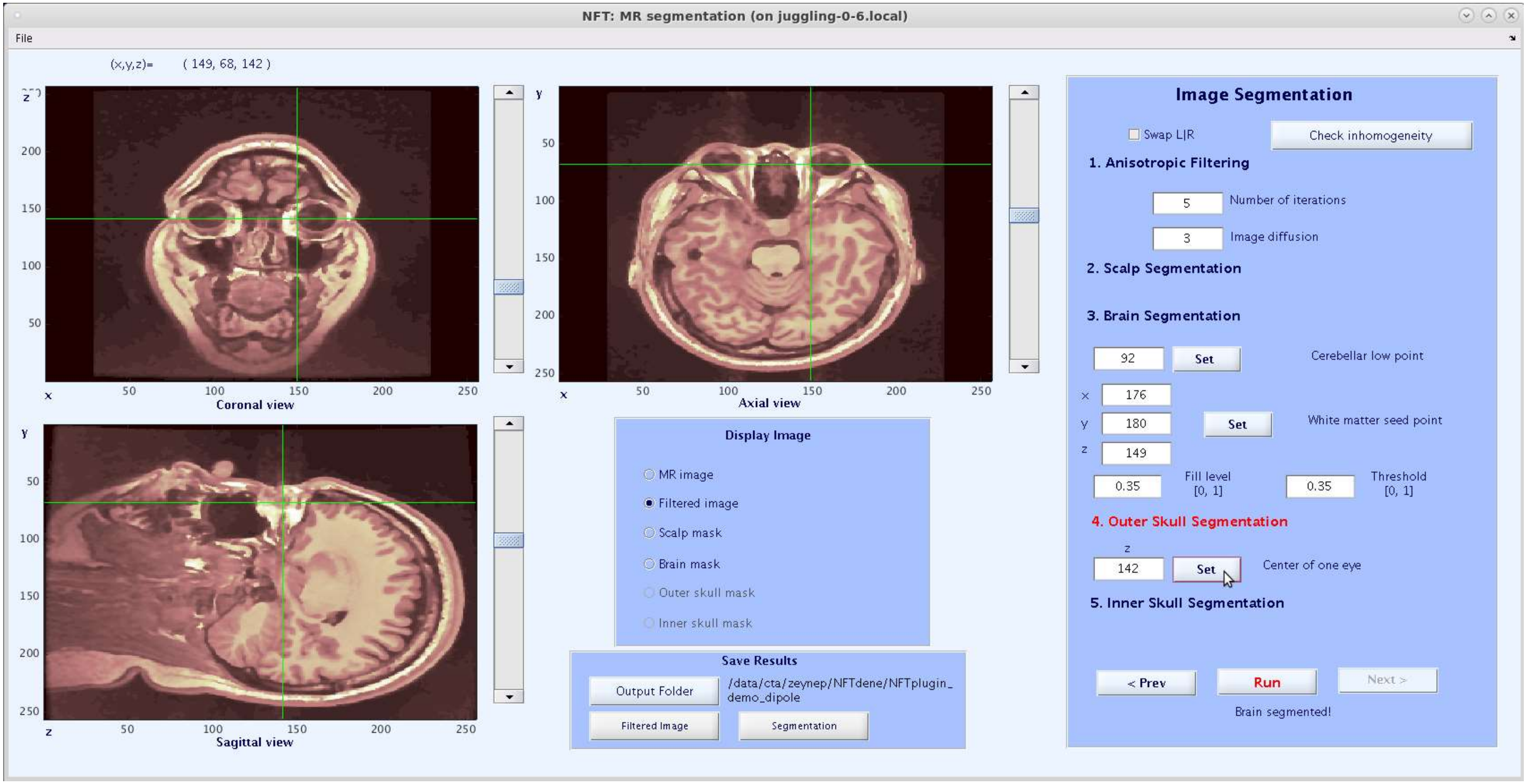
Segmentation

Click ‘Next’ for skull segmentation



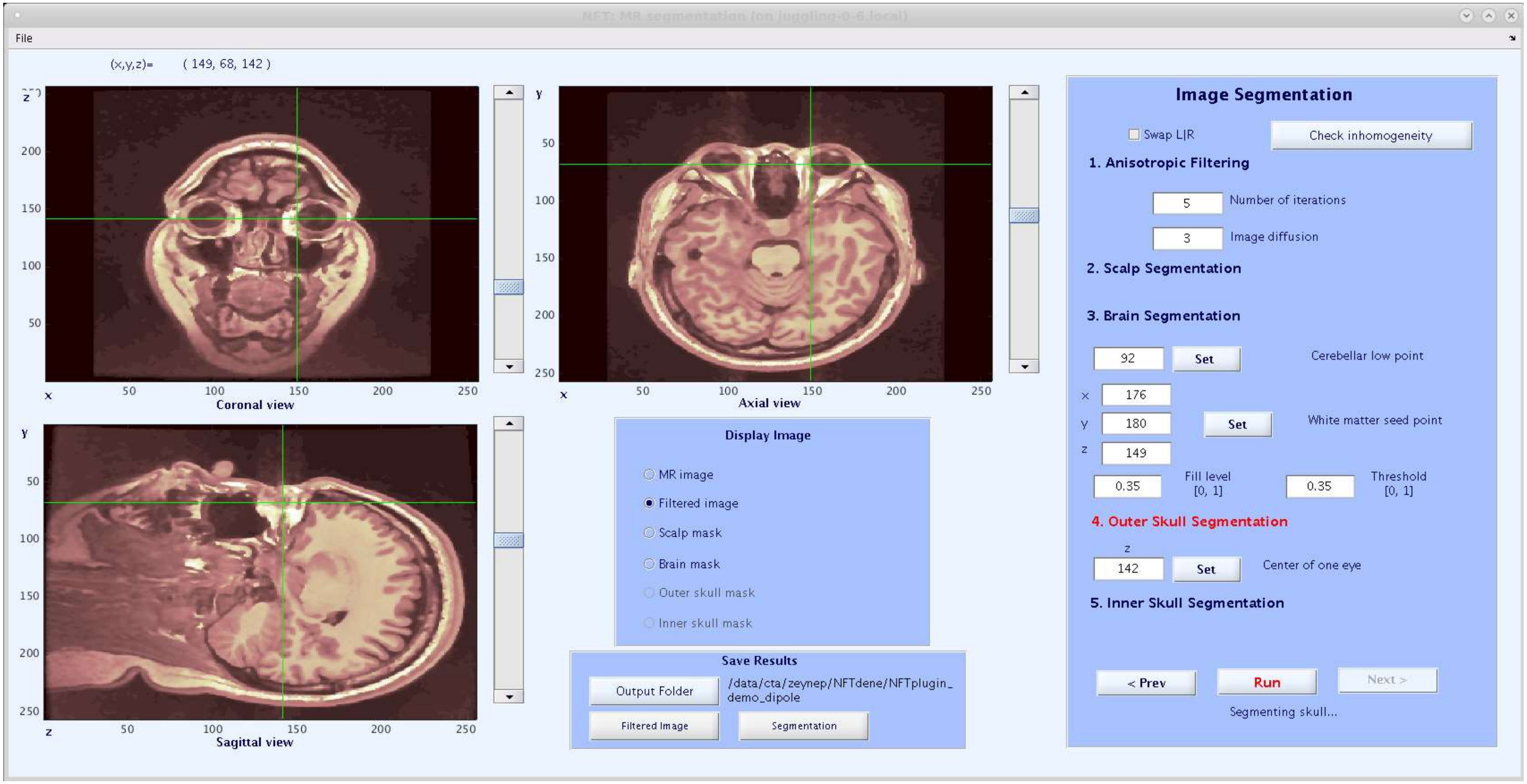
Segmentation

Select a slice for eyes and click ‘set’



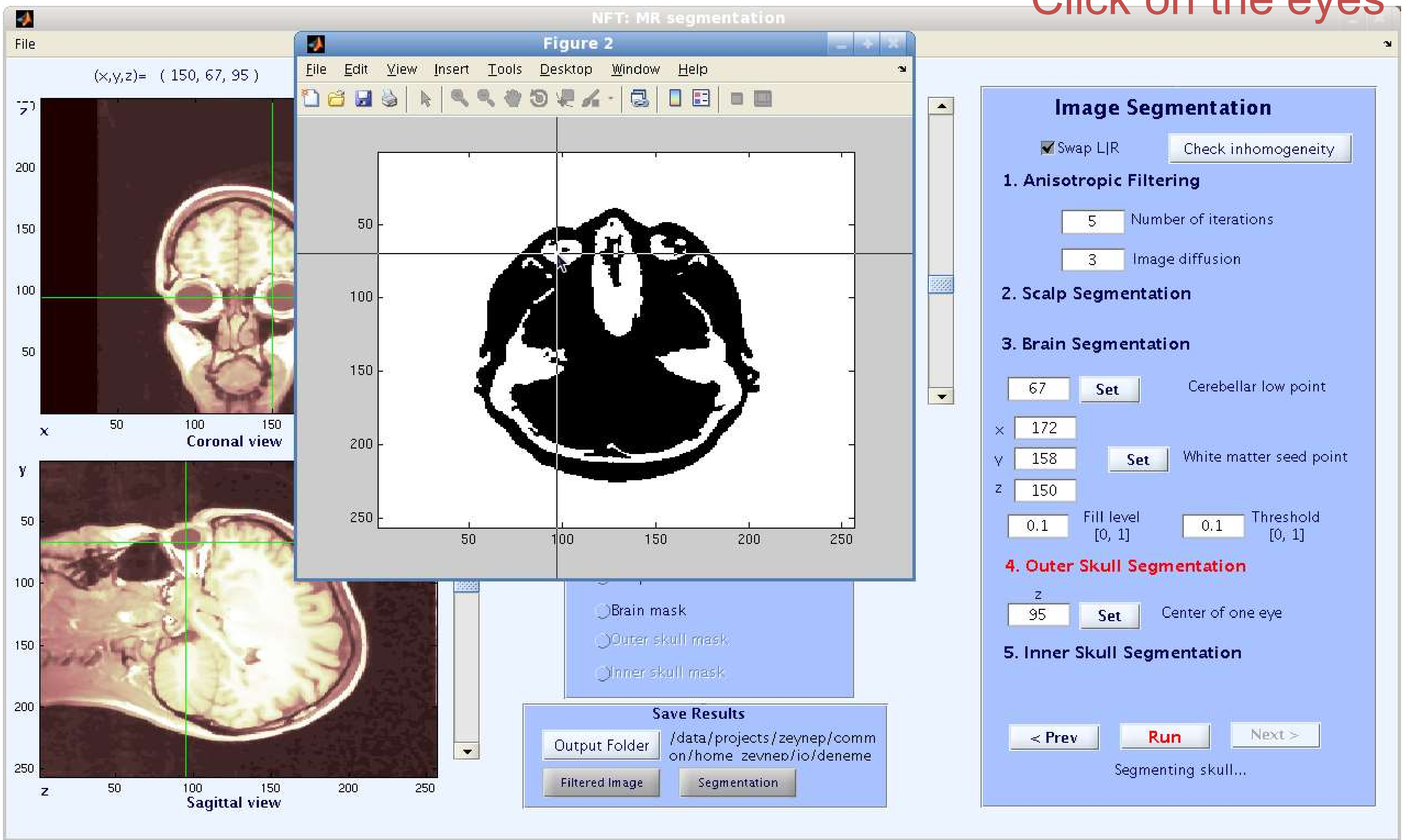
Segmentation

Click 'Run' for skull segmentation

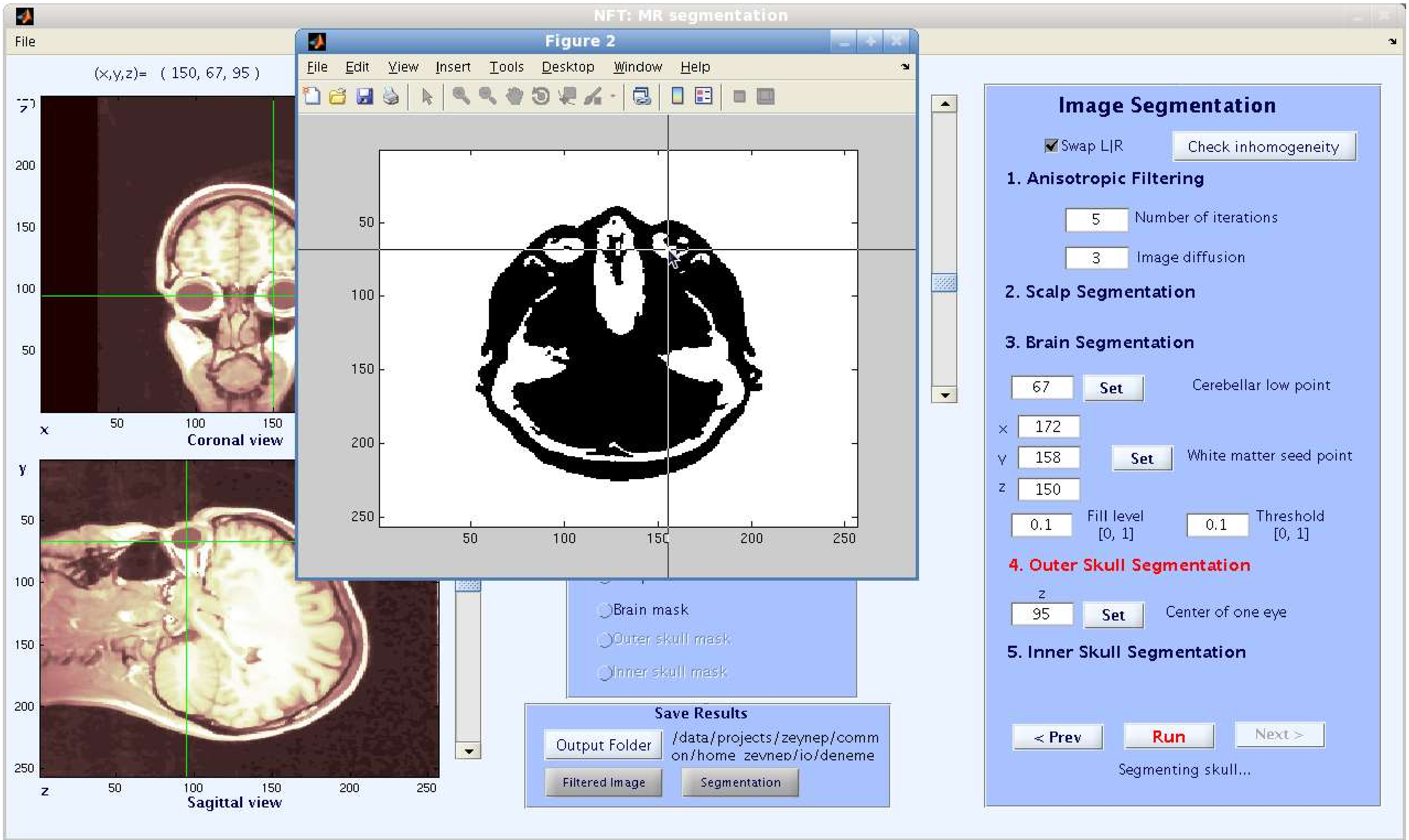


Segmentation

Click on the eyes

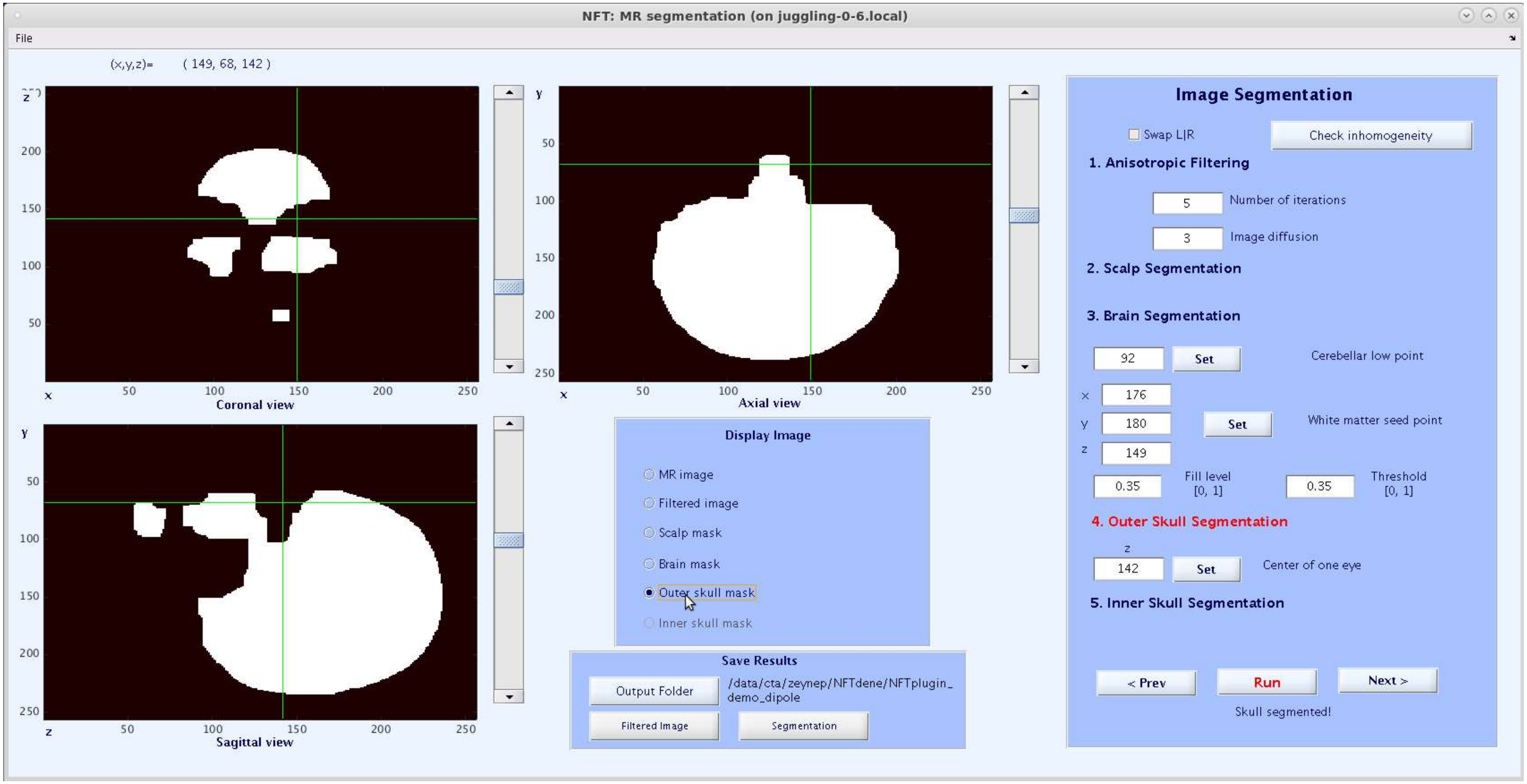


Segmentation



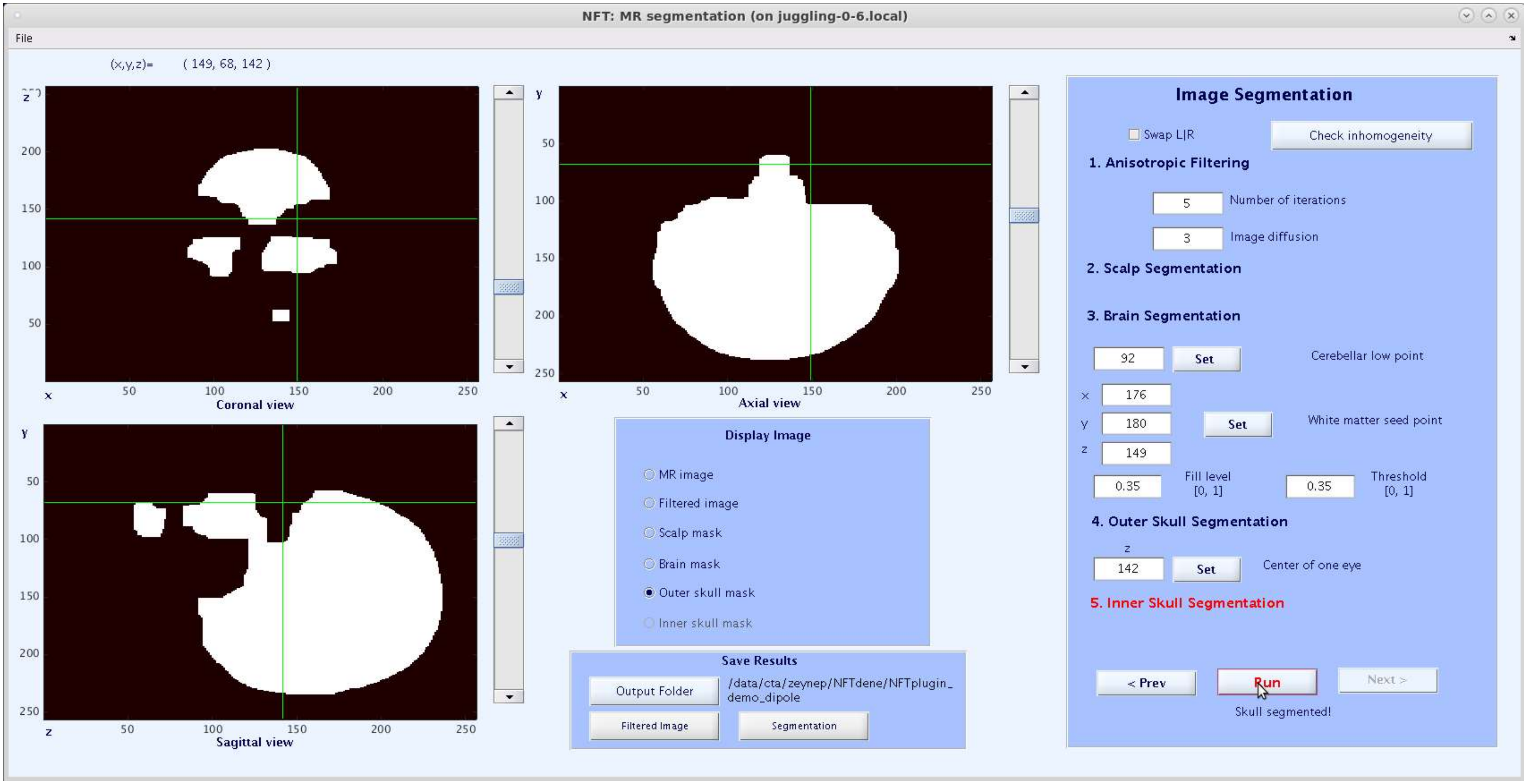
Segmentation

View skull segmentation



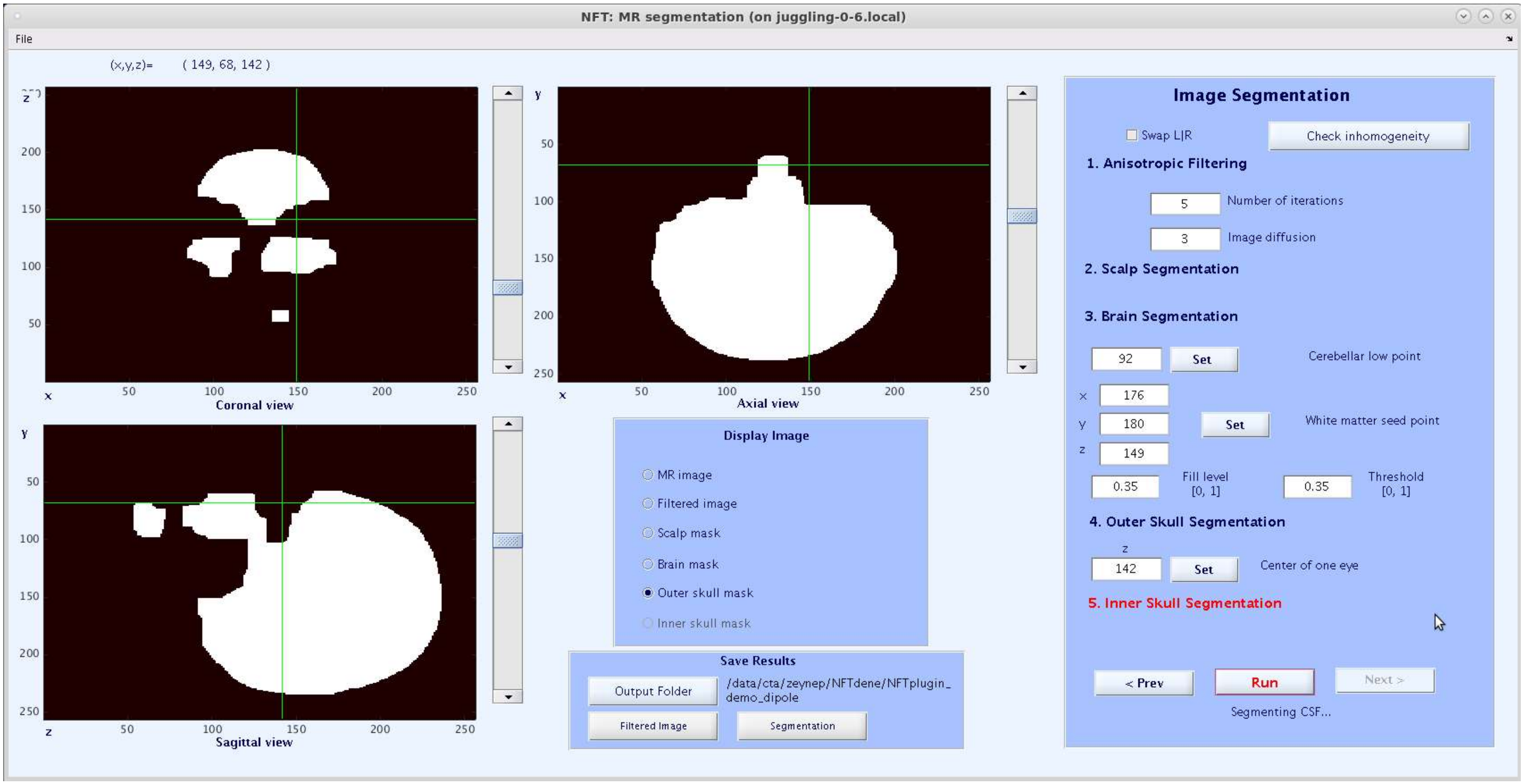
Segmentation

Click 'Next' for CSF segmentation

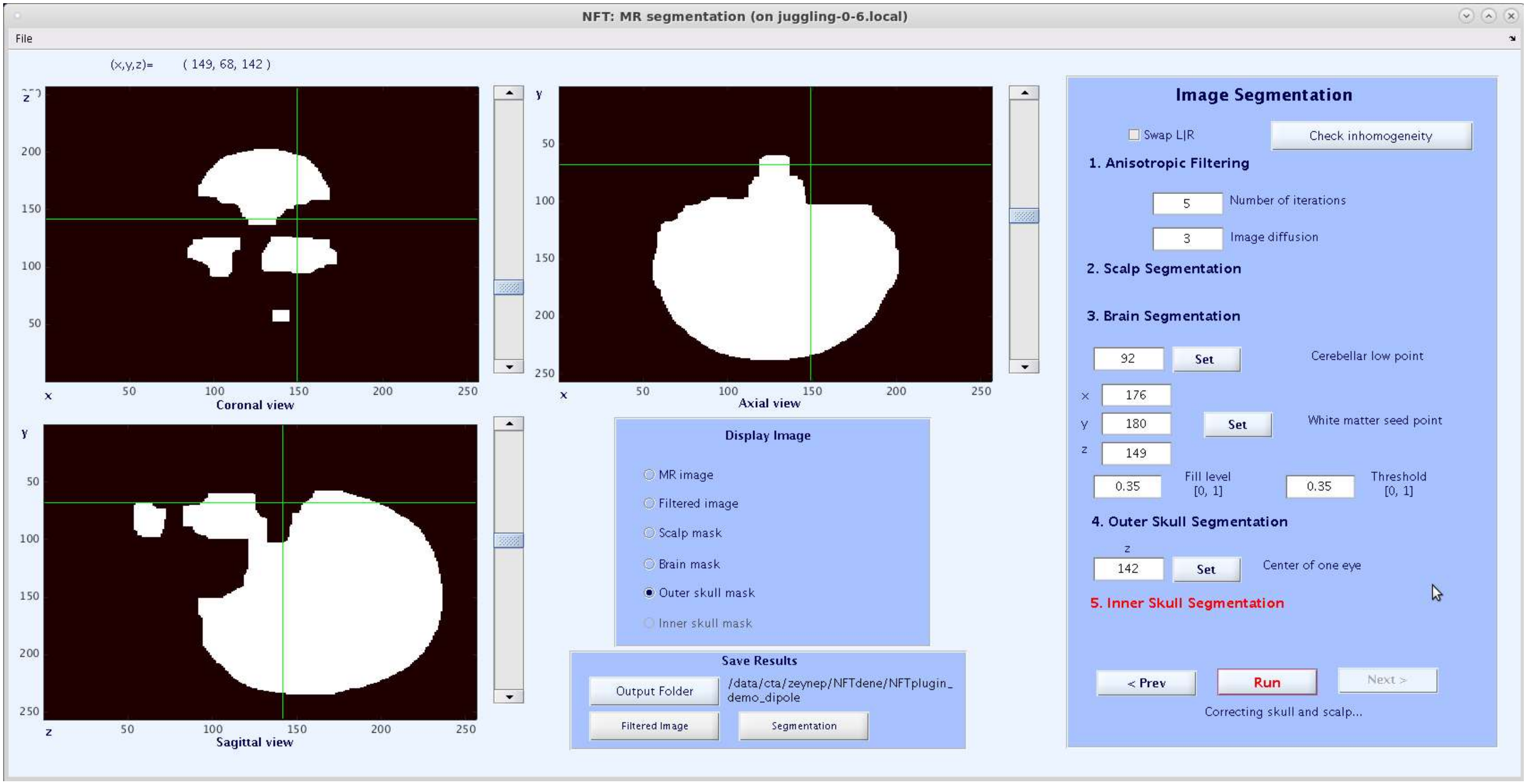


Segmentation

Click ‘Run’ for CSF segmentation

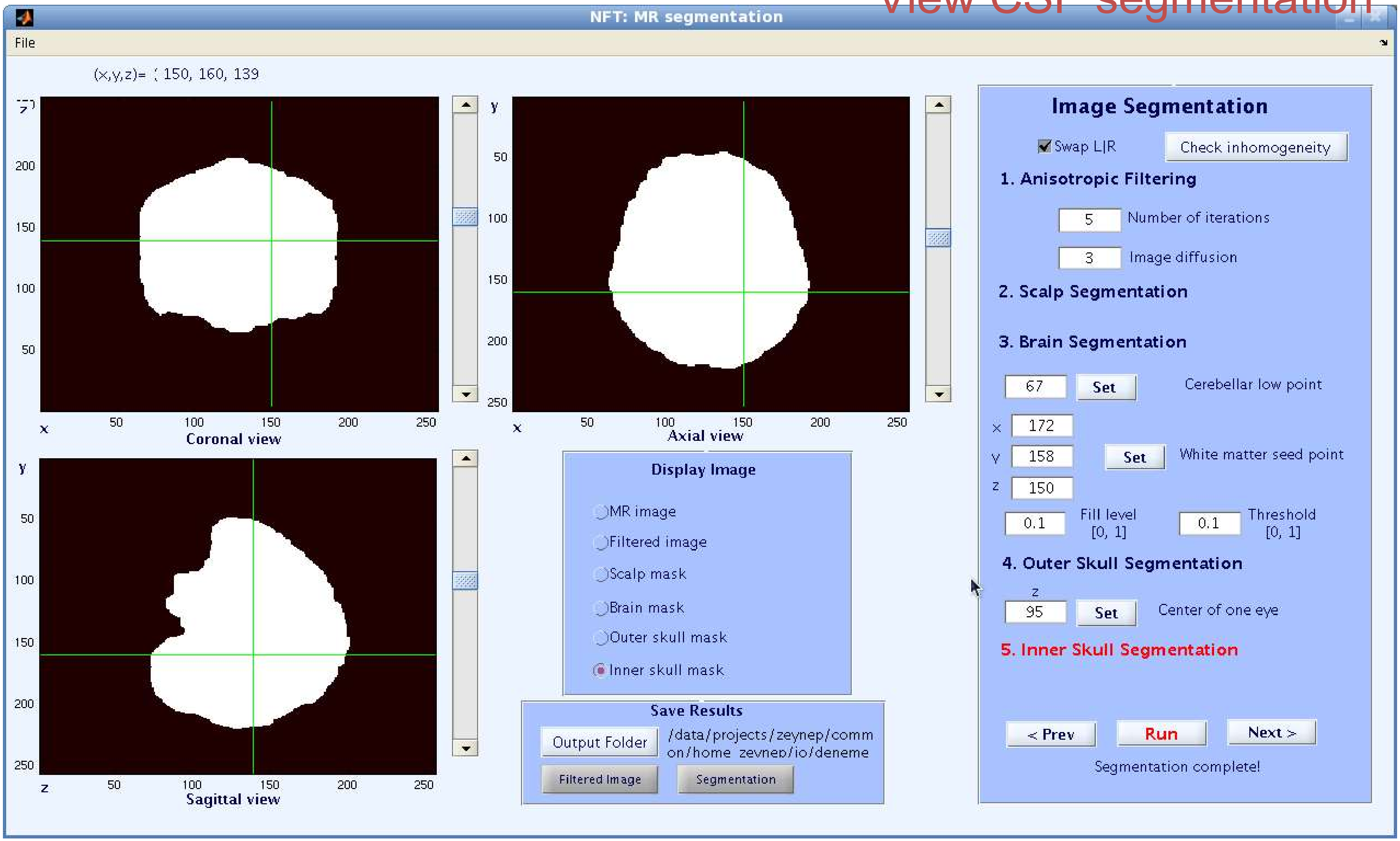


Segmentation



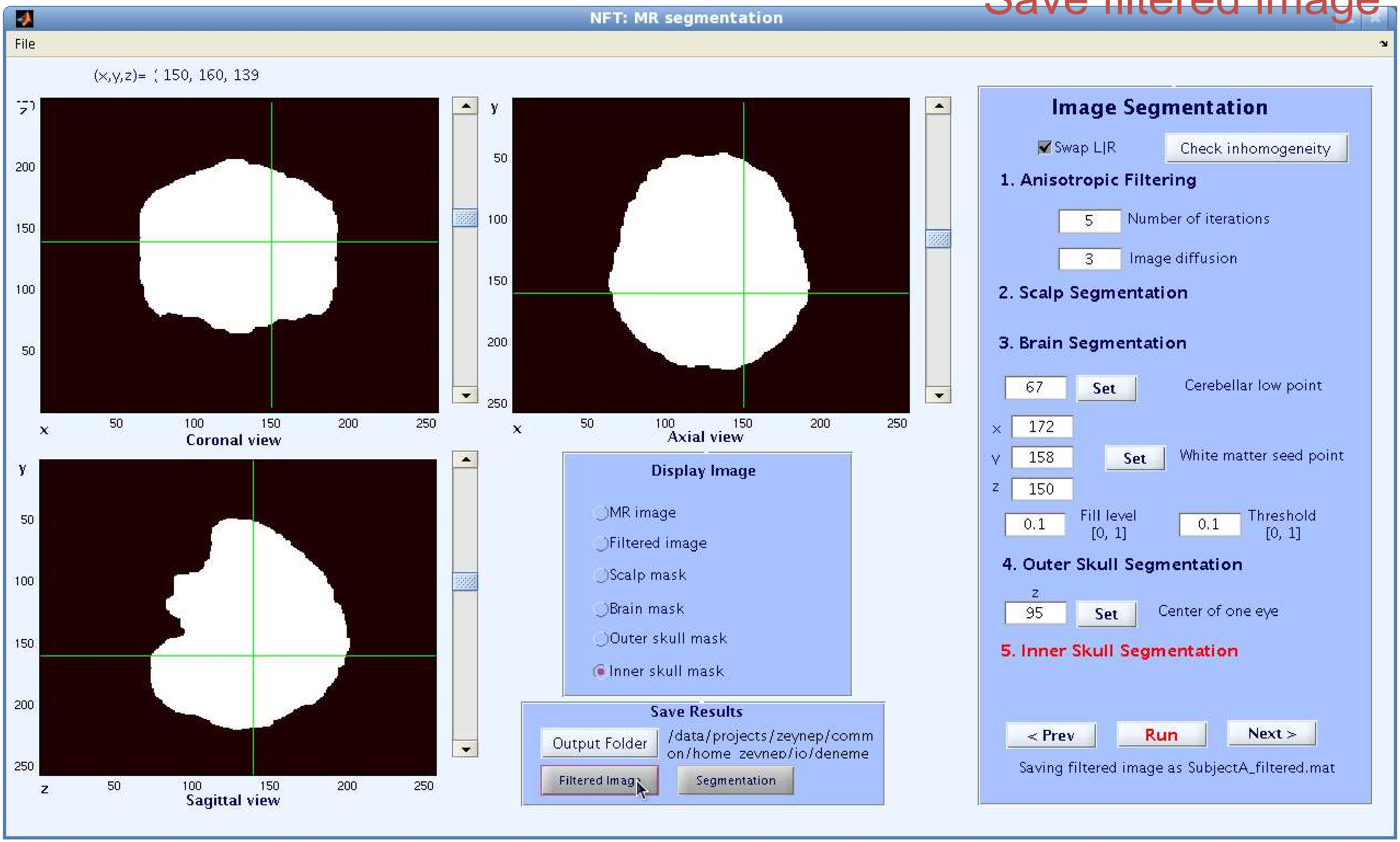
Segmentation

View CSF segmentation

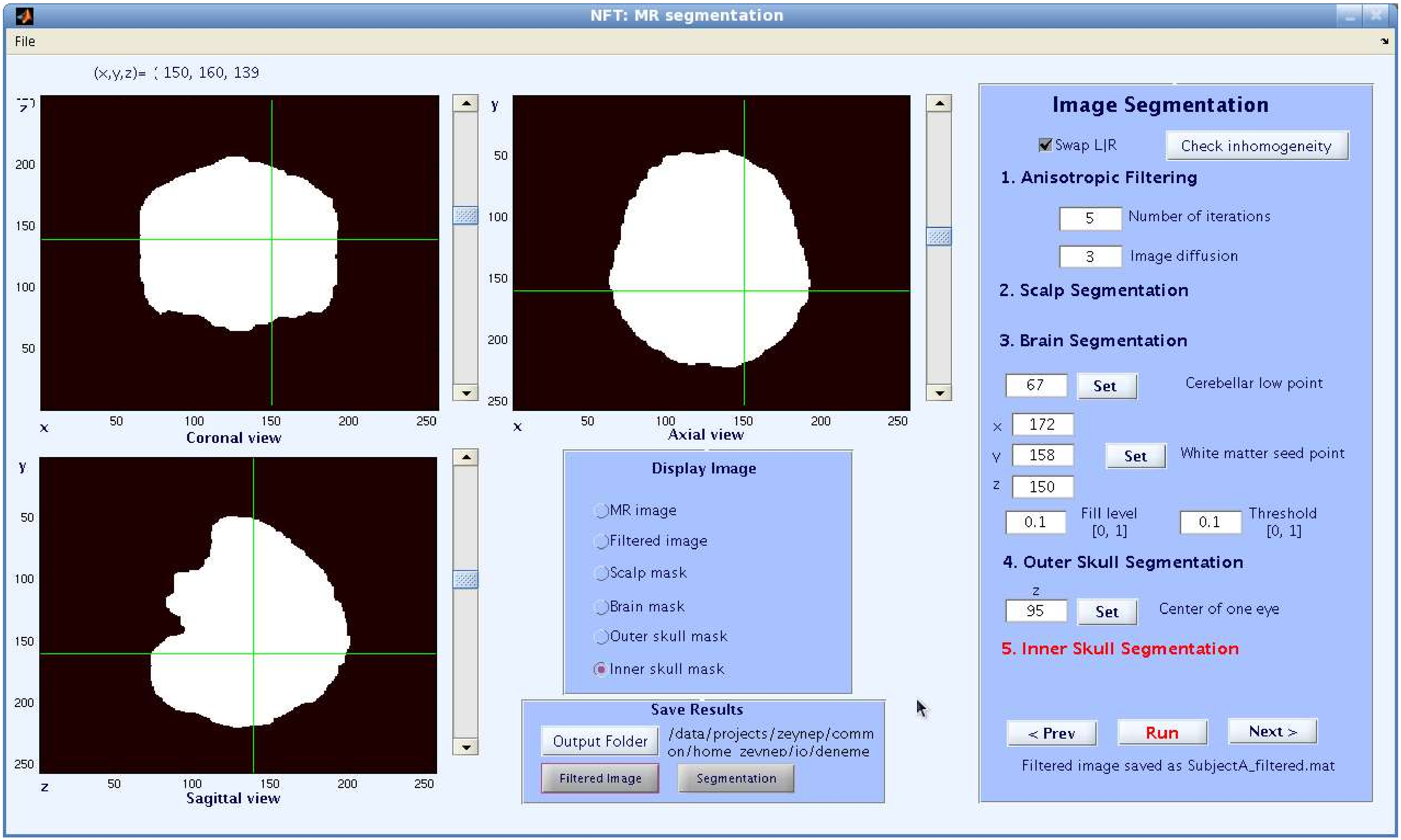


Segmentation

Save filtered image

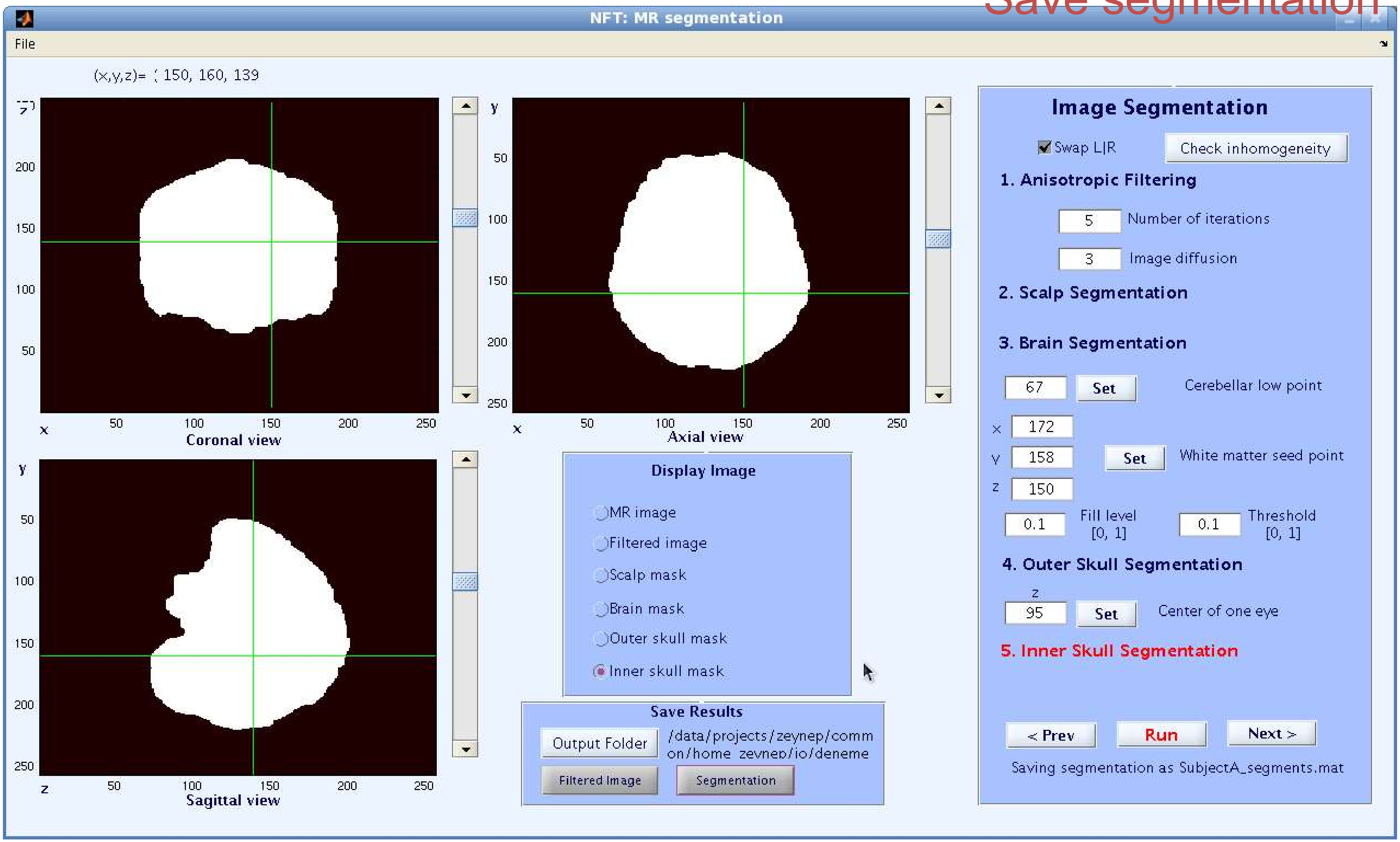


Segmentation



Segmentation

Save segmentation



Segmentation

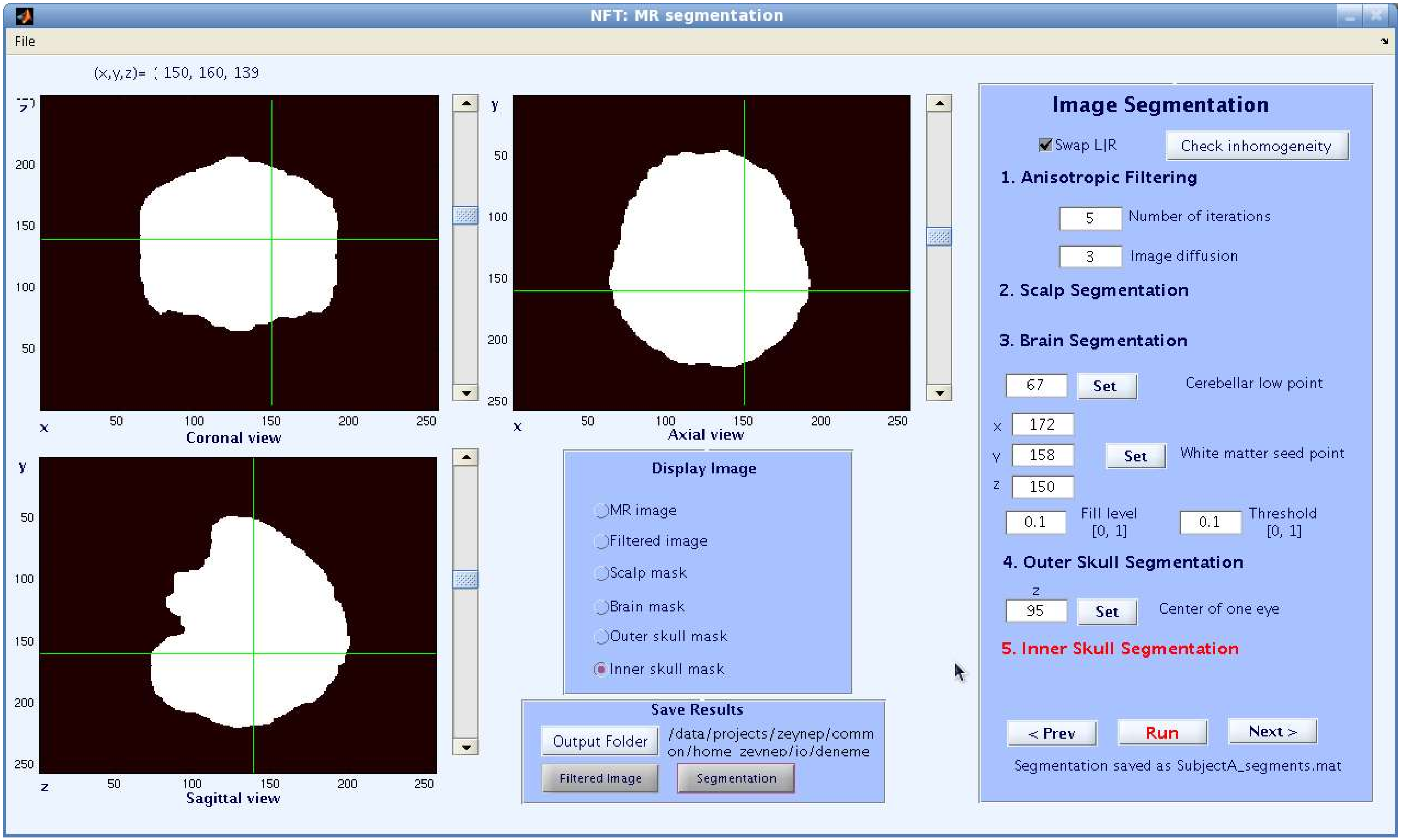
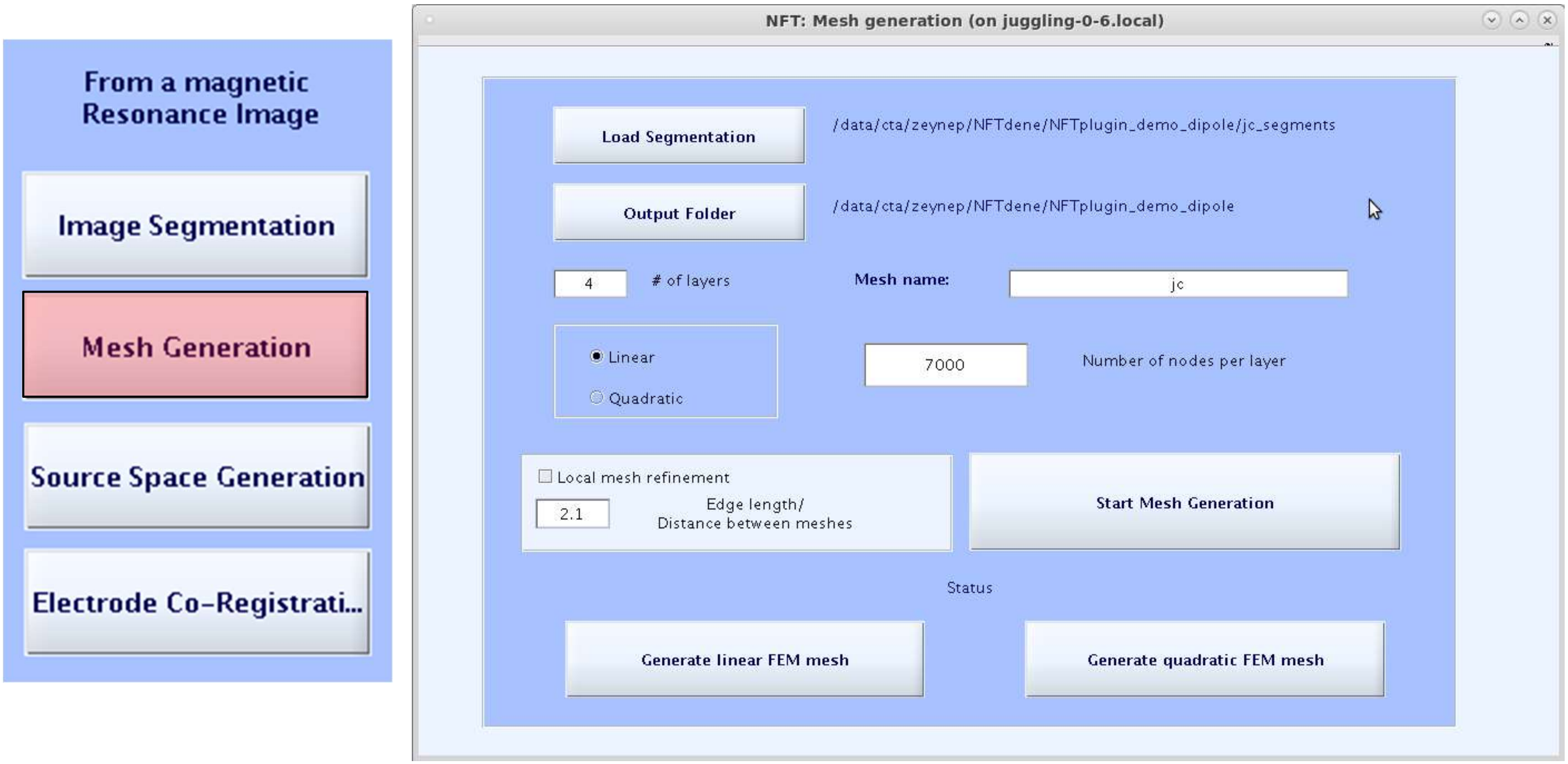


Image Segmentation

Command Window

```
>> load jc_mri
>> mri

mri = 

  struct with fields:

          dim: [256 256 256]
        xgrid: [1×256 double]
        ygrid: [1×256 double]
        zgrid: [1×256 double]
      anatomy: [256×256×256 double]
    transform: [4×4 double]
          hdr: 'test2c.hdr'

>> load jc_segments
>> Segm

Segm = 

  struct with fields:

        scalpmask: [256×256×256 logical]
        brainmask: [256×256×256 logical]
    outerskullmask: [256×256×256 logical]
    innerskullmask: [256×256×256 logical]
       parameters: [1×1 struct]
```

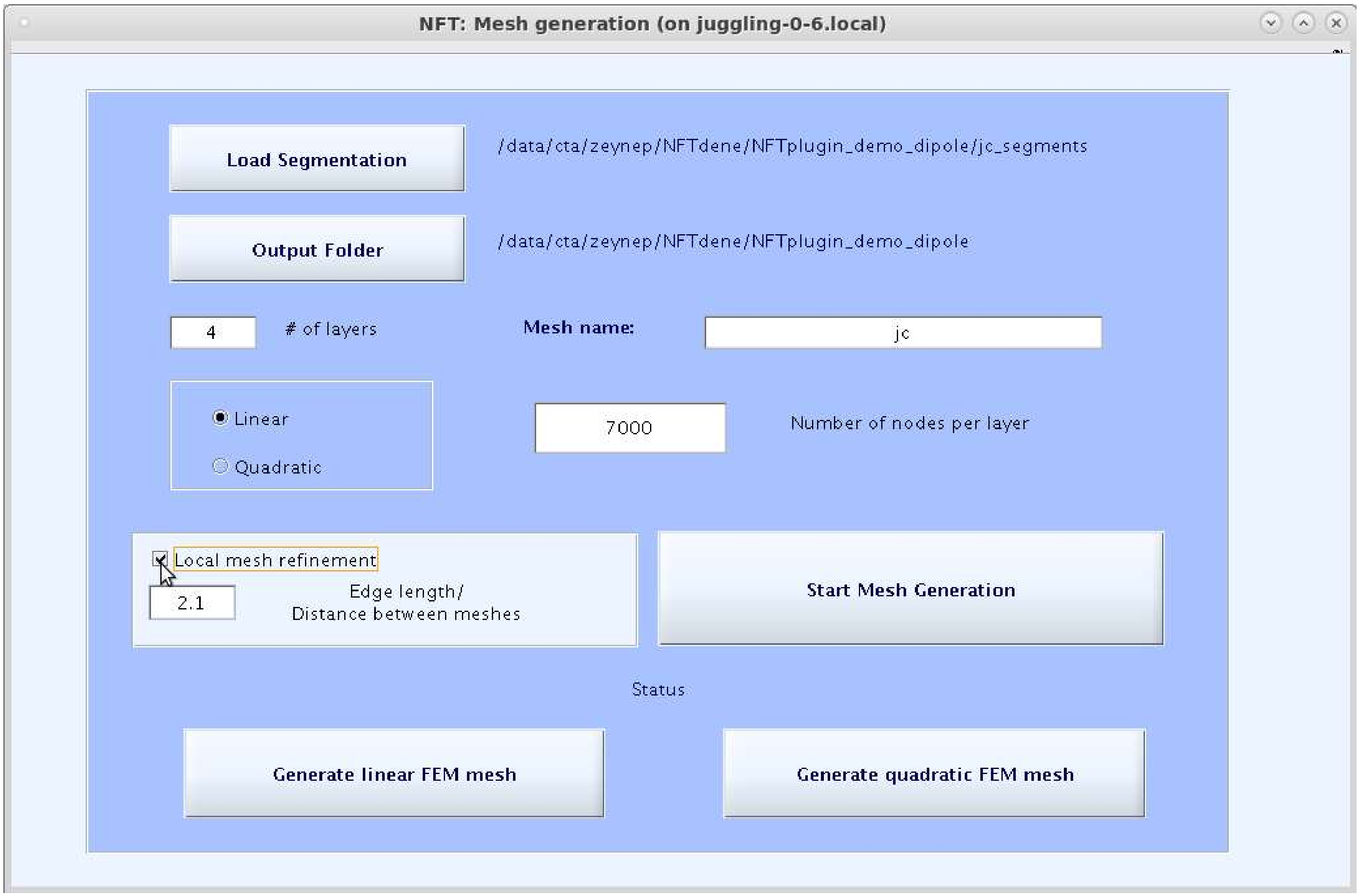
Mesh Generation



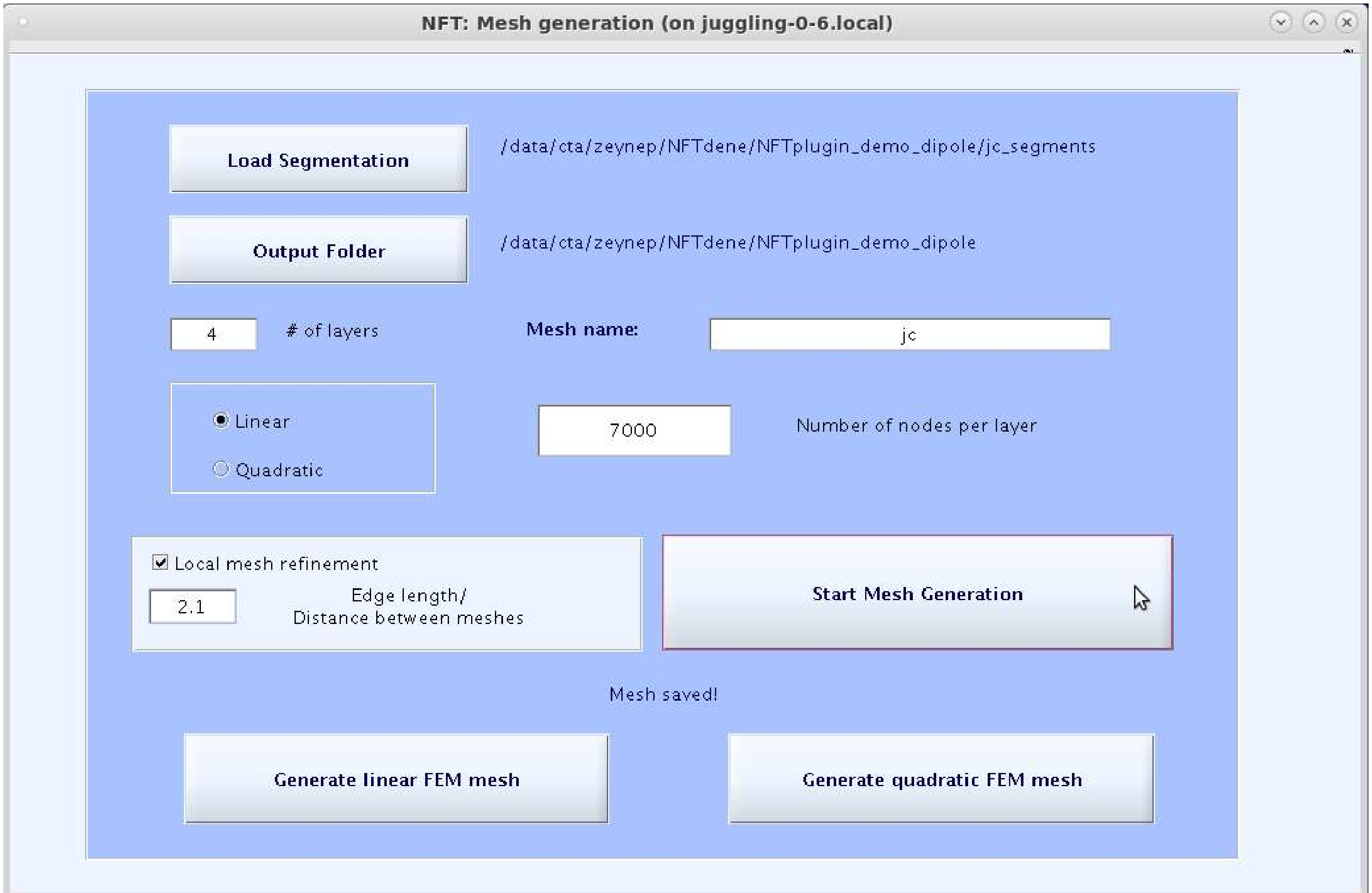
Generate Mesh for a 3 or 4 layer head model

Mesh generation

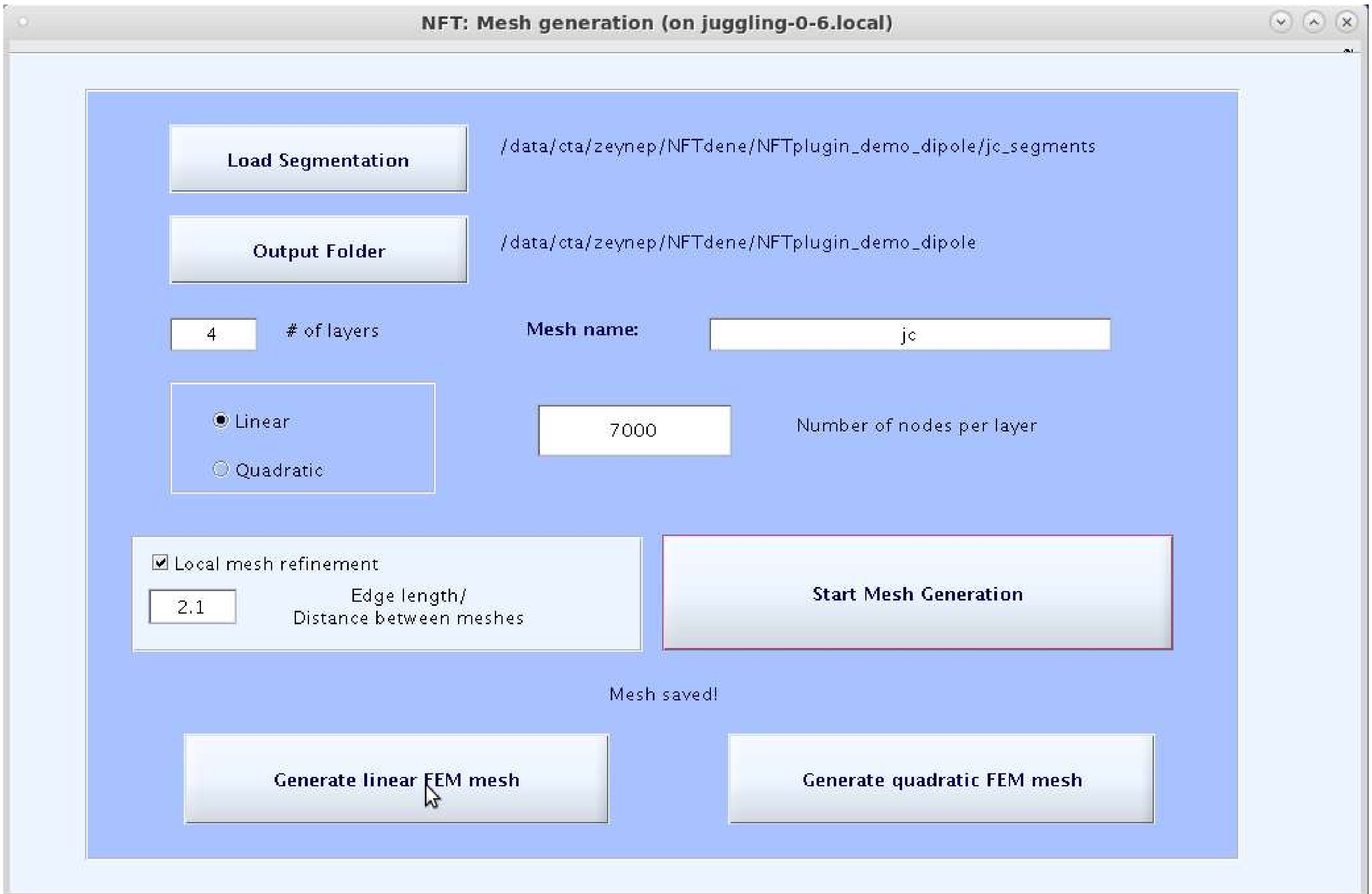
Click local mesh refinement



Mesh generation

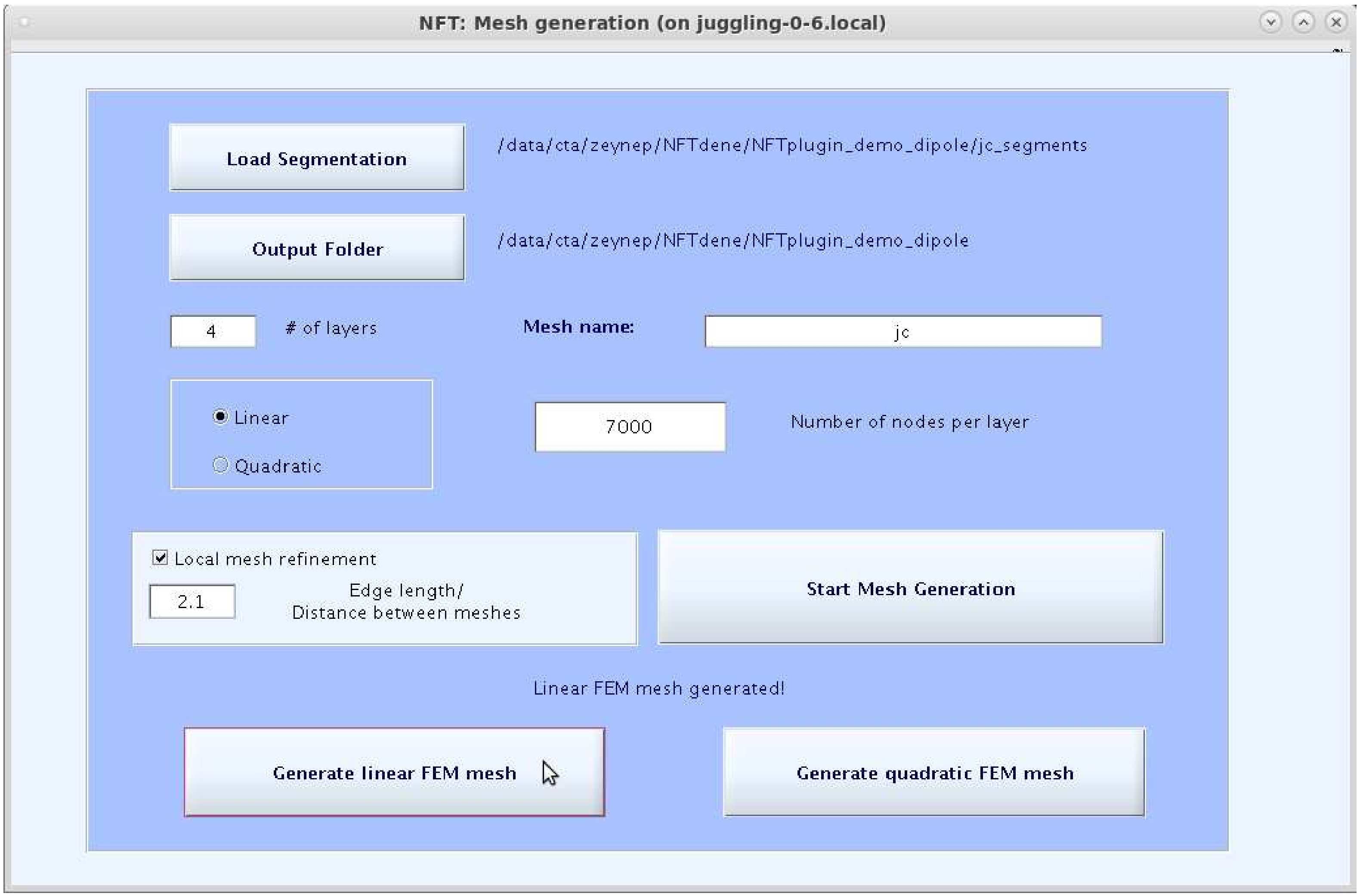


Mesh generation

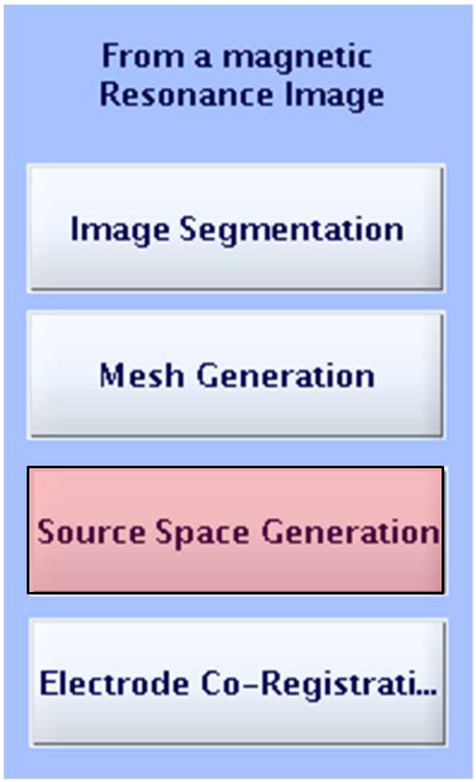


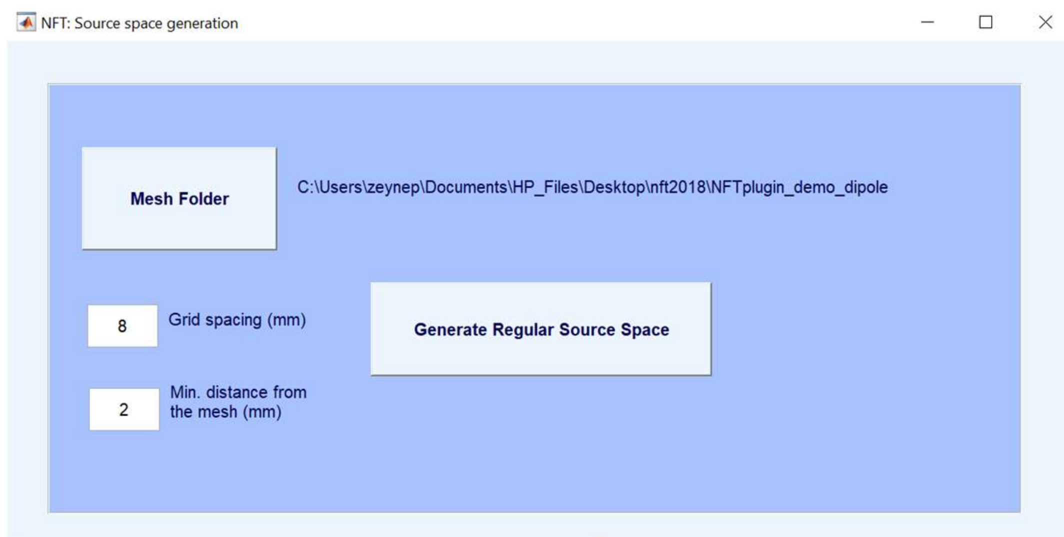
- [C,E] = ReadSMF('Scalp.smf',0,0,0,1);
- Plotmesh(E(:,2:4),C(:,2:4))

Mesh generation



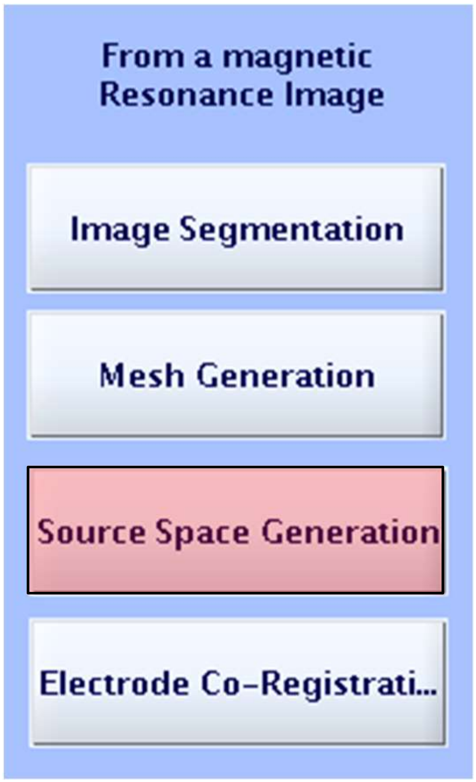
Source Space Generation



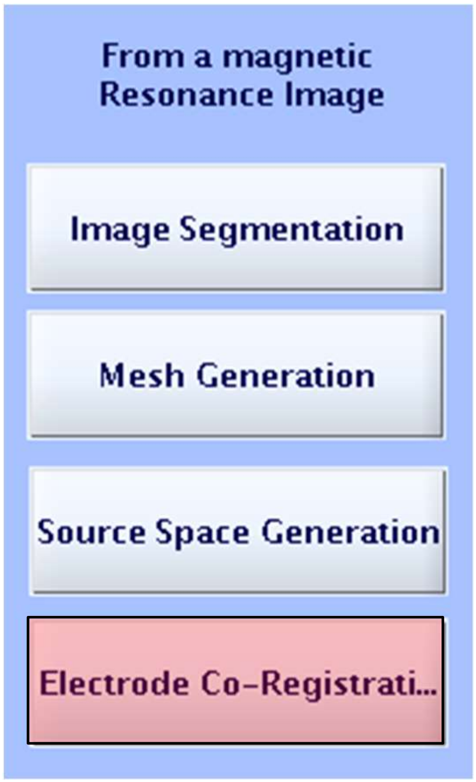


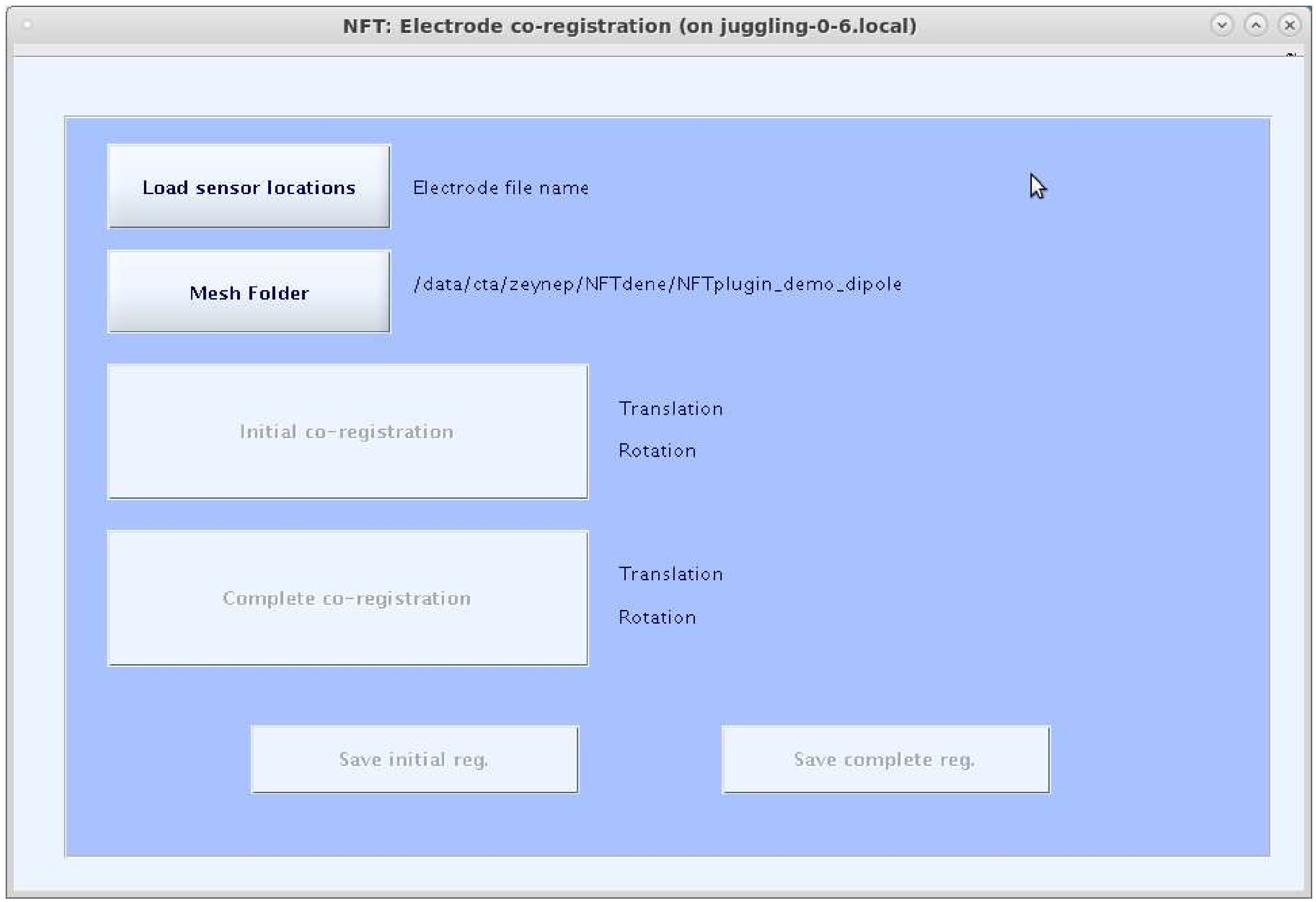
Generates a simple source space:
Regular Grid inside the brain
With a given spacing and distance to the mesh

Source Space Generation

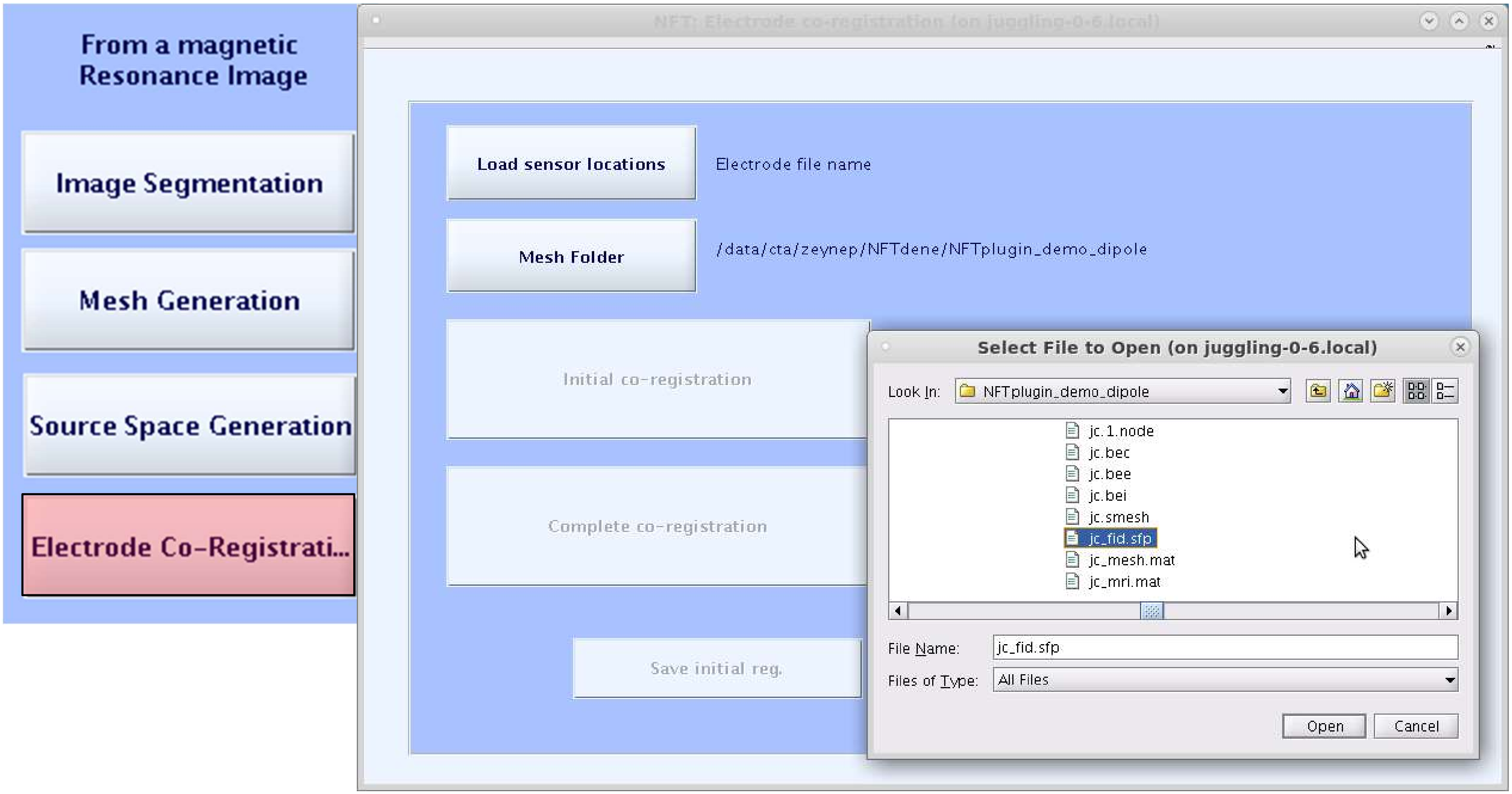


Electrode Co-registration

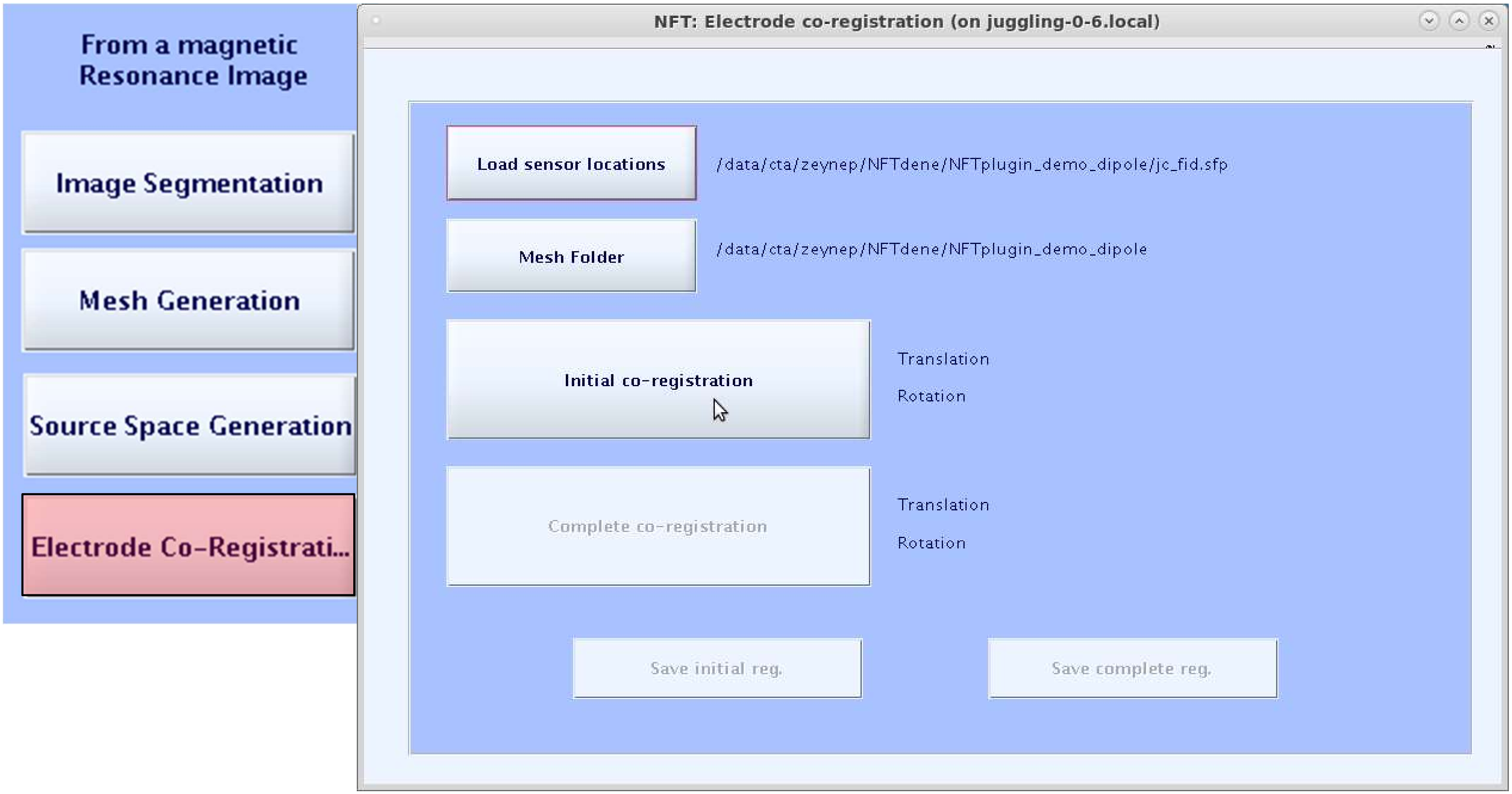




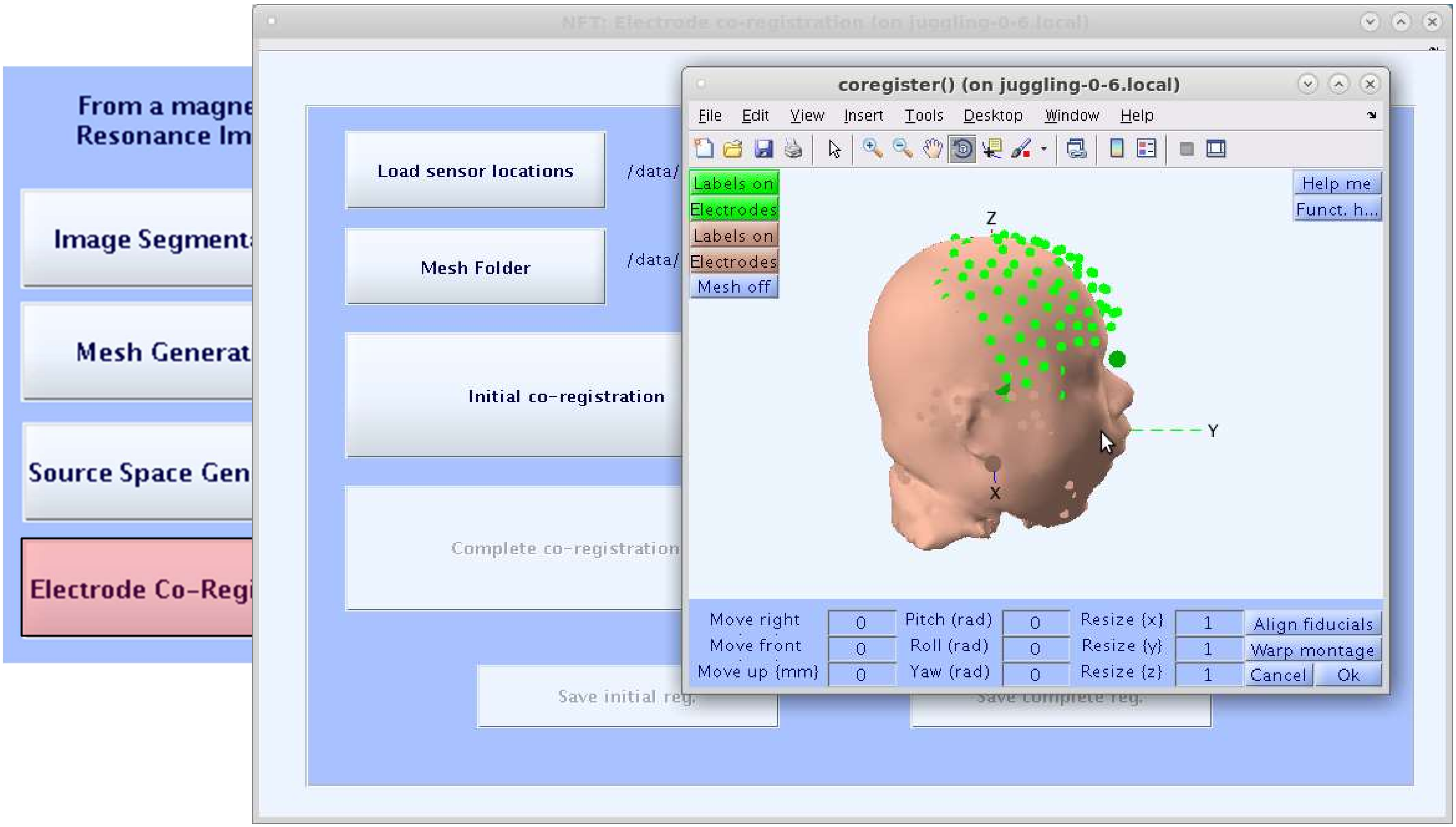
Electrode Co-registration



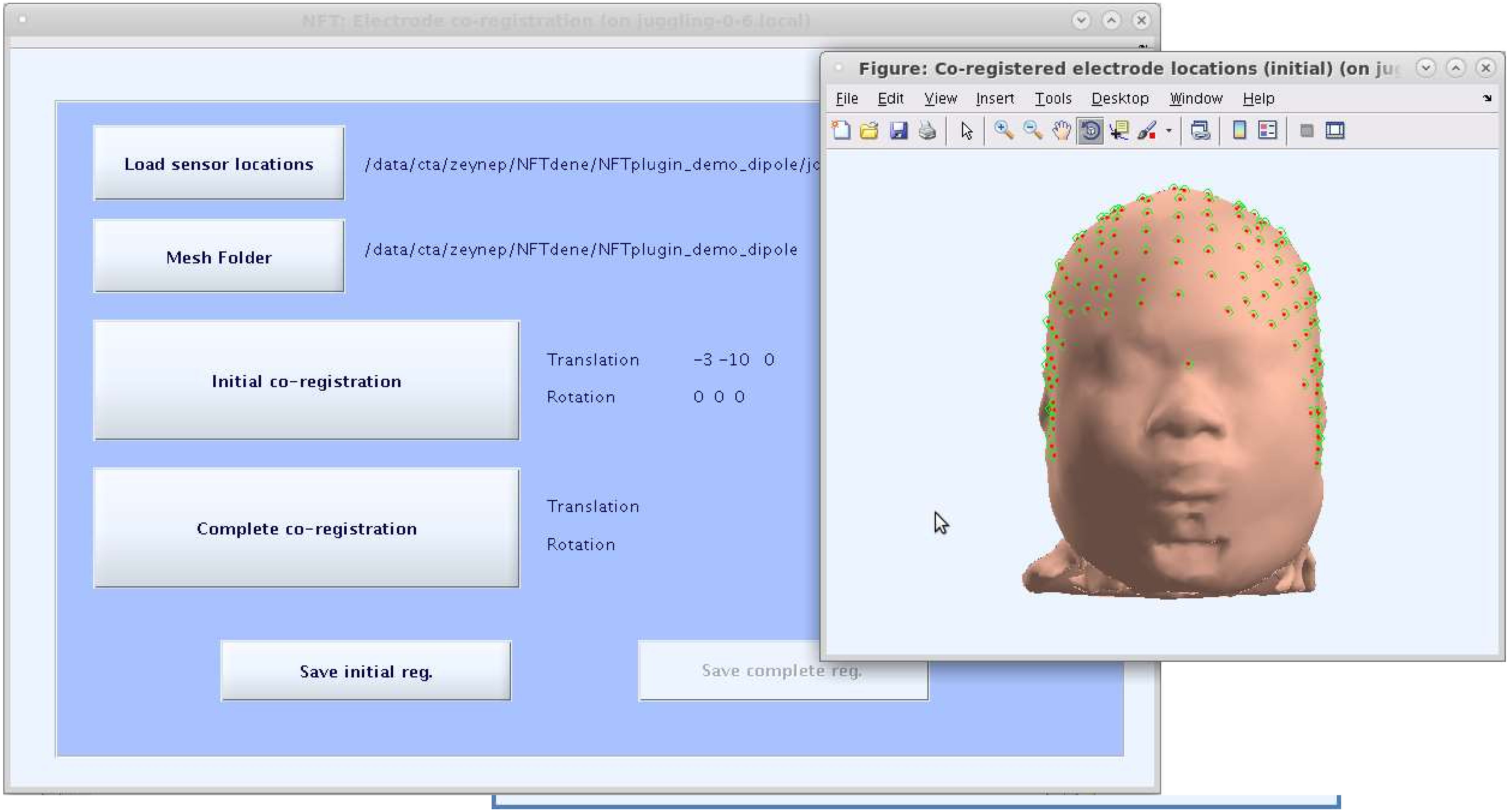
Electrode Co-registration



Electrode Co-registration

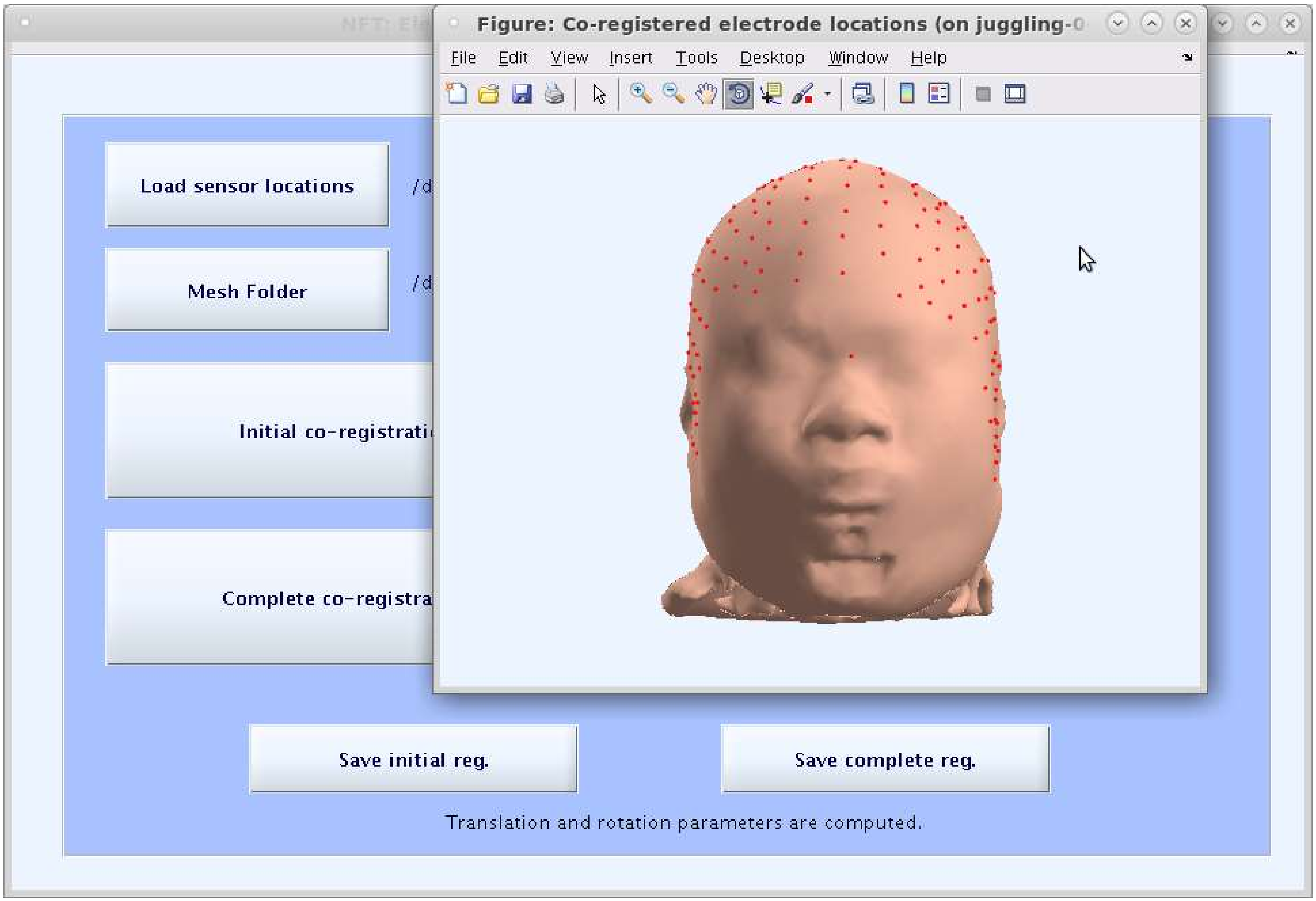


Electrode Co-registration

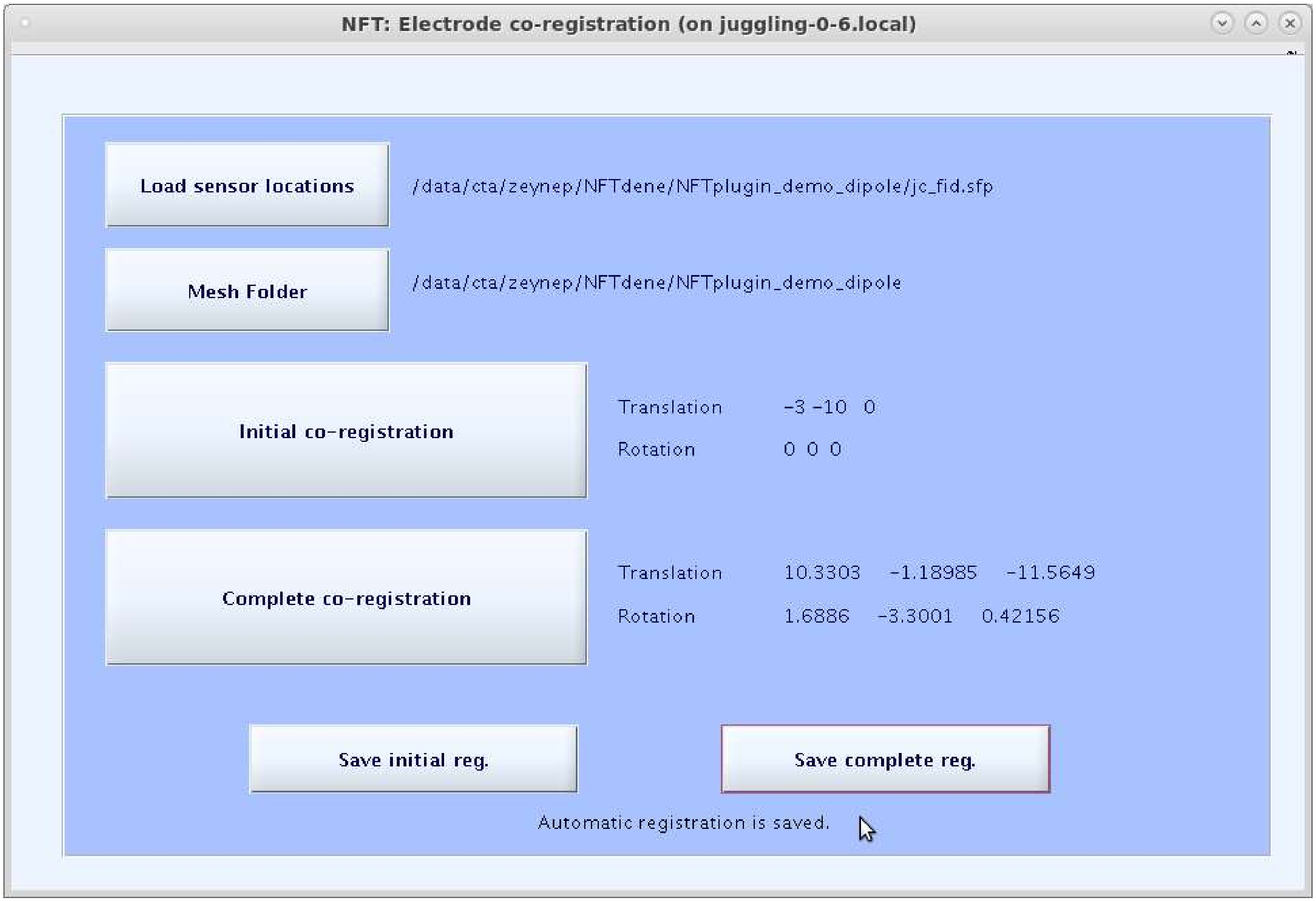


Electrode co-registration

Electrode co-registration

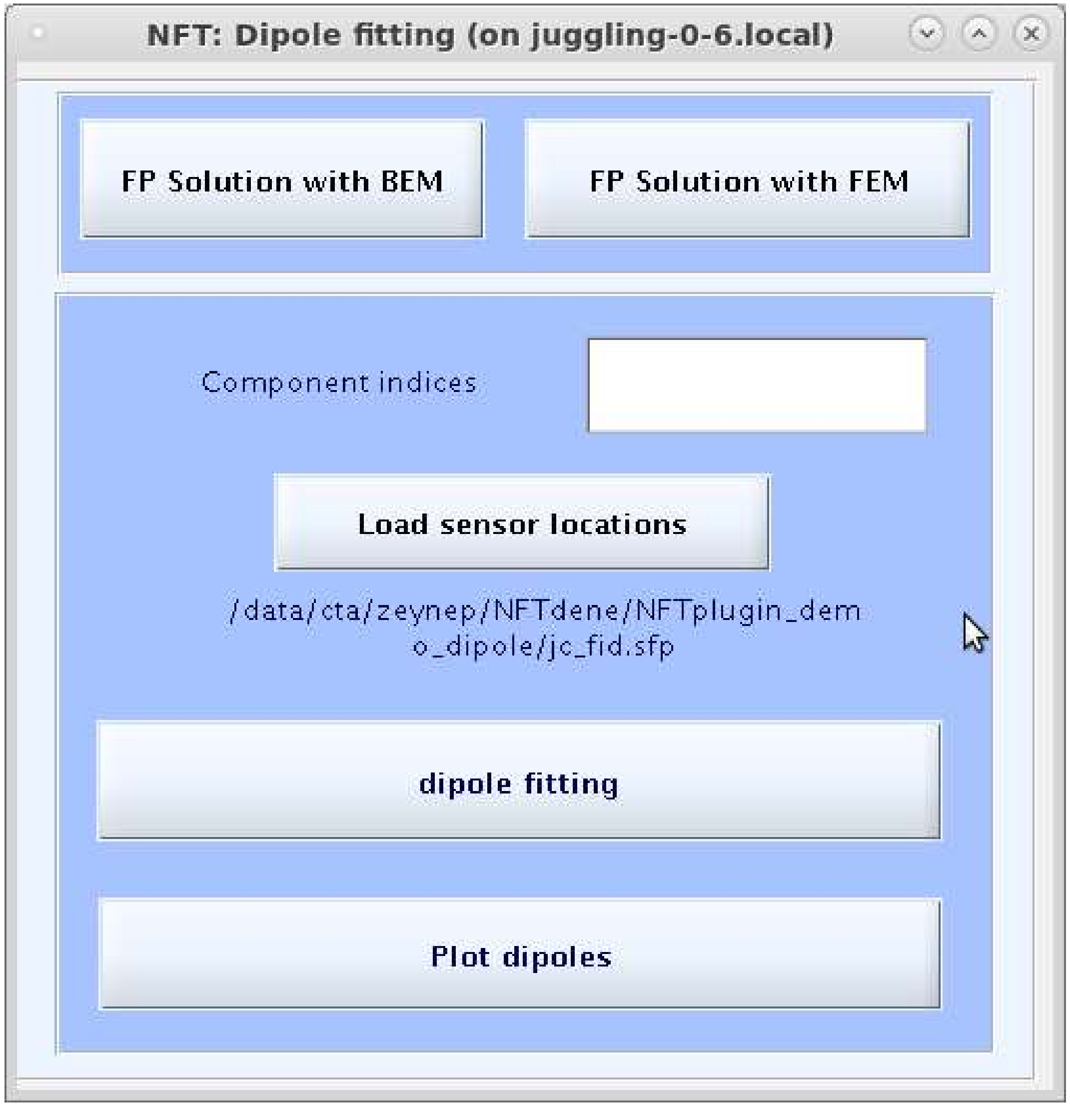


Mesh generation

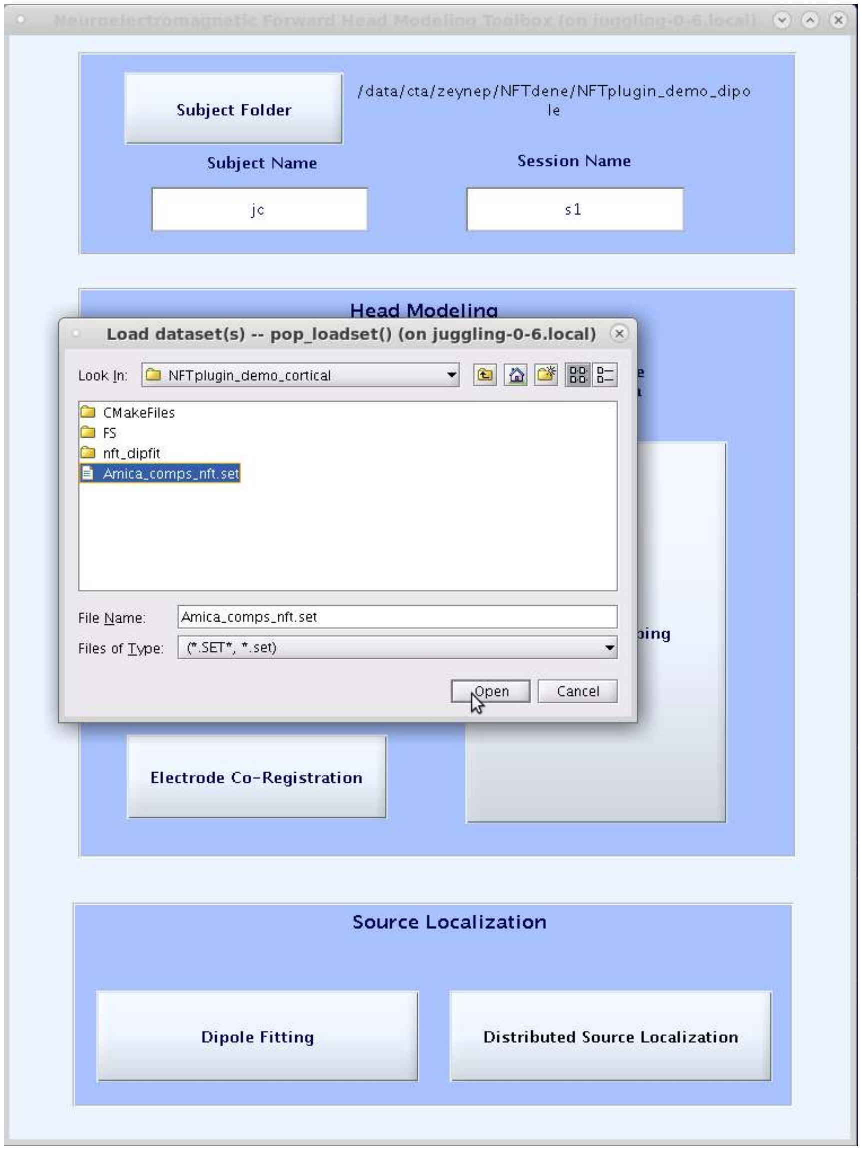


```
>> sens=load('jc_s1.sensors','-mat')

sens =

       fn: '/data/cta/zeynep/NFTdene/JC/jc_fid.sfp'
     eloc: [1x208 struct]
      pnt: [208x3 double]
      ind: [1x208 double]
    param: [1x1 struct]

>> sens.param

ans =

    init: [-3 -10 0 0 0 0 1 1 1]
    auto: [10.3303 -1.1899 -11.5649 1.6886 -3.3001 0.4216]

>> |
```

Dipole source localization



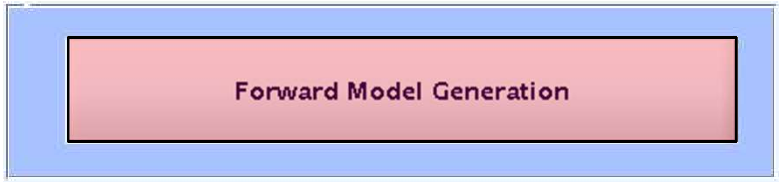
Select EEG data

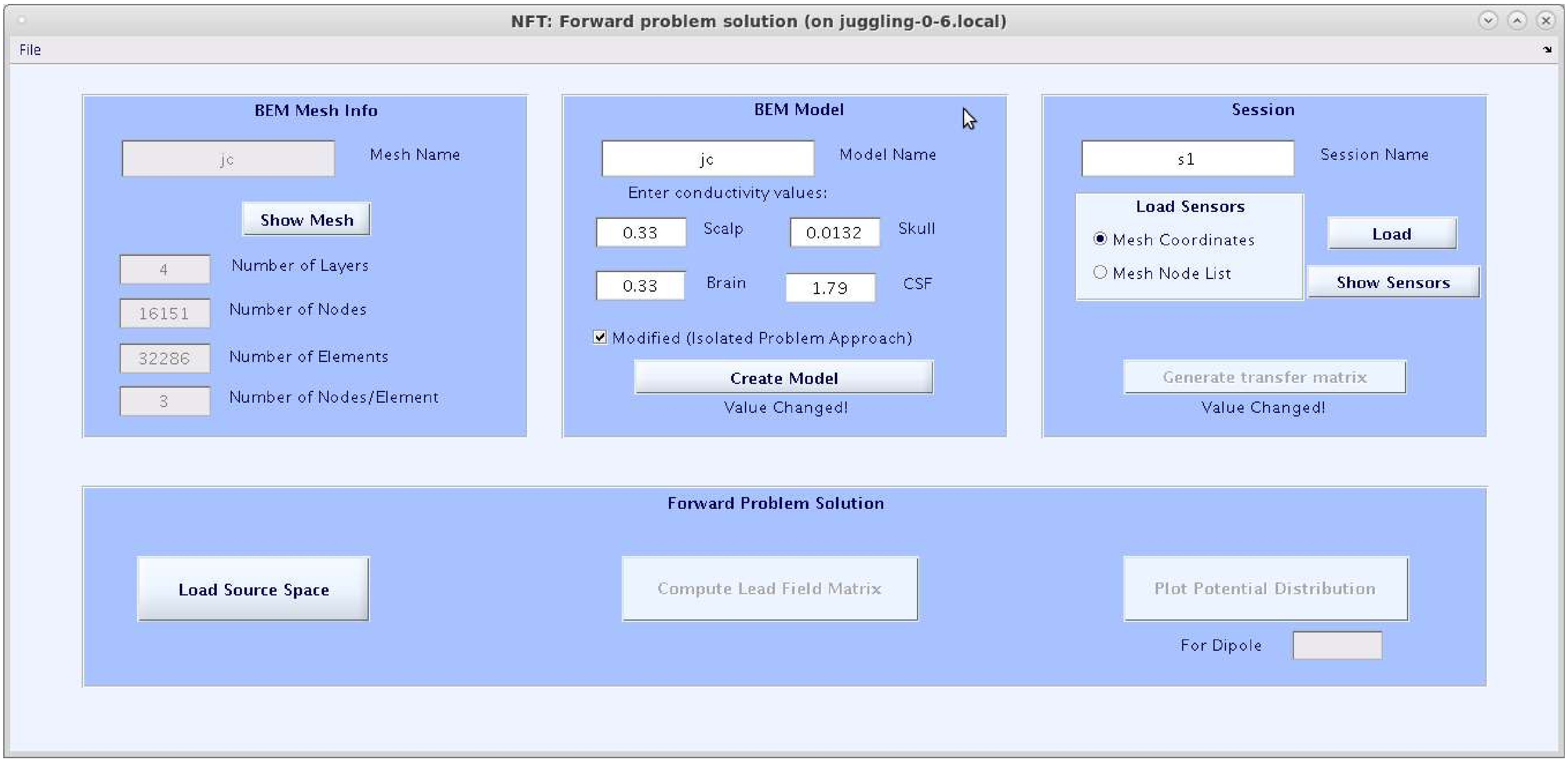


Forward Problem Solver

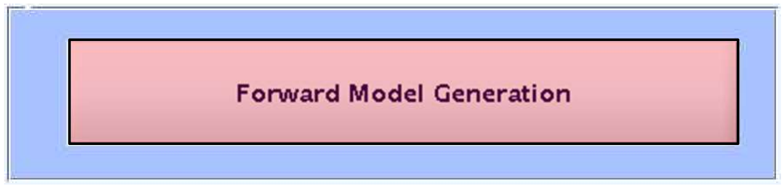
- MATLAB interface to numerical solvers
- Boundary Element Method or Finite Element Method
  - EEG Only (for now)
  - Interfaces to the Matrix generator executable written in C++
- Other computation done in MATLAB
- Generated matrices are stored on disk for future use.

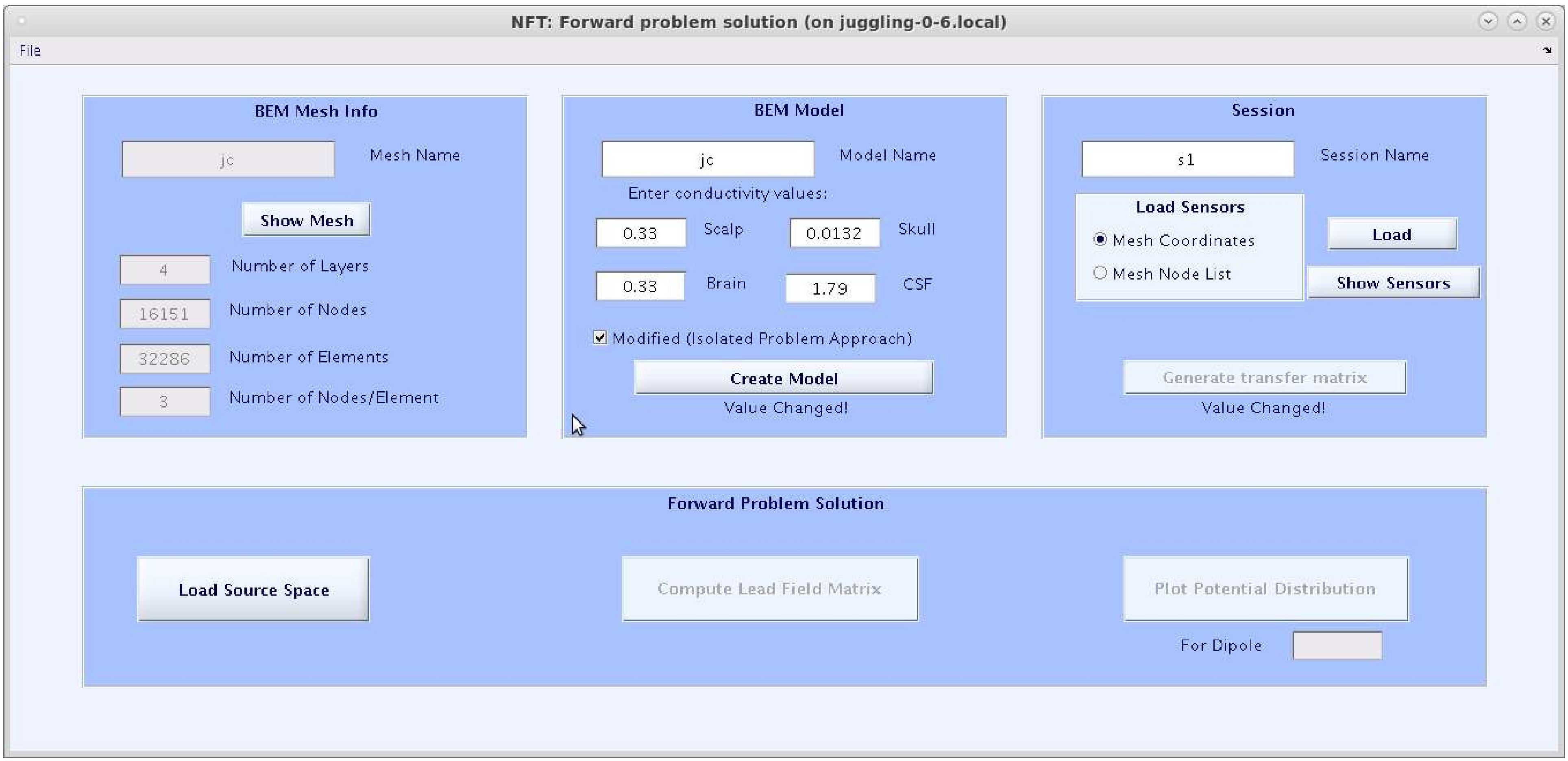
Forward Problem Solution with BEM





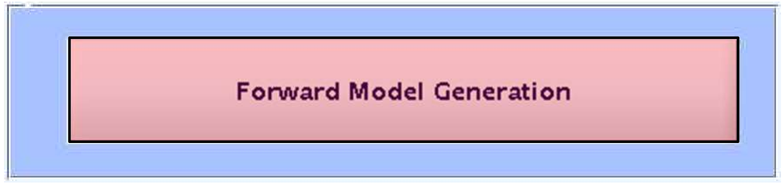
Forward Problem Solution with BEM

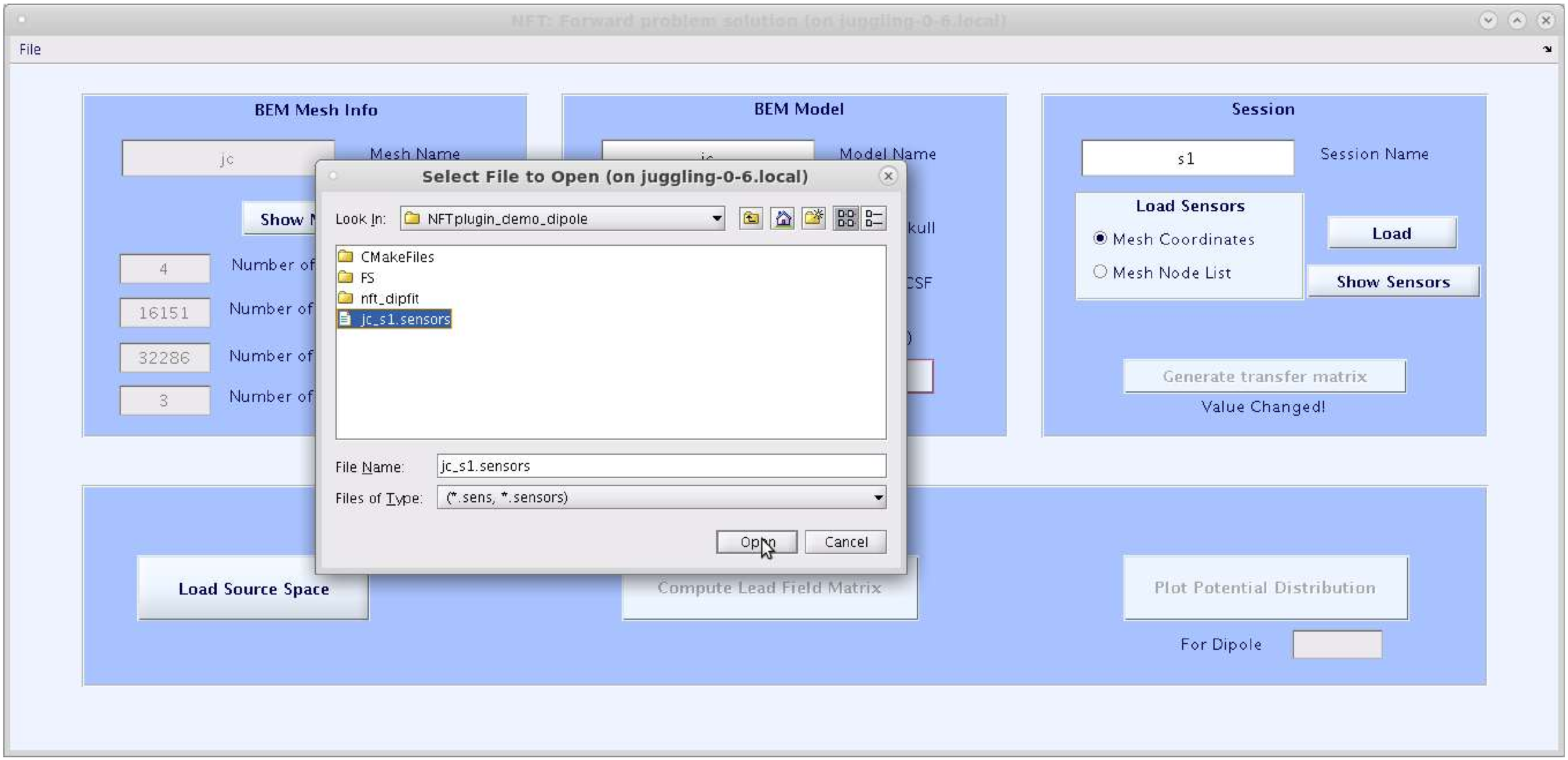




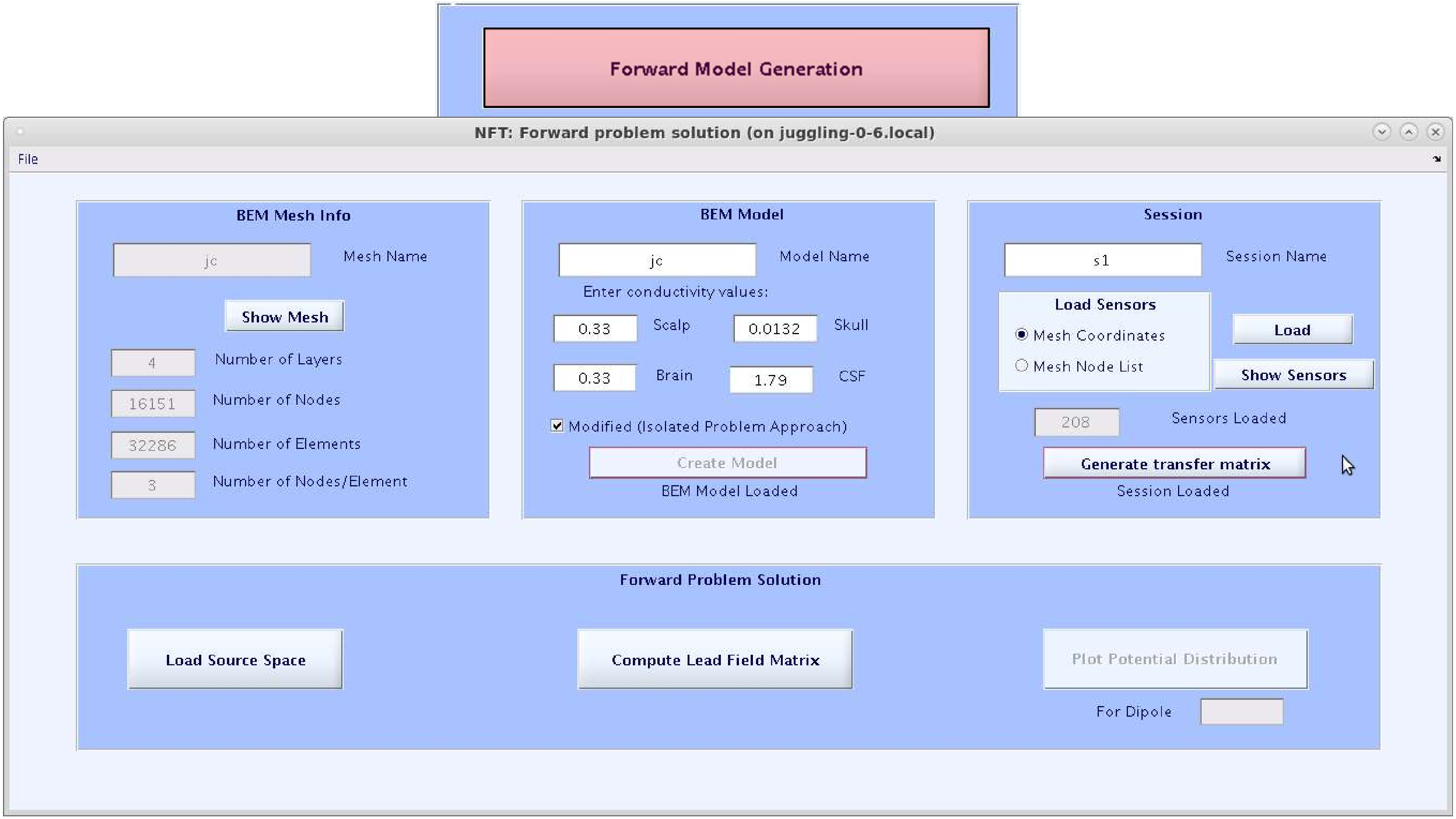
Forward Problem Solution with BEM

Forward Problem Solution with BEM

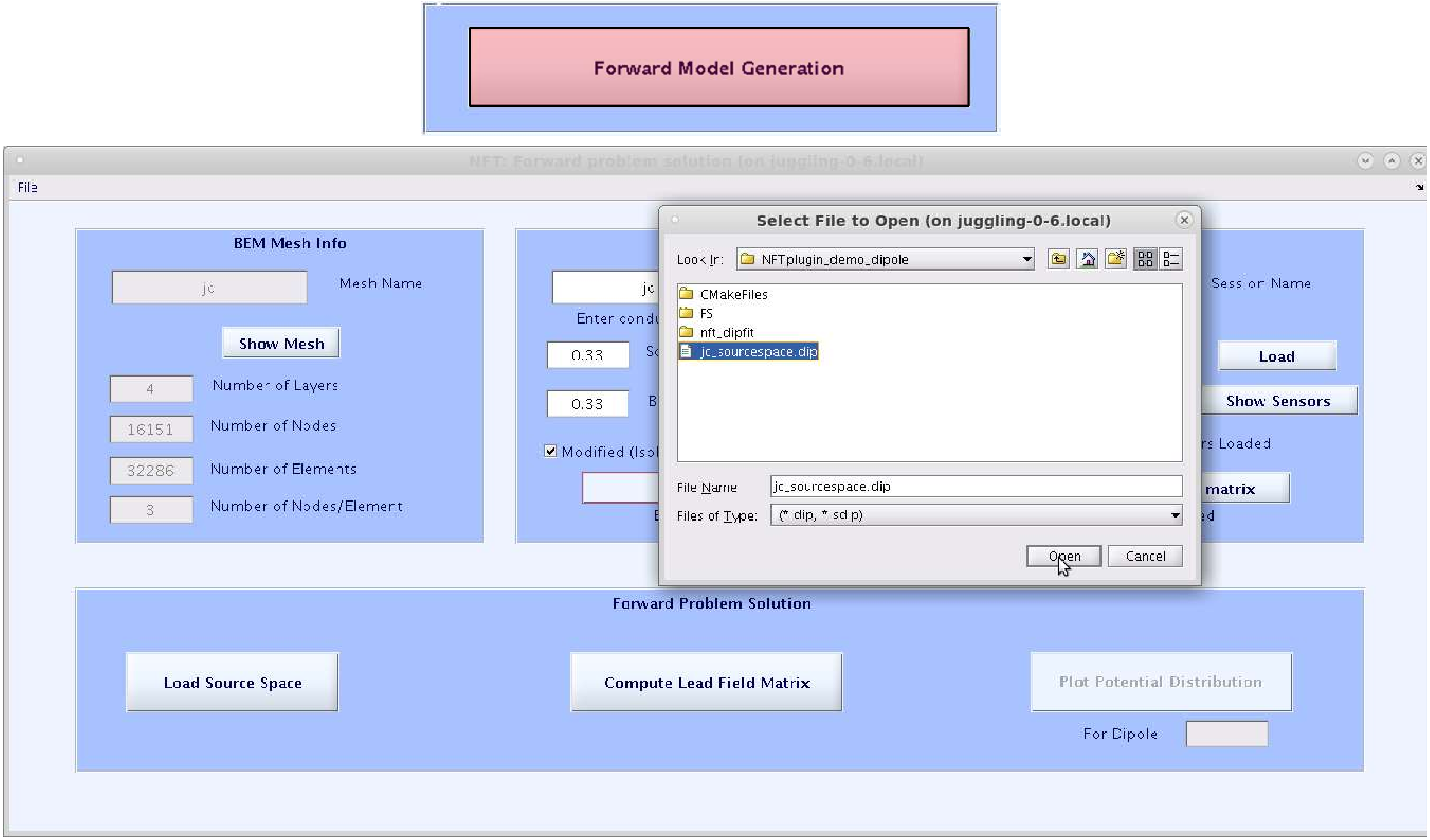




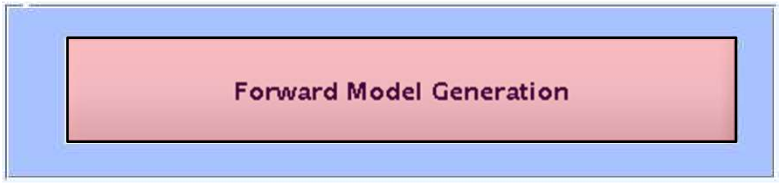
Forward Problem Solution with BEM

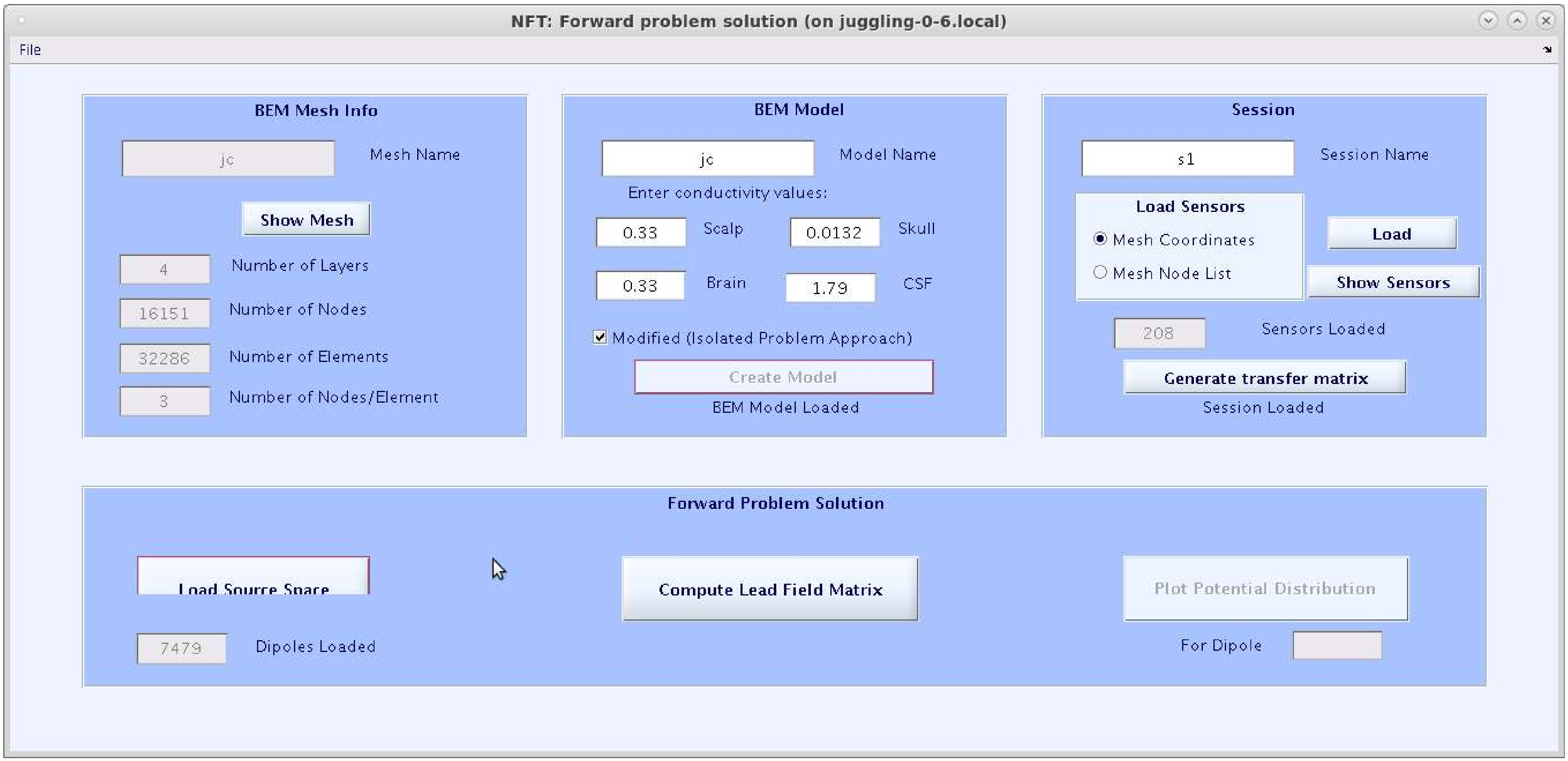


Forward Problem Solution with BEM

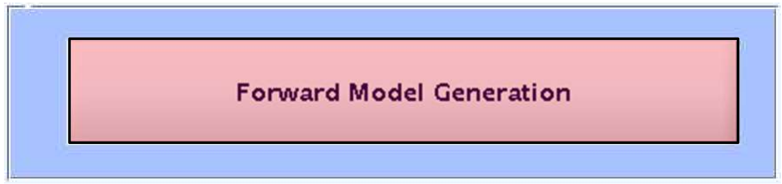


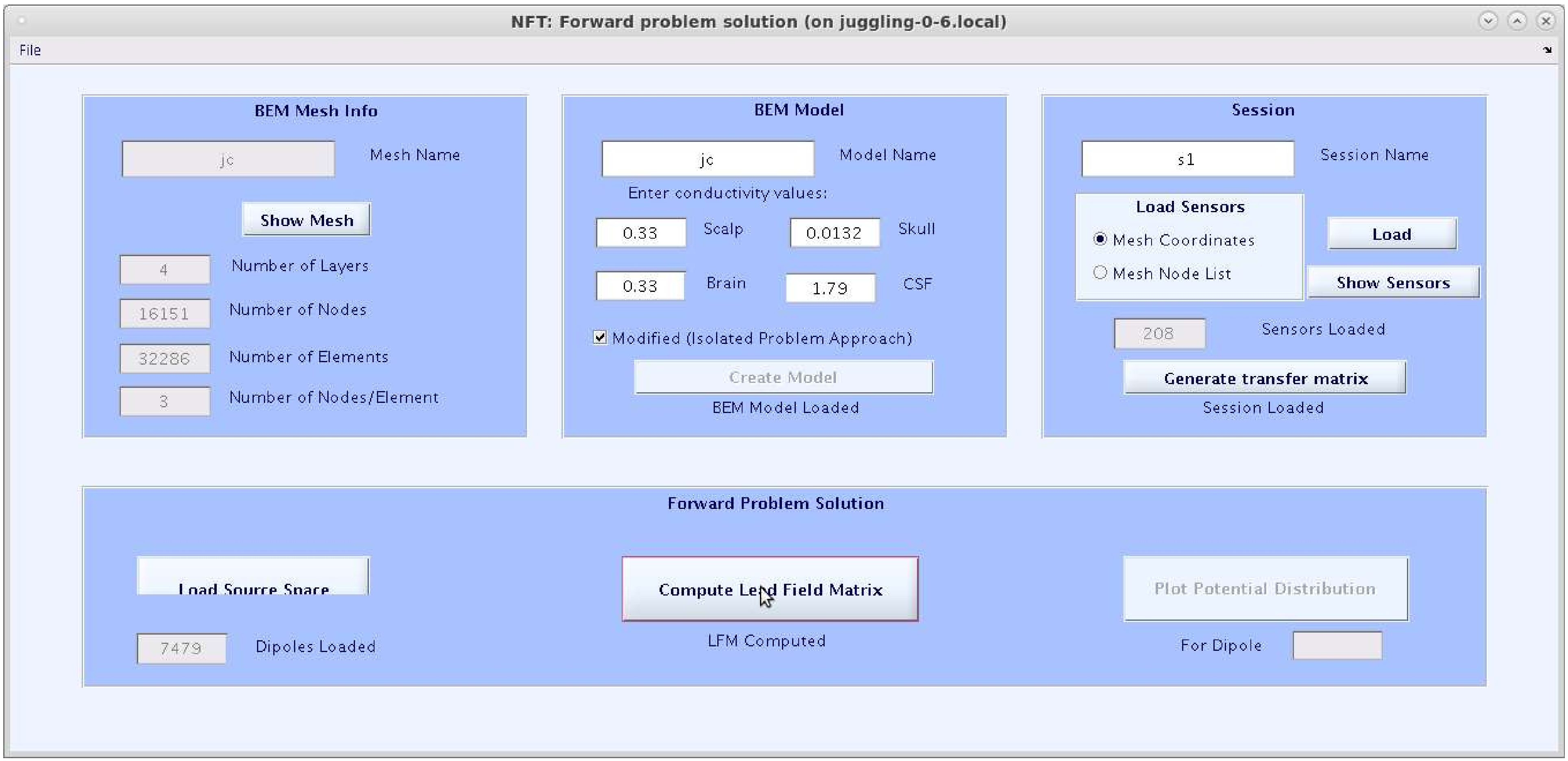
Forward Problem Solution with BEM



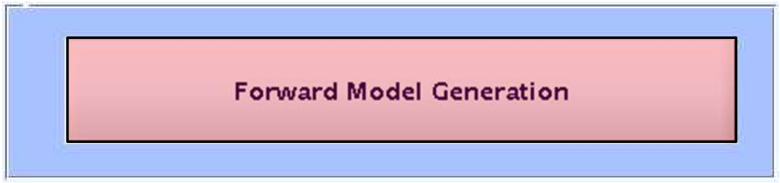


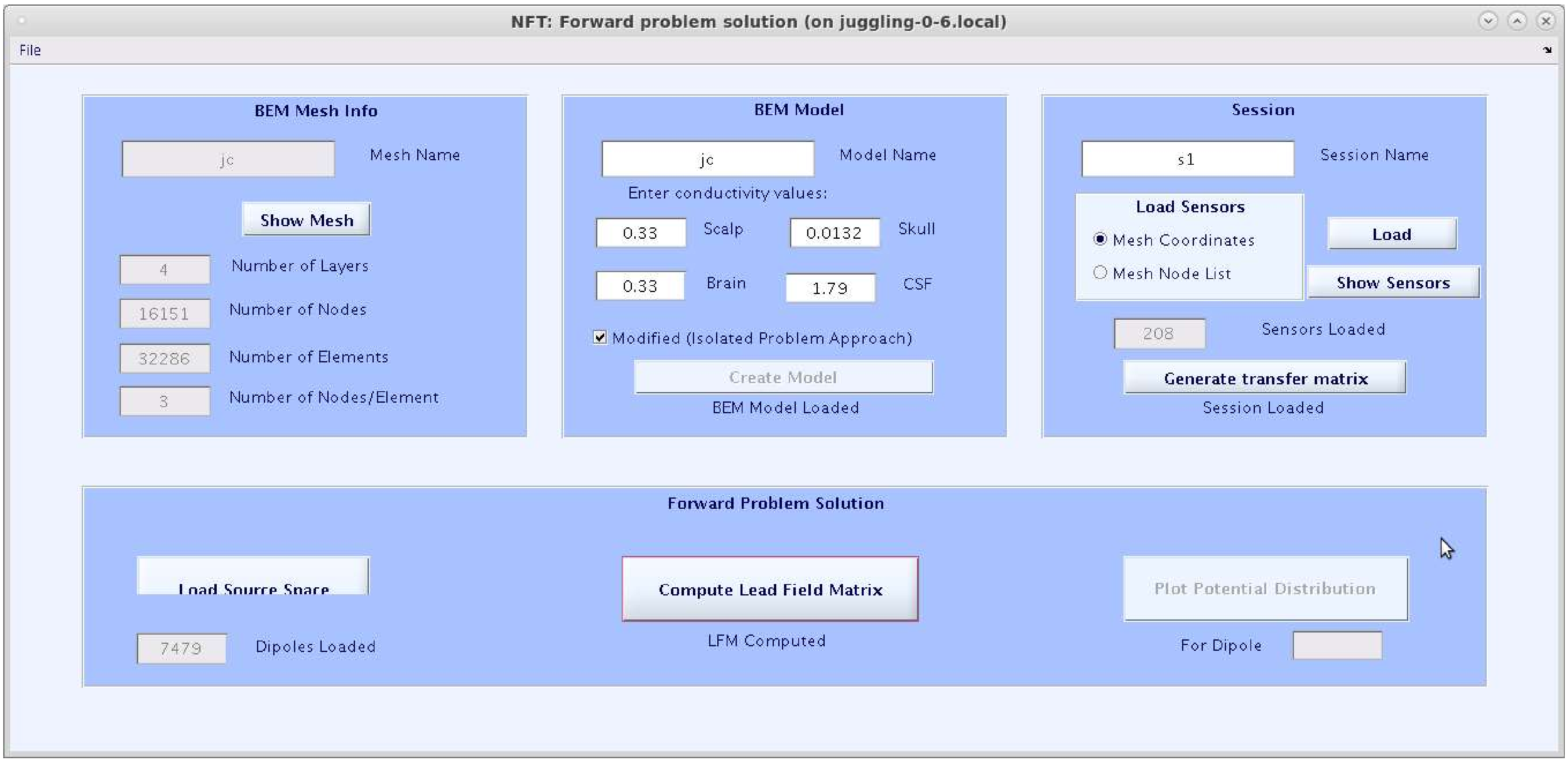
Forward Problem Solution with BEM



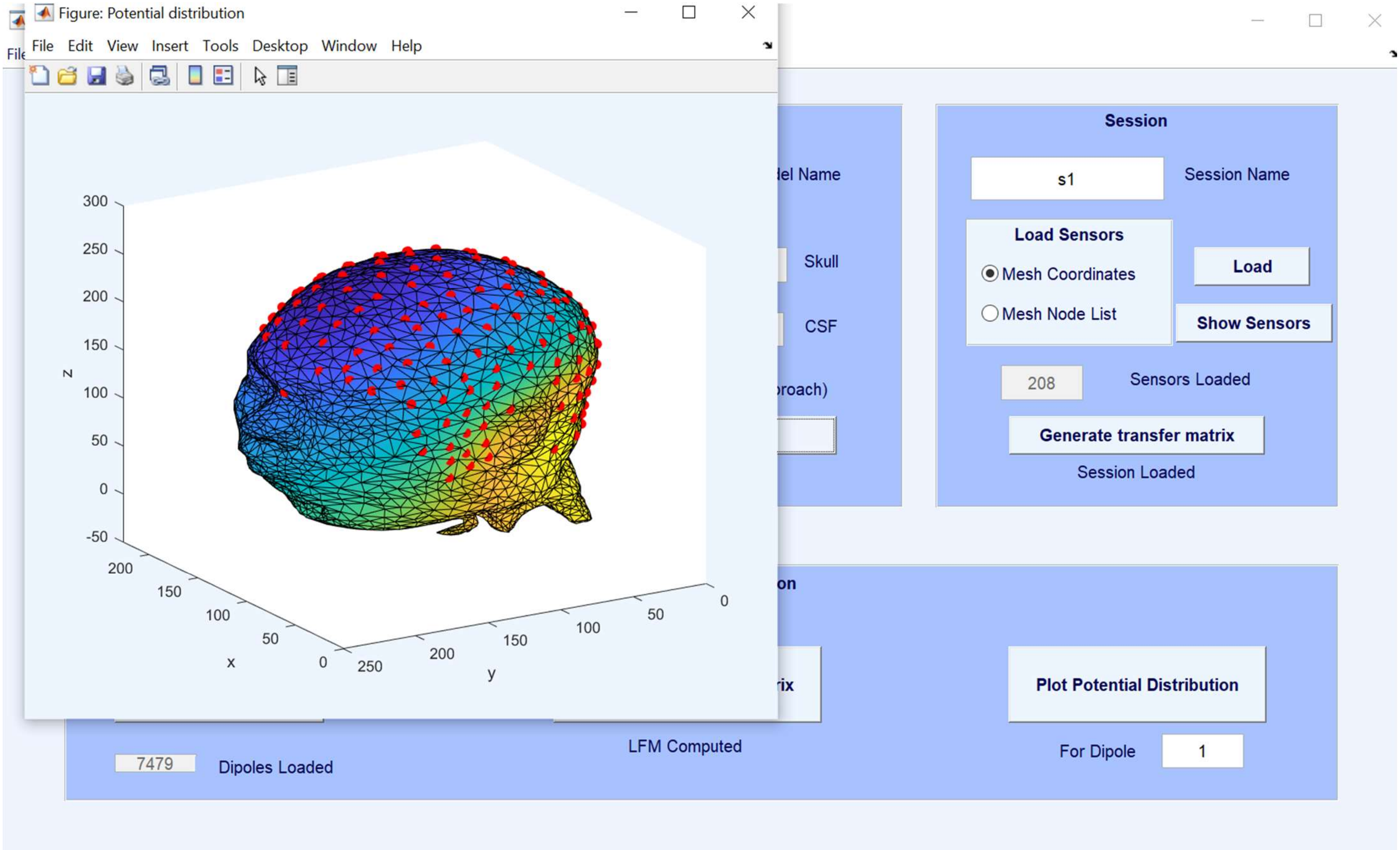


Forward Problem Solution with BEM

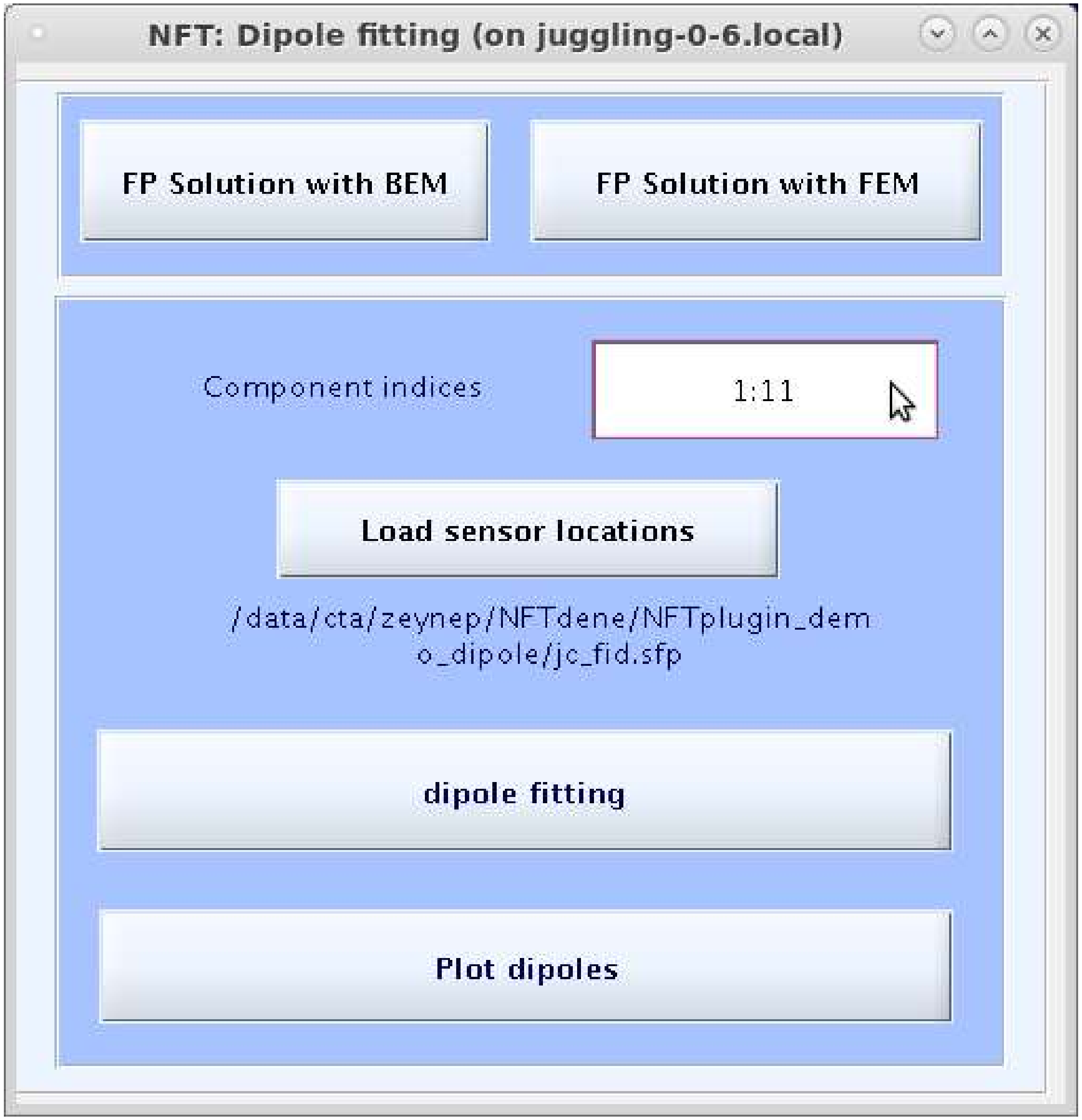




Forward Problem Solution with BEM



Inverse Problem Solution with BEM



Output

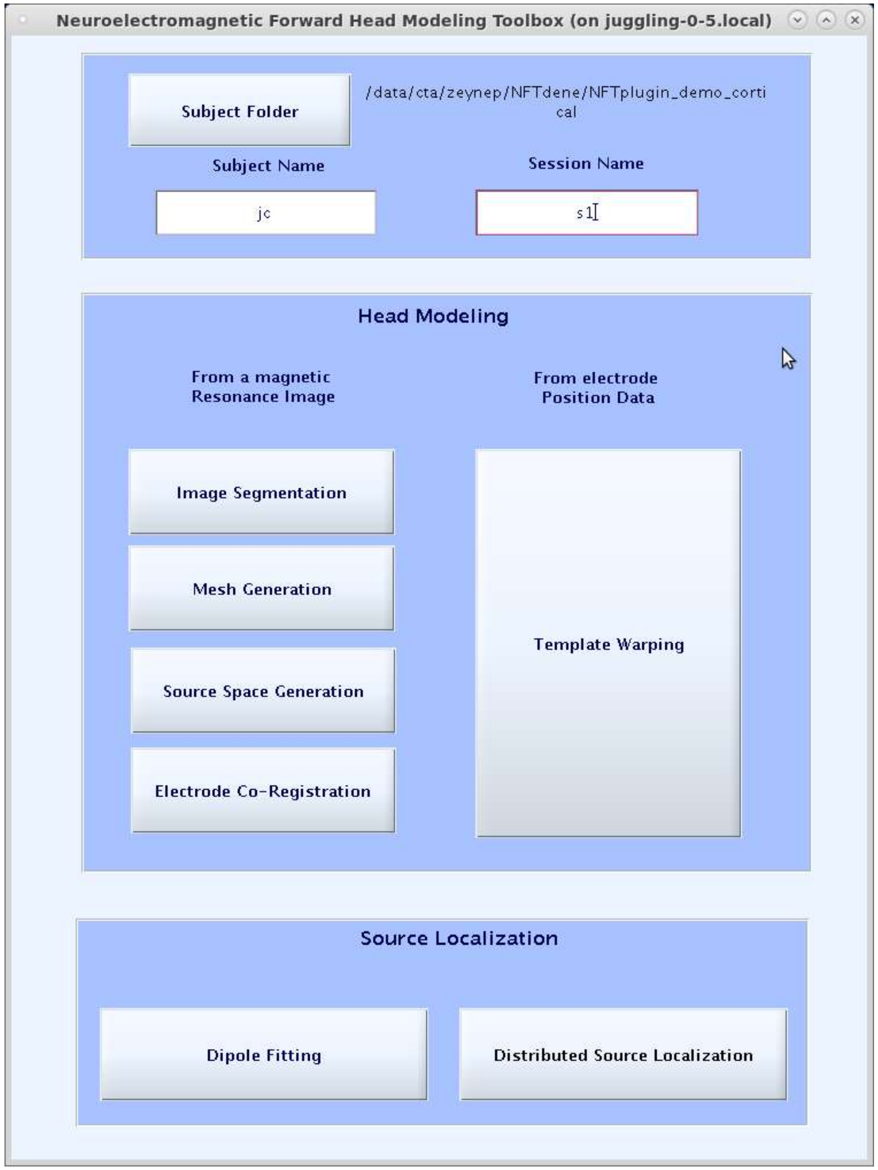
- Dipole source localization is saved in EEG structure, under EEG.etc.nft.
- After source localization with NFT, you can continue using EEGLAB;

EEG.dipfit.model = EEG.etc.nft.model;

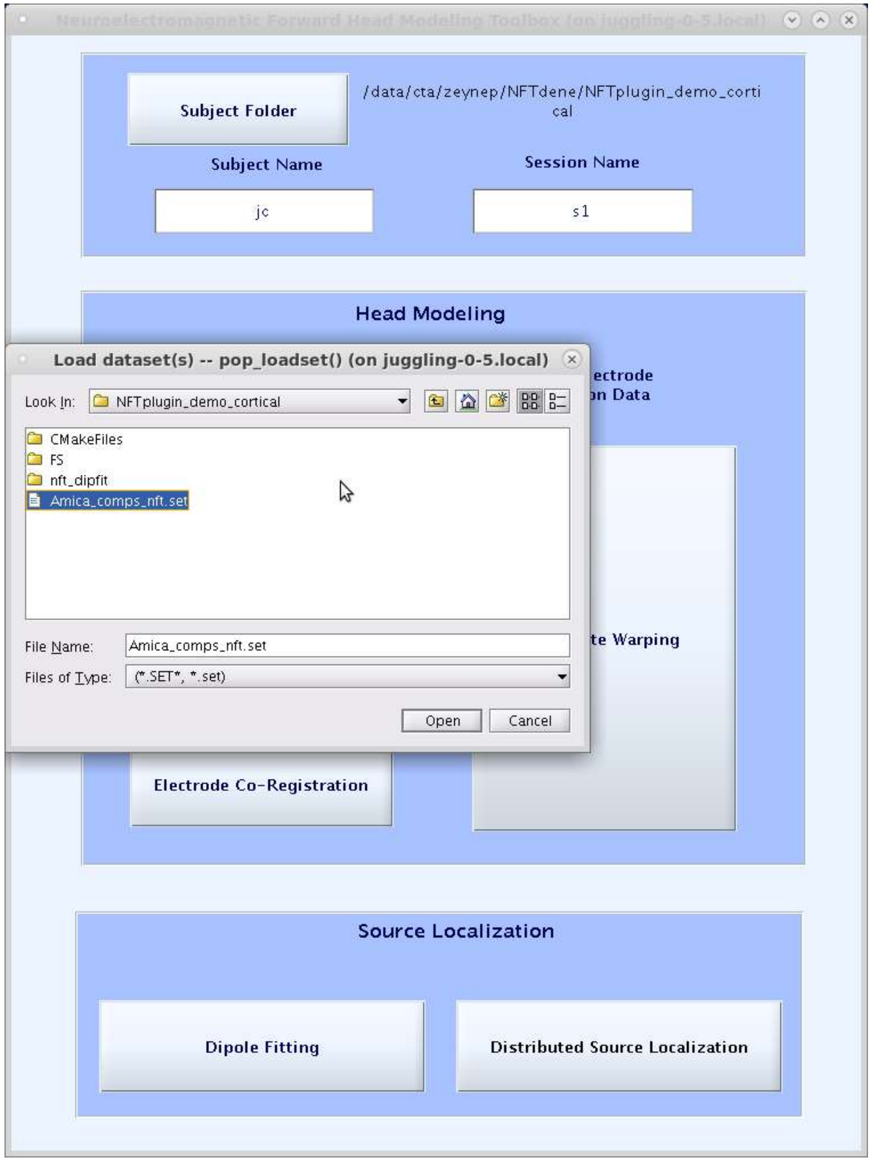
Distributed Source localization

Go to the folder NFTplugin_demo_cortical

Distributed Source Localization



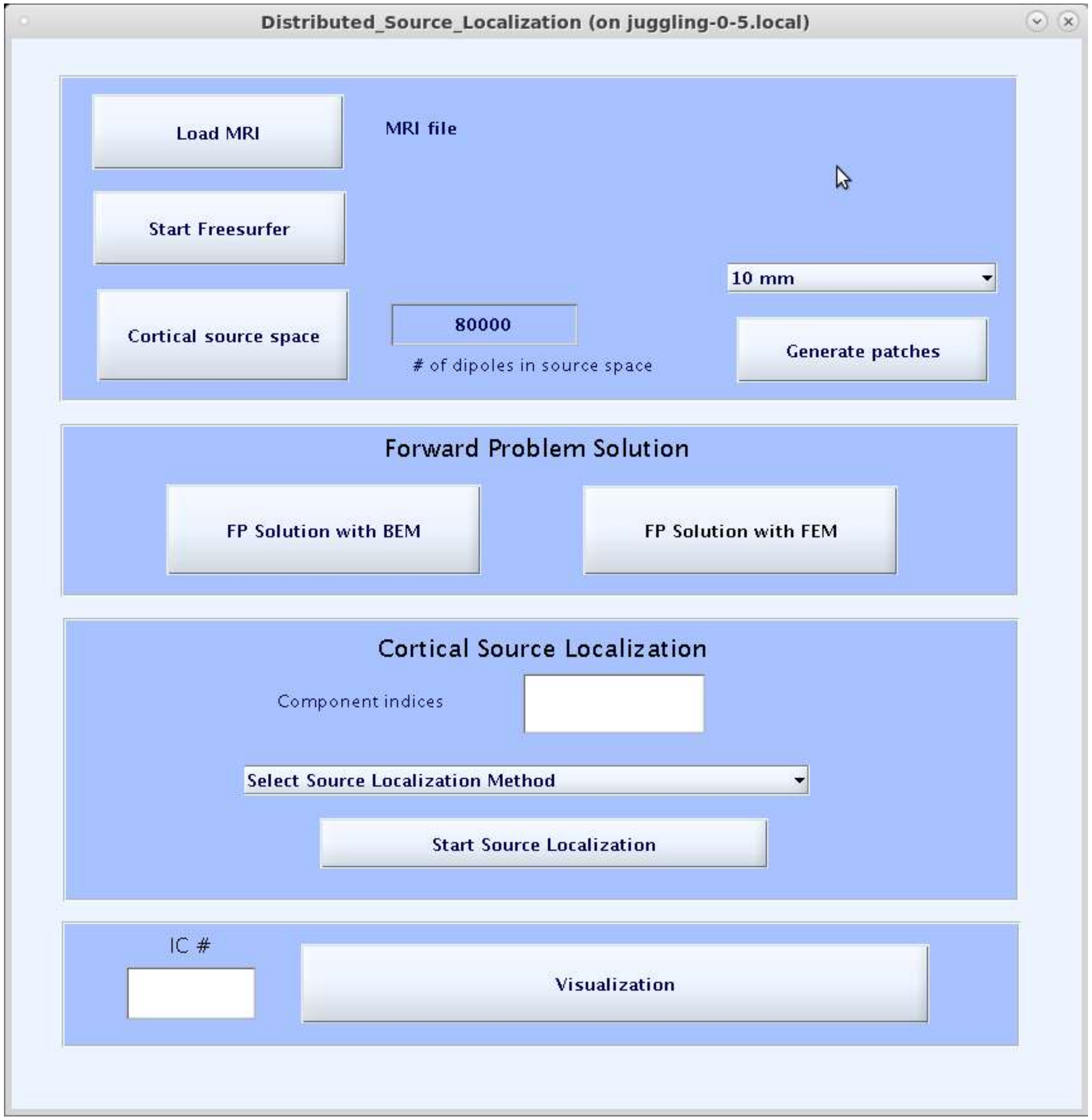
Select EEG data



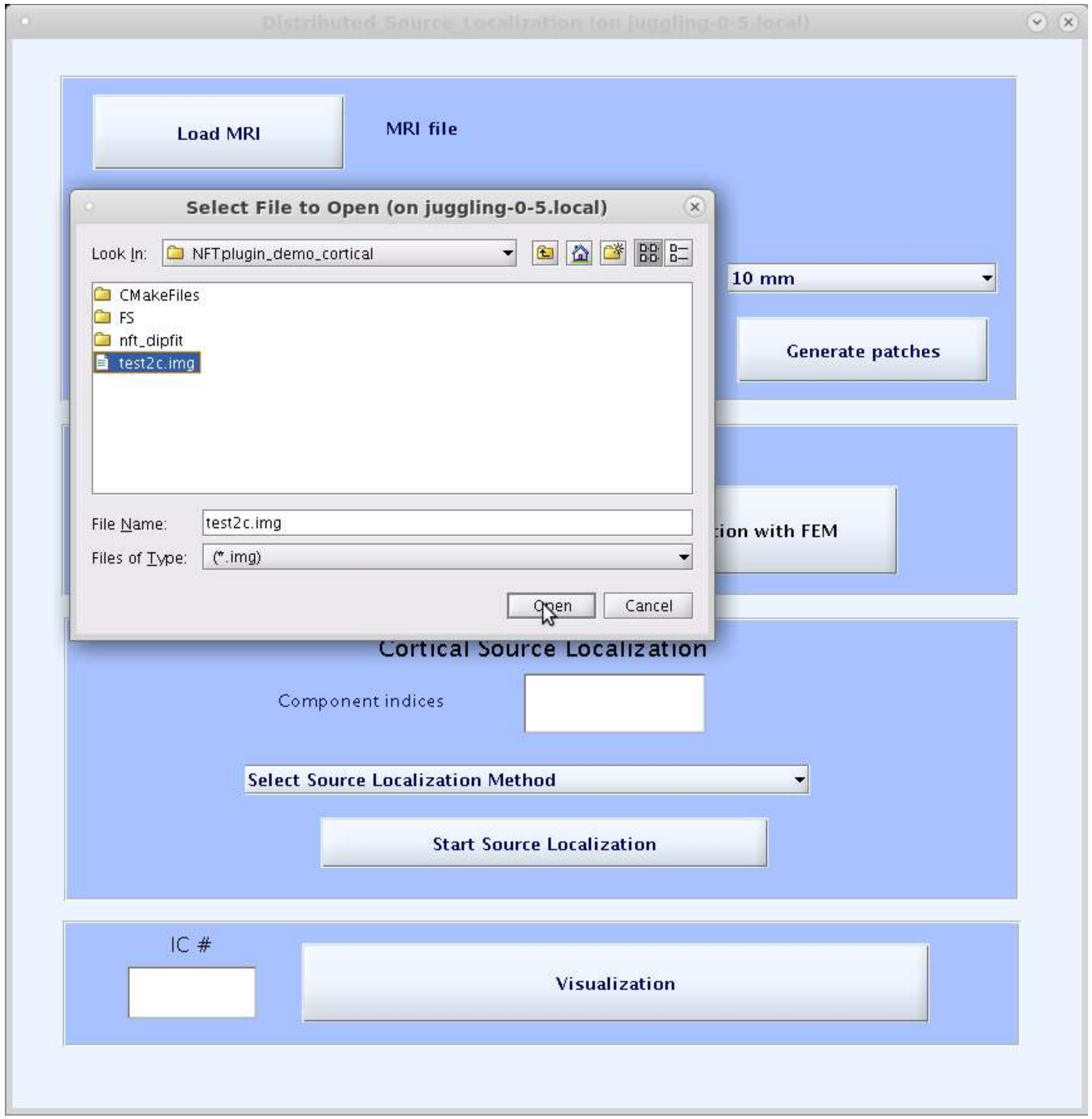
NIST – Generation of a cortical source space

- Load Freesurface cortical surface
- Downsample to 80,000 vertices
- Co-register with the NFT brain surface
- Re-generate NFT head model
- Calculate normals for each vertex on the cortical surface
- Save the cortical source space as: Subject_name FS_ss.dip
- Calculate node area of each vertex for source localization, save as Node_area
- Check if the sensor locations need to be updated according to the new NFT model.

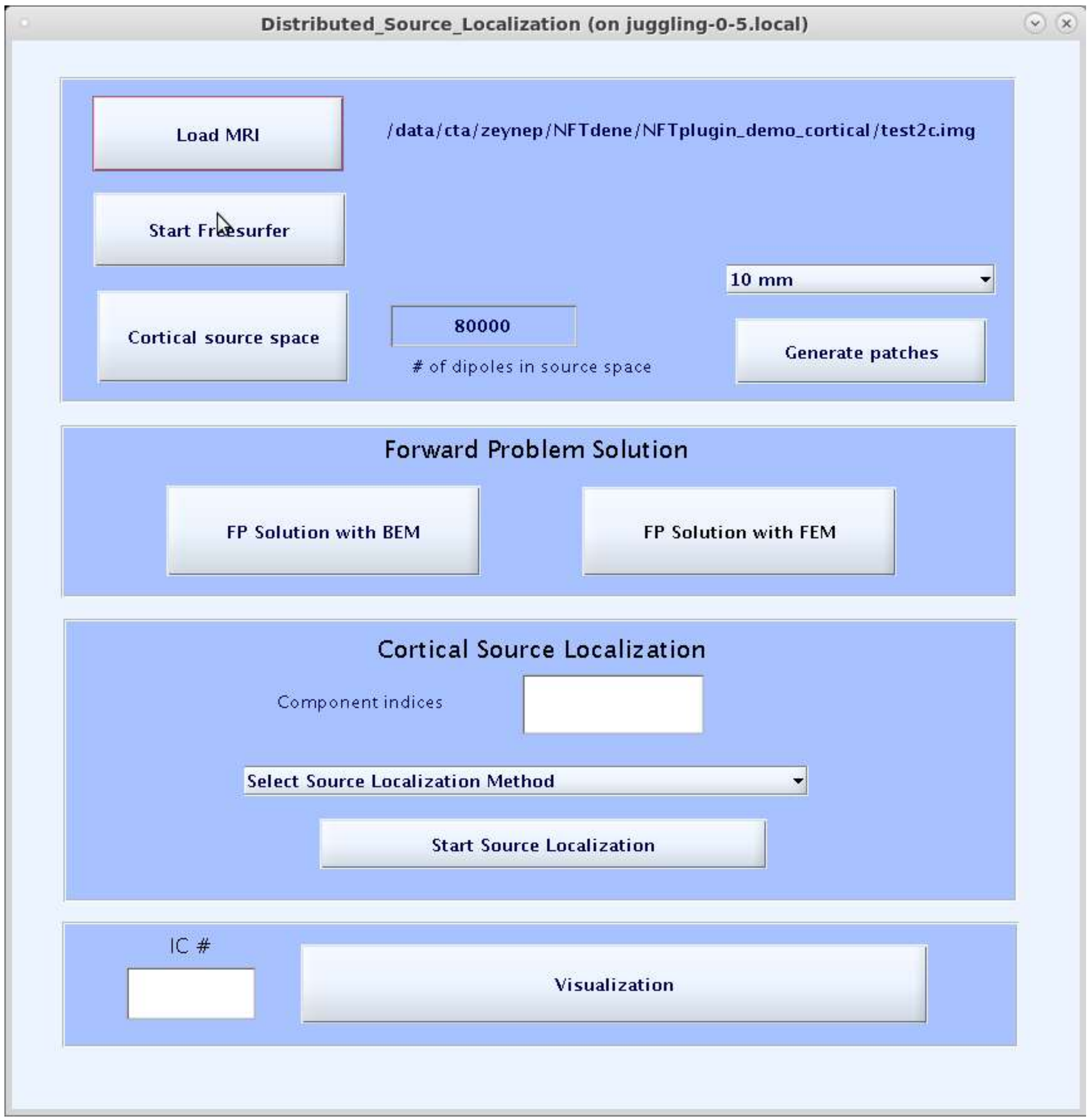
Distributed Source Localization



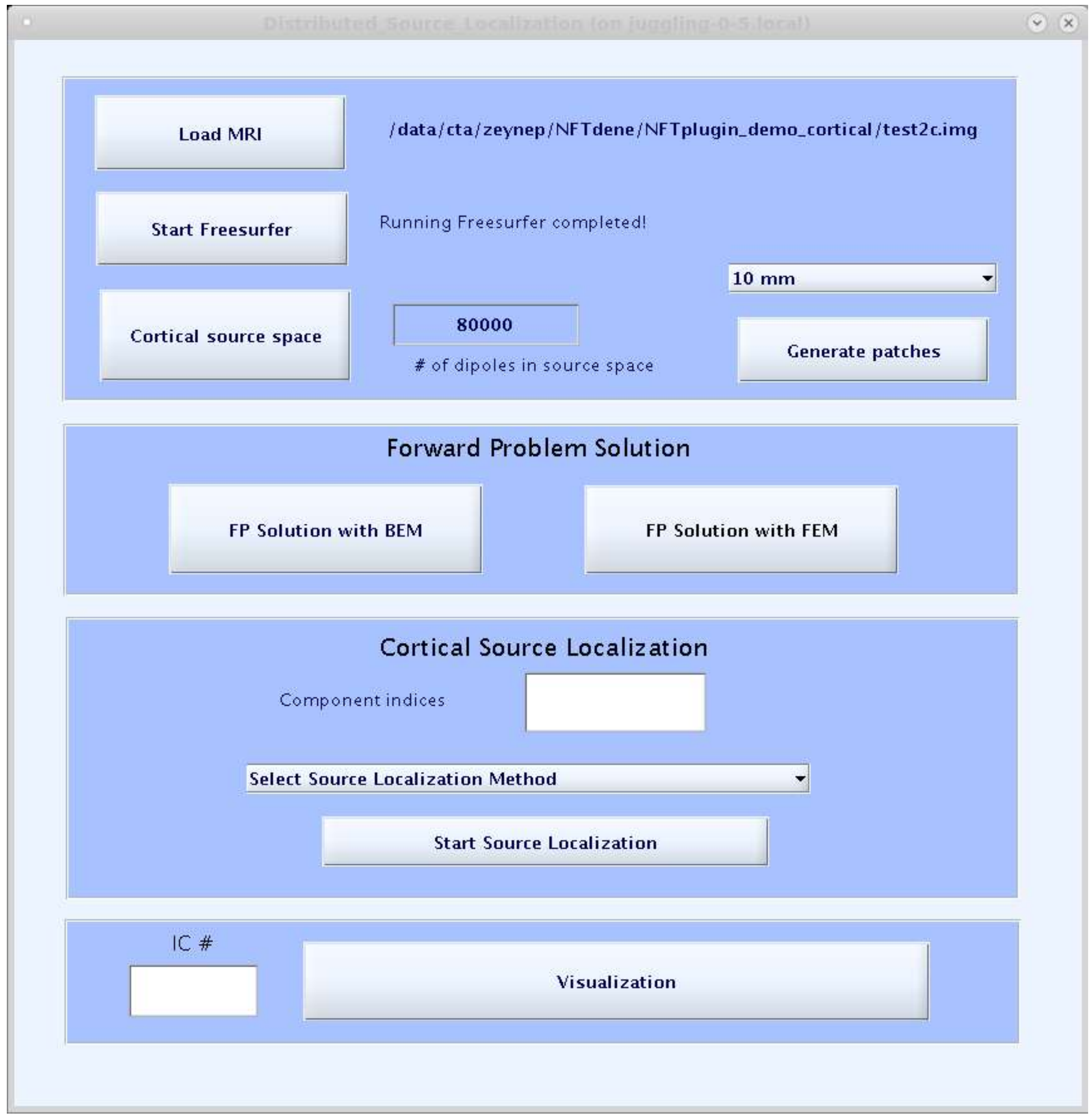
Load MRI


Load MRI
MRI file
Select File to Open (on juggling-0-5.local)
Look In: NFTplugin_demo_cortical
CMakeFiles
FS
nft_dipfit
test2c.img
File Name: test2c.img
Files of Type: (*.img)
Open
Cancel
10 mm
Generate patches
tion with FEM
Cortical Source Localization
Component indices
Select Source Localization Method
Start Source Localization
IC #
Visualization

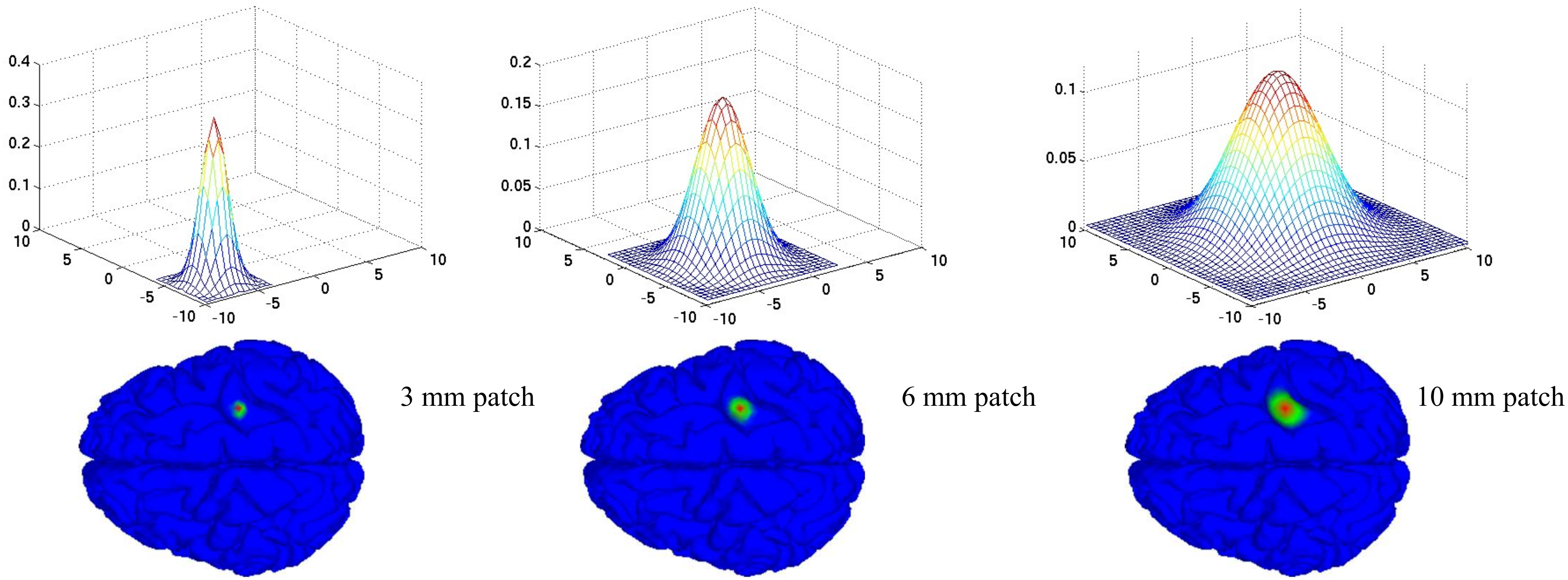
Run Freesurfer



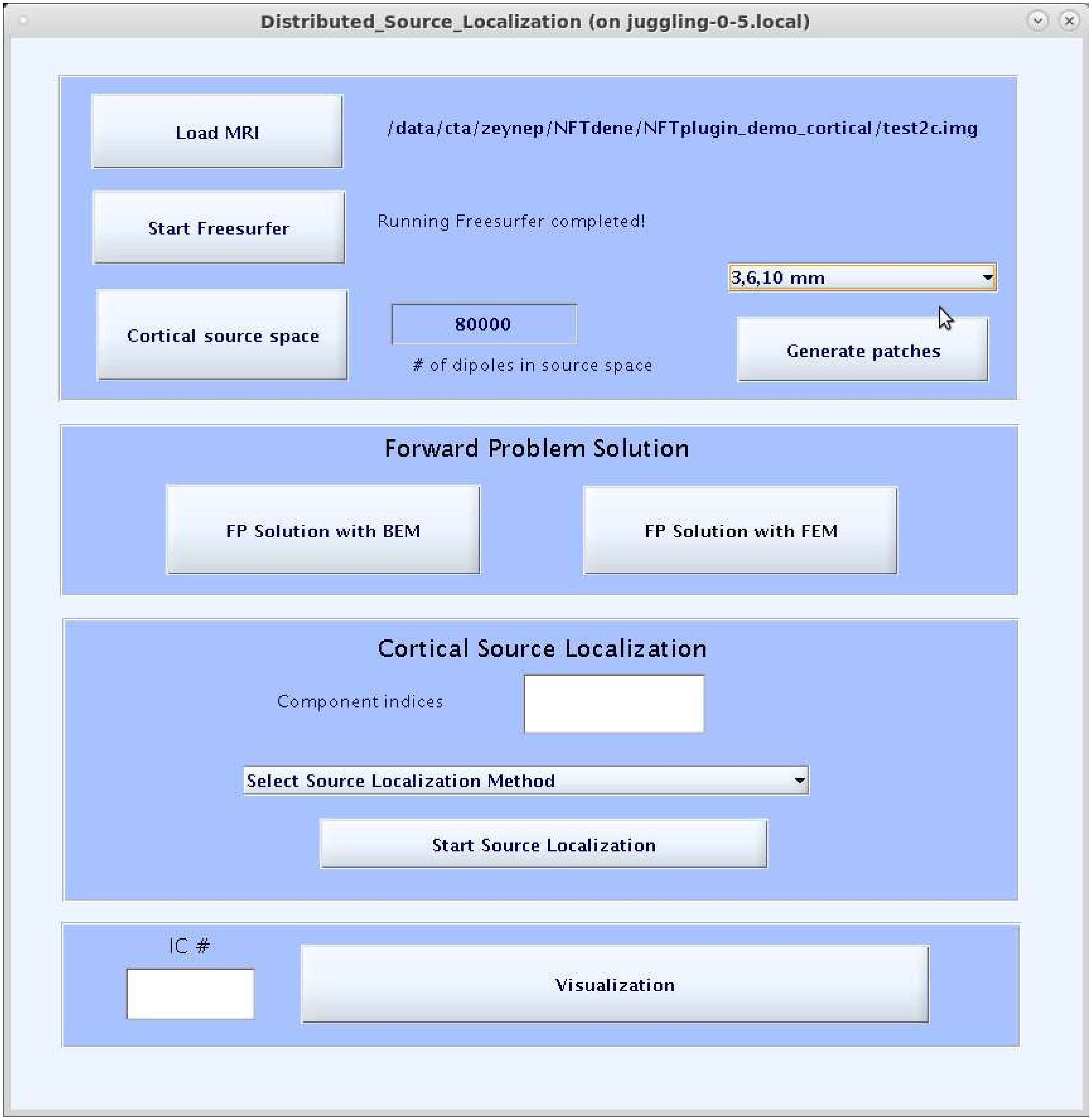
Freesurfer completed



NIST – Patch generation

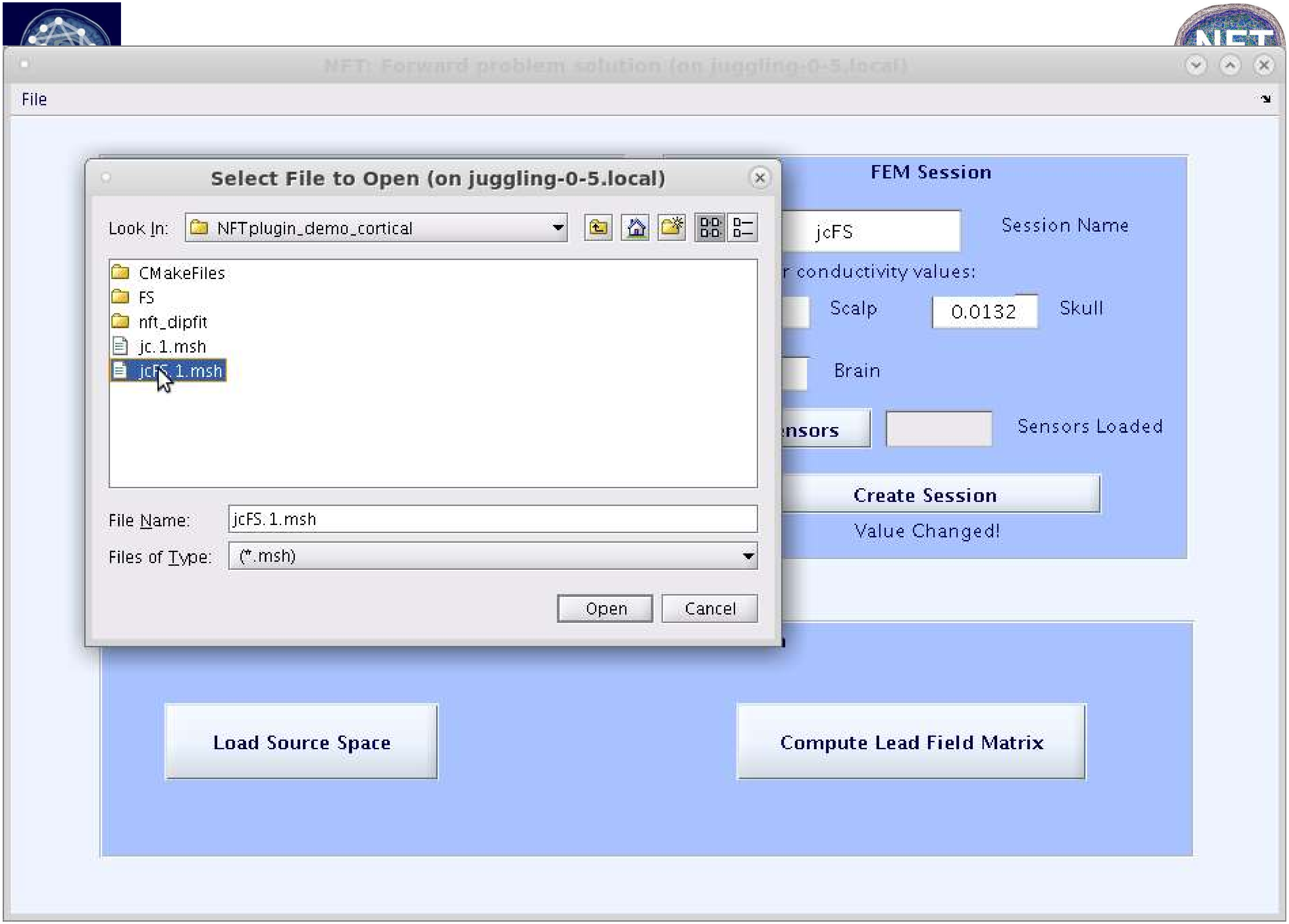


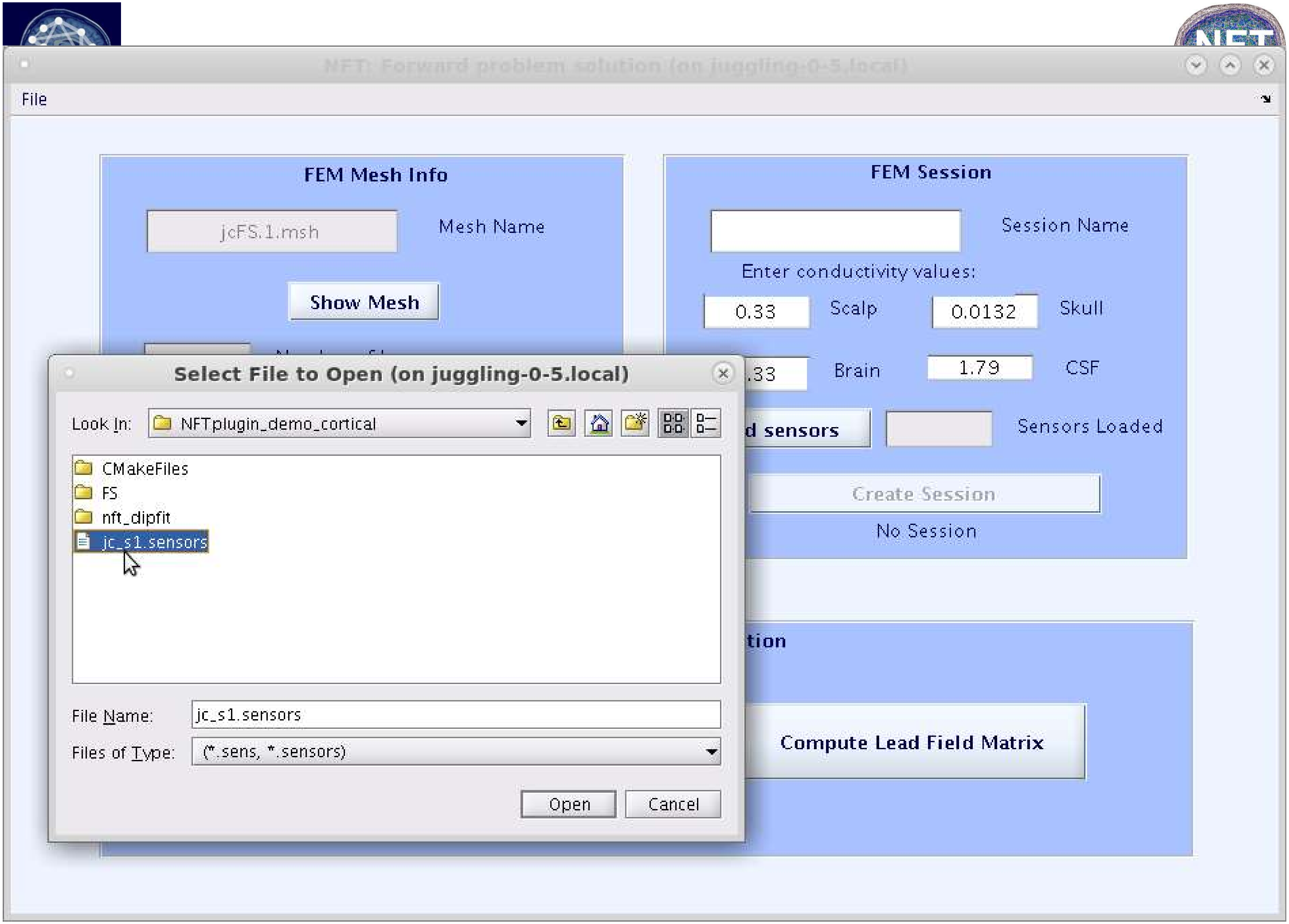
NIST has options to generate Gaussian patches with 10 mm, 6 mm, 3 mm in radius.

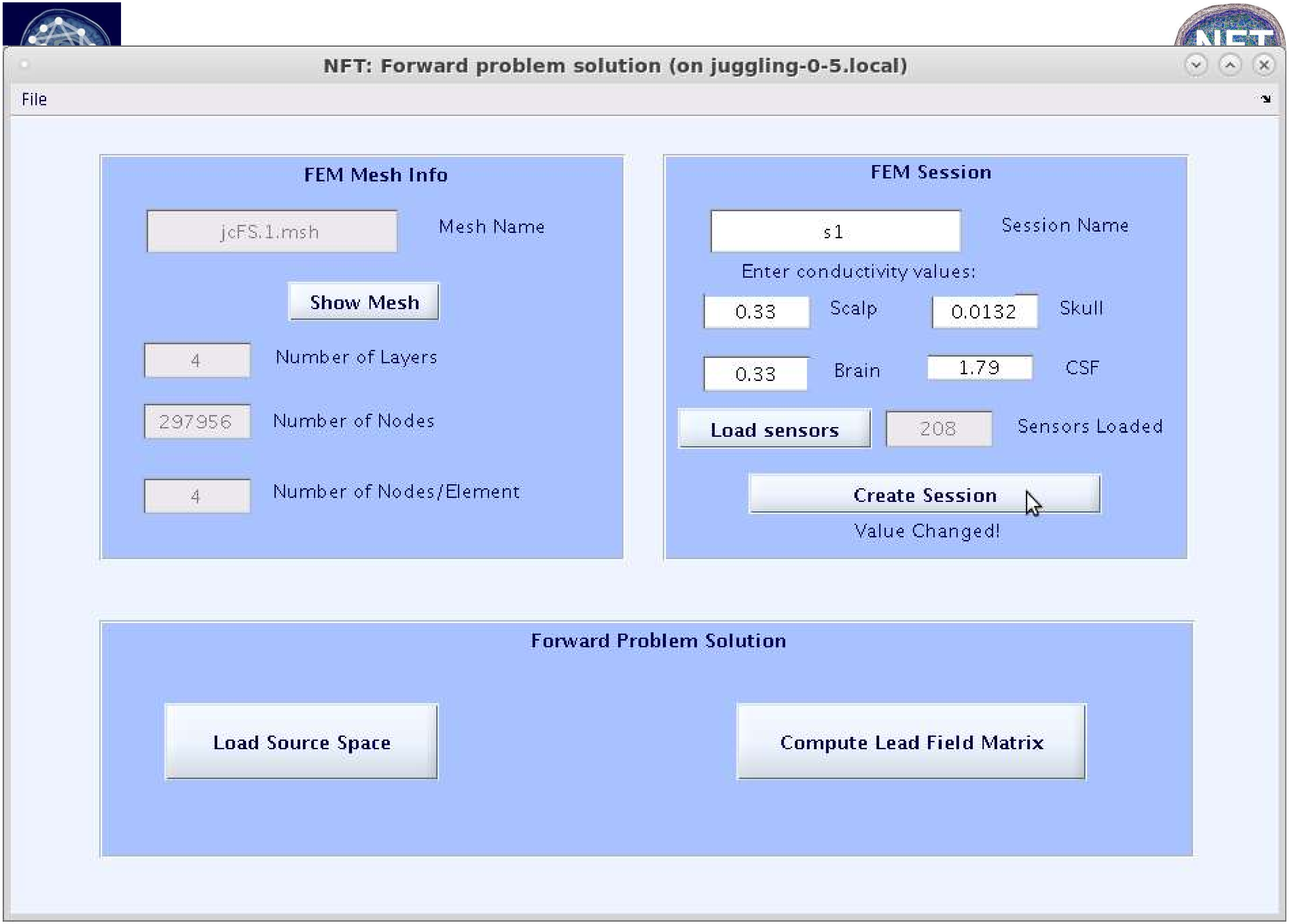

Distributed_Source_Localization (on juggling-0-5.local)
Load MRI
/data/cta/zeynep/NFTdene/NFTplugin_demo_cortical/test2c.img
Start Freesurfer
Running Freesurfer completed!
3,6,10 mm
Cortical source space
80000
# of dipoles in source space
Generate patches
Forward Problem Solution
FP Solution with BEM
FP Solution with FEM
Cortical Source Localization
Component indices
Select Source Localization Method
Start Source Localization
IC #
Visualization

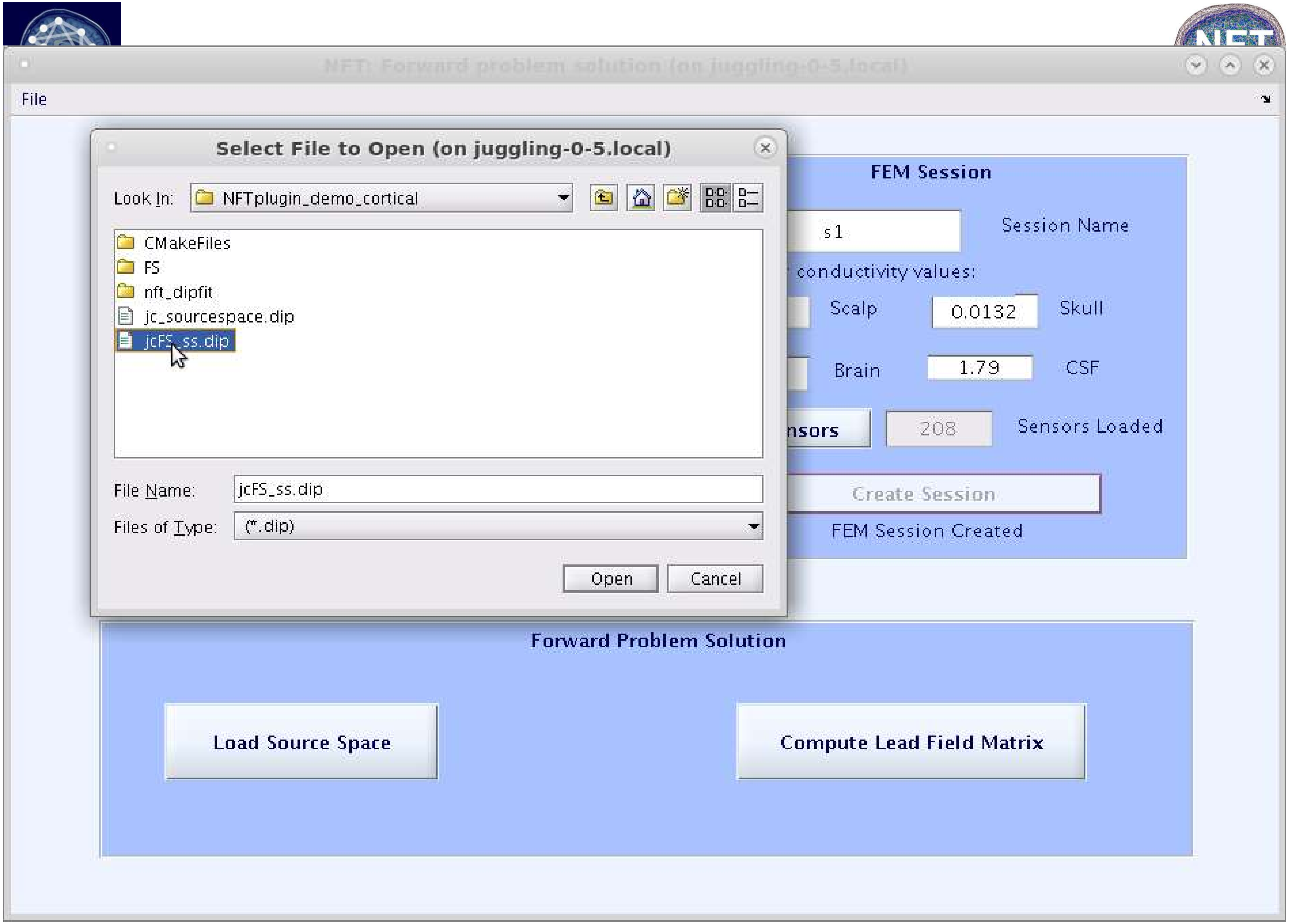
Forward Problem Solution with FEM

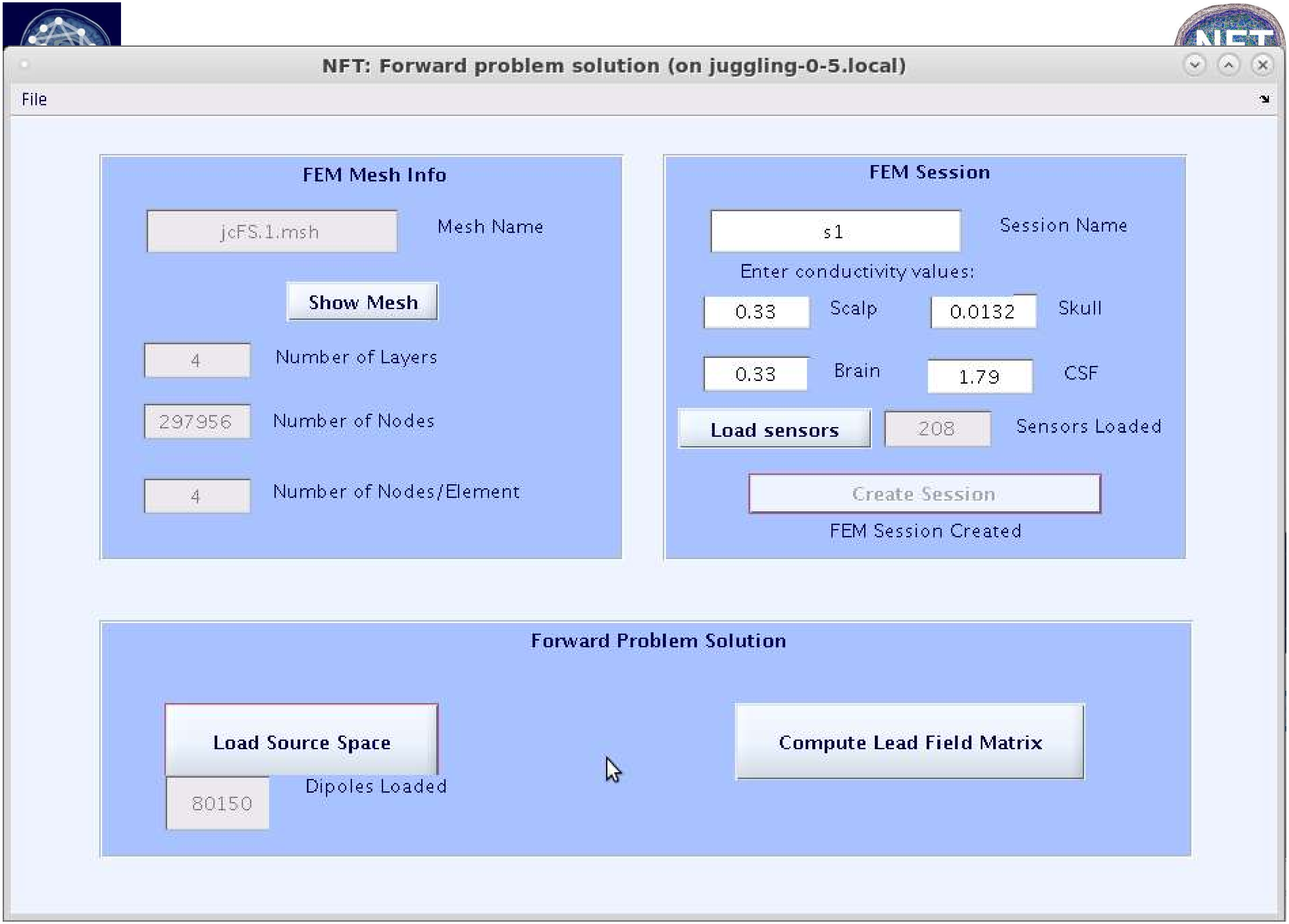
- Tetgen for mesh generation
  - Uses BEM meshes as boundaries
- METU-FEM to generate transfer matrix
  - Compiled from source
  - Requires PETSc for matrix operations
- metufem .mex file for forward solutions in MATLAB
- Instructions available under README.FEM file.

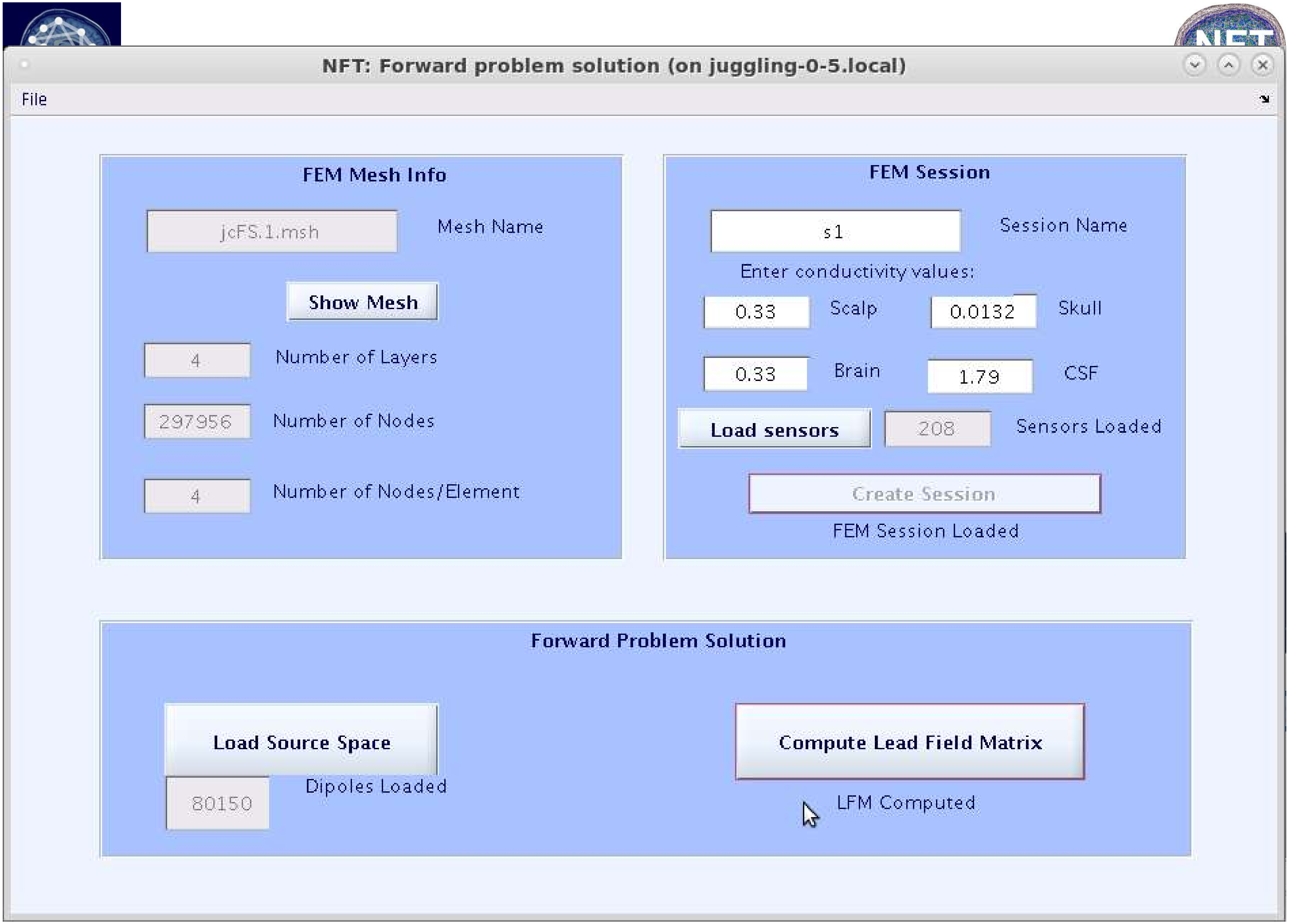

NFT: Forward problem solution (on juggling-0-5.local)
File
Select File to Open (on juggling-0-5.local)
Look In: NFTplugin_demo_cortical
CMakeFiles
FS
nft_dipfit
jc.1.msh
jcFS.1.msh
File Name: jcFS.1.msh
Files of Type: (*.msh)
Open
Cancel
FEM Session
jcFS
Session Name
r conductivity values:
Scalp
0.0132
Skull
Brain
nsors
Sensors Loaded
Create Session
Value Changed!
Load Source Space
Compute Lead Field Matrix

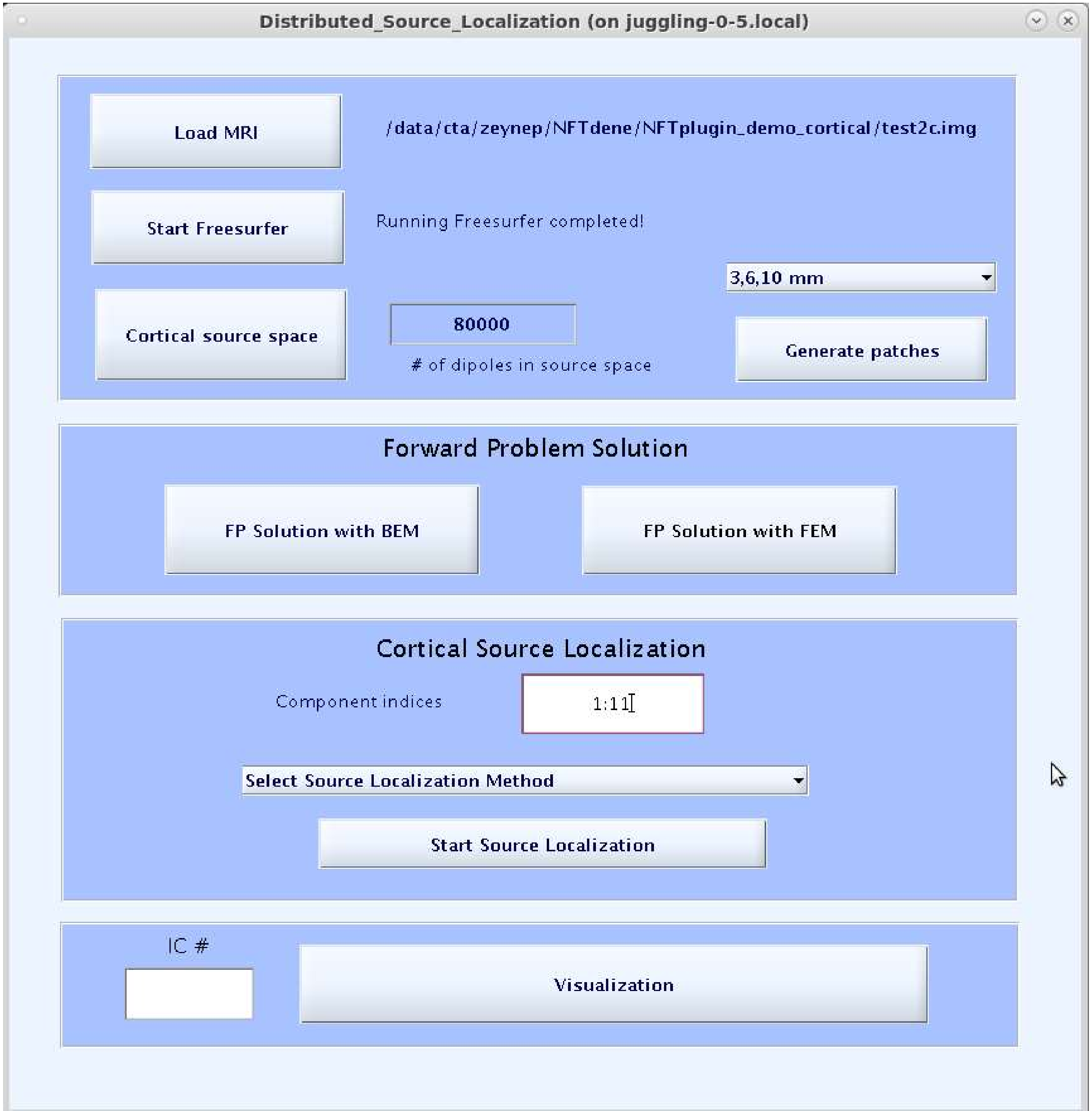

File
FEM Mesh Info
jcFS.1.msh
Mesh Name
Show Mesh
FEM Session
Session Name
Enter conductivity values:
0.33
Scalp
0.0132
Skull
.33
Brain
1.79
CSF
d sensors
Sensors Loaded
Create Session
No Session
tion
Compute Lead Field Matrix
Select File to Open (on juggling-0-5.local)
Look In:
NFTplugin_demo_cortical
CMakeFiles
FS
nft_dipfit
jc_s1.sensors
File Name:
jc_s1.sensors
Files of Type:
(*.sens, *.sensors)
Open
Cancel

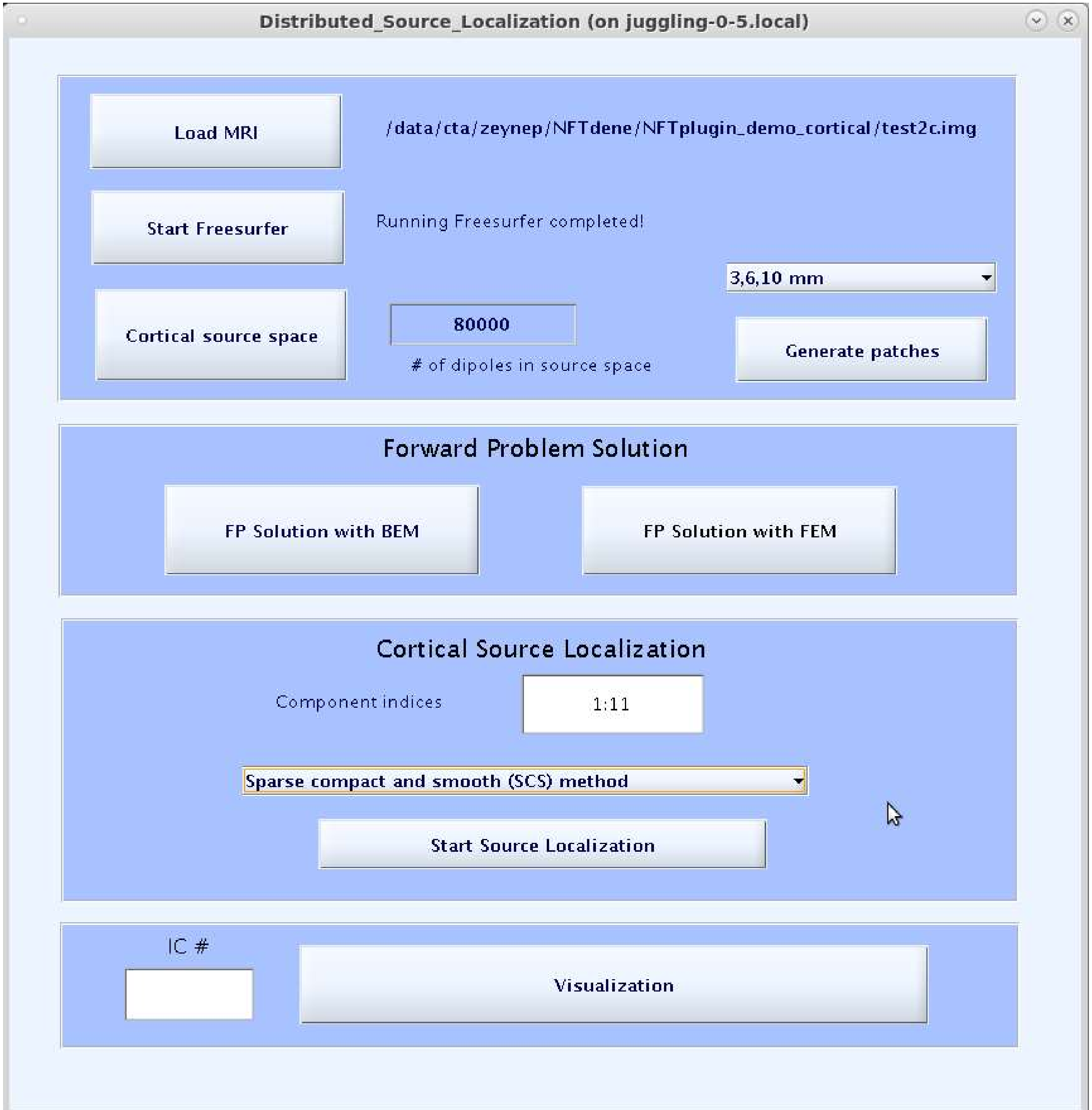

NFT: Forward problem solution (on juggling-0-5.local)
File
FEM Mesh Info
jcFS.1.msh
Mesh Name
Show Mesh
4
Number of Layers
297956
Number of Nodes
4
Number of Nodes/Element
FEM Session
s1
Session Name
Enter conductivity values:
0.33
Scalp
0.0132
Skull
0.33
Brain
1.79
CSF
Load sensors
208
Sensors Loaded
Create Session
Value Changed!
Forward Problem Solution
Load Source Space
Compute Lead Field Matrix


NFT: Forward problem solution (on juggling-0-5.local)
File
Select File to Open (on juggling-0-5.local)
Look In:
NFTplugin_demo_cortical
CMakeFiles
FS
nft_dipfit
jc_sourcespace.dip
jcFS_ss.dip
File Name:
jcFS_ss.dip
Files of Type:
(*.dip)
Open
Cancel
FEM Session
s1
Session Name
conductivity values:
Scalp
0.0132
Skull
Brain
1.79
CSF
nsors
208
Sensors Loaded
Create Session
FEM Session Created
Forward Problem Solution
Load Source Space
Compute Lead Field Matrix

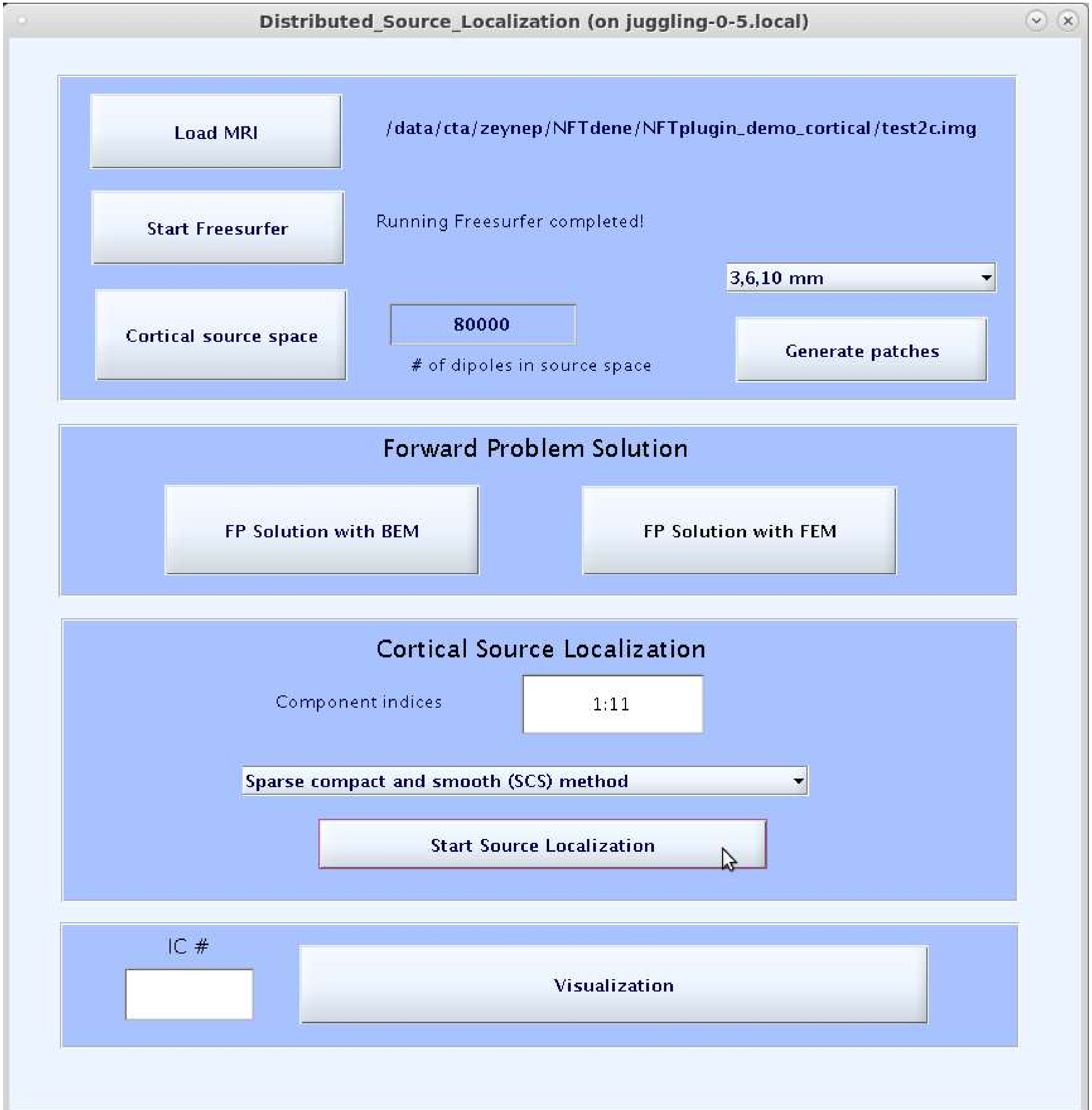

NFT: Forward problem solution (on juggling-0-5.local)
File
FEM Mesh Info
jcFS.1.msh
Mesh Name
Show Mesh
4
Number of Layers
297956
Number of Nodes
4
Number of Nodes/Element
FEM Session
s1
Session Name
Enter conductivity values:
0.33
Scalp
0.0132
Skull
0.33
Brain
1.79
CSF
Load sensors
208
Sensors Loaded
Create Session
FEM Session Created
Forward Problem Solution
Load Source Space
80150
Dipoles Loaded
Compute Lead Field Matrix

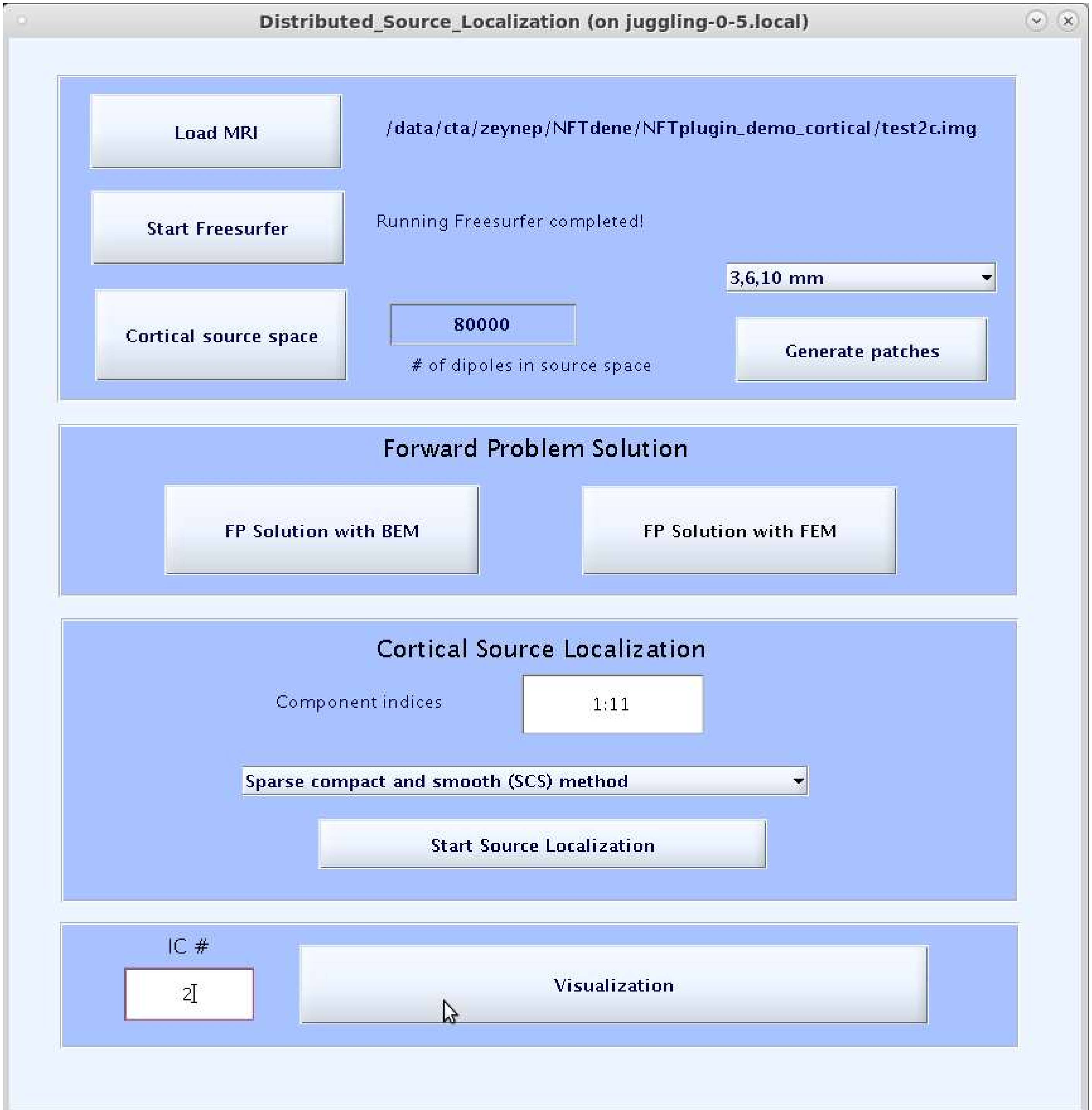

NFT: Forward problem solution (on juggling-0-5.local)
File
FEM Mesh Info
jcFS.1.msh
Mesh Name
Show Mesh
4
Number of Layers
297956
Number of Nodes
4
Number of Nodes/Element
FEM Session
s1
Session Name
Enter conductivity values:
0.33
Scalp
0.0132
Skull
0.33
Brain
1.79
CSF
Load sensors
208
Sensors Loaded
Create Session
FEM Session Loaded
Forward Problem Solution
Load Source Space
80150
Dipoles Loaded
Compute Lead Field Matrix
LFM Computed


Distributed_Source_Localization (on juggling-0-5.local)
Load MRI
/data/cta/zeynep/NFTdene/NFTplugin_demo_cortical/test2c.img
Start Freesurfer
Running Freesurfer completed!
3,6,10 mm
Cortical source space
80000
# of dipoles in source space
Generate patches
Forward Problem Solution
FP Solution with BEM
FP Solution with FEM
Cortical Source Localization
Component indices
1:11
Select Source Localization Method
Start Source Localization
IC #
Visualization



Distributed_Source_Localization (on juggling-0-5.local)
Load MRI
/data/cta/zeynep/NFTdene/NFTplugin_demo_cortical/test2c.img
Start Freesurfer
Running Freesurfer completed!
3,6,10 mm
Cortical source space
80000
# of dipoles in source space
Generate patches
Forward Problem Solution
FP Solution with BEM
FP Solution with FEM
Cortical Source Localization
Component indices
1:11
Sparse compact and smooth (SCS) method
Start Source Localization
IC #
Visualization


Distributed_Source_Localization (on juggling-0-5.local)
Load MRI
/data/cta/zeynep/NFTdene/NFTplugin_demo_cortical/test2c.img
Start Freesurfer
Running Freesurfer completed!
3,6,10 mm
Cortical source space
80000
# of dipoles in source space
Generate patches
Forward Problem Solution
FP Solution with BEM
FP Solution with FEM
Cortical Source Localization
Component indices
1:11
Sparse compact and smooth (SCS) method
Start Source Localization
IC #
Visualization


Distributed_Source_Localization (on juggling-0-5.local)
Load MRI
/data/cta/zeynep/NFTdene/NFTplugin_demo_cortical/test2c.img
Start Freesurfer
Running Freesurfer completed!
3,6,10 mm
Cortical source space
80000
# of dipoles in source space
Generate patches
Forward Problem Solution
FP Solution with BEM
FP Solution with FEM
Cortical Source Localization
Component indices
1:11
Sparse compact and smooth (SCS) method
Start Source Localization
IC #
2
Visualization

Output

- Source estimates are saved as cortex_source_scs and/or cortex_source_sbl

Visualization

- NIST uses Showmesh for visualization of potentials on the cortical surface.
- Showmesh loads a mesh in .smf format,
- Loads potential distribution.
- There are options to load a point set, zoom in, out, rotate, take snapshots.

SHOWMESH TUTORIAL

NFT Matlab Scripts

- Test_script.m

- Start EEGLAB and set your parameters:

```
eeglab
EEG = pop_loadset('filename',eeg_file,'filepath',eeg_path);
[ALLEEG, EEG, CURRENTSET] = eeg_store( ALLEEG, EEG, 0 );
% set 'of' (output folder), subject_name,
%   session_name, and elec_file
subject_name = 'SubjectA';
session_name = 's1';
nl = 4;   % number of layers
plotting = 1;
comp_index = 1:20; % component index for source localization
```

NFT Matlab Scripts

- Realistic modeling from MRI

  % Do segmentation using the GUI

  nft_mesh_generation(subject_name, of, nl)

  nft_source_space_generation(subject_name, of)

  % Do co-registration using the GUI

  nft_forward_problem_solution(subject_name, session_name, of);

  dip1 = nft_inverse_problem_solution(subject_name, session_name, of, EEG, comp_index, plotting, elec_file)

NFT Matlab Scripts

- BEM warping mesh

```
nft_warping_mesh(subject_name, session_name,
   elec_file, nl, of,0,0);
nft_forward_problem_solution(subject_name,
   session_name, of);
dip1 = nft_inverse_problem_solution(subject_name,
   session_name, of, EEG, comp_index, plotting,
   elec_file)
```

NFT Matlab Scripts

- FEM warping mesh

```
session_name='s1_fem';
nft_warping_mesh(subject_name, session_name,
   elec_file, nl, of,0,1);
nft_fem_forward_problem_solution(subject_name,
   session_name, of);
dip1 = nft_inverse_problem_solution(subject_name,
   session_name, of, EEG, comp_index, plotting,
   elec_file)
```

NFT Matlab Scripts

- Set NFT dipole structure to EEGLAB dipole structure

```
eeglab_folder = dirname(which('eeglab'));
mri_file = [eeglab_folder
   '/plugins/dipfit2.2/standard_BEM/standard_mri.m
   at'];
EEG.dipfit.mrifile =  mri_file;
EEG.dipfit.model = EEG.etc.nft.model;
```

Distributed source localization

% generation of forward model:

nft_dsl_forward_model_generation(subject_name, of, mri_file)

% BEM forward problem solution

nft_forward_problem_solution(subject_name, session_name, of, 'ss_name', [subject_name 'FS_ss.dip’])

%FEM forward problem solution

nft_fem_forward_problem_solution(subject_name, session_name, of, 'ss_name', [subject_name 'FS_ss.dip’])

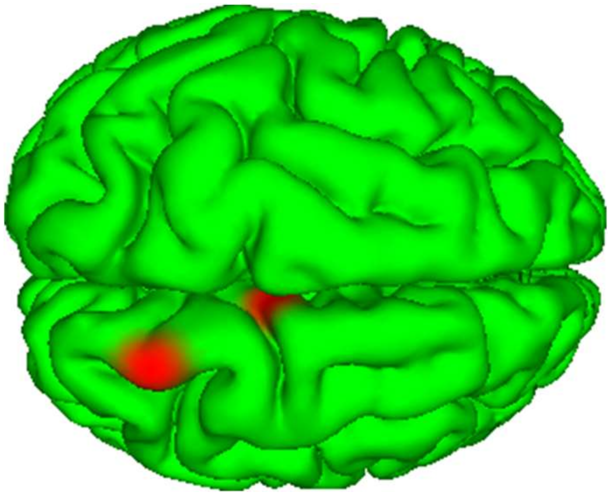
%cortical source localization

nft_dsl_inverse_problem_solution(subject_name, session_name, of, EEG, comp_index, selection)

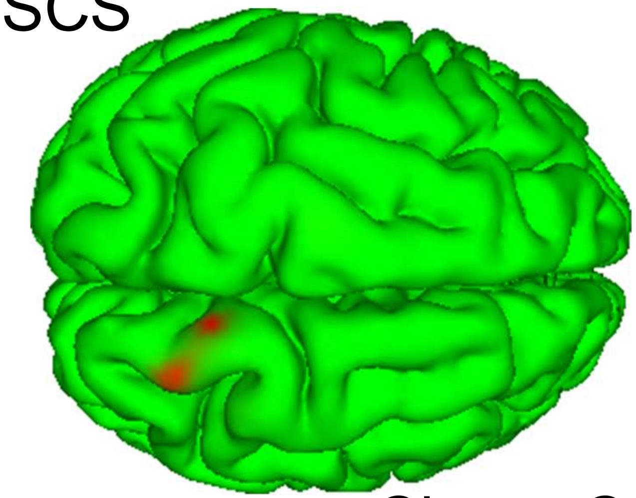
EXAMPLES

Source localization results

Simulated source

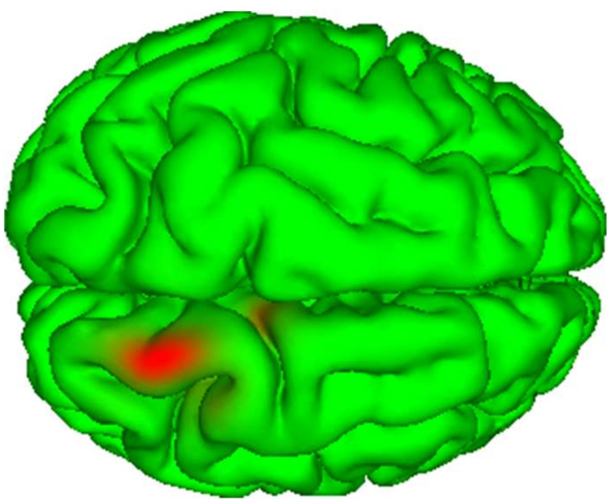


SCS

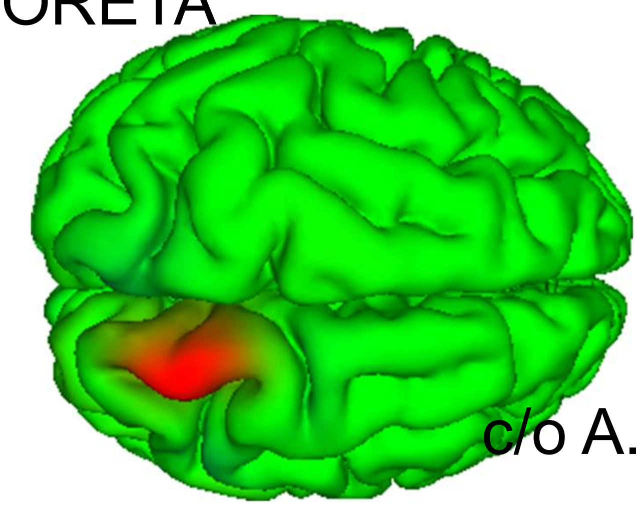


Cheng Cao, 2012

Patch-based SBL

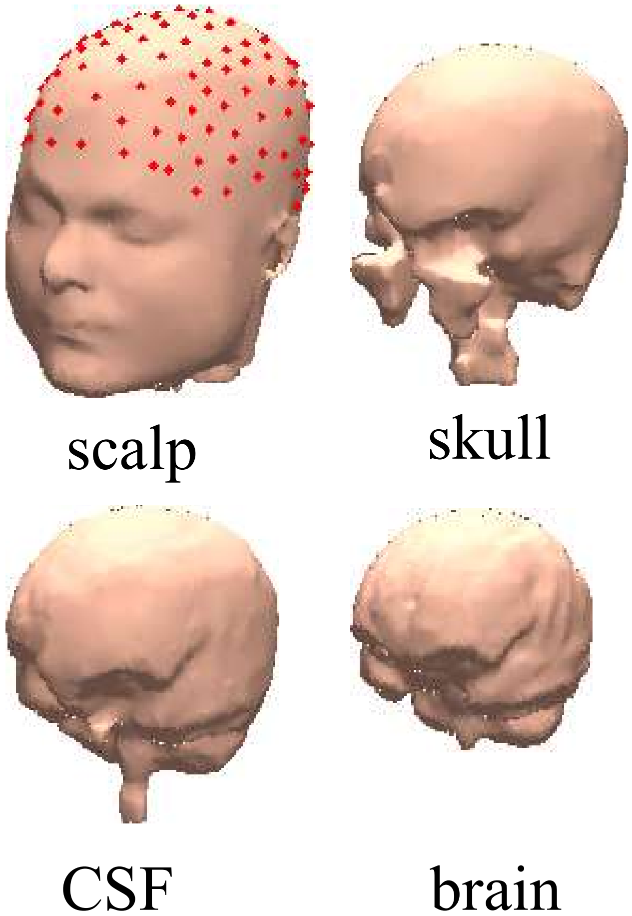


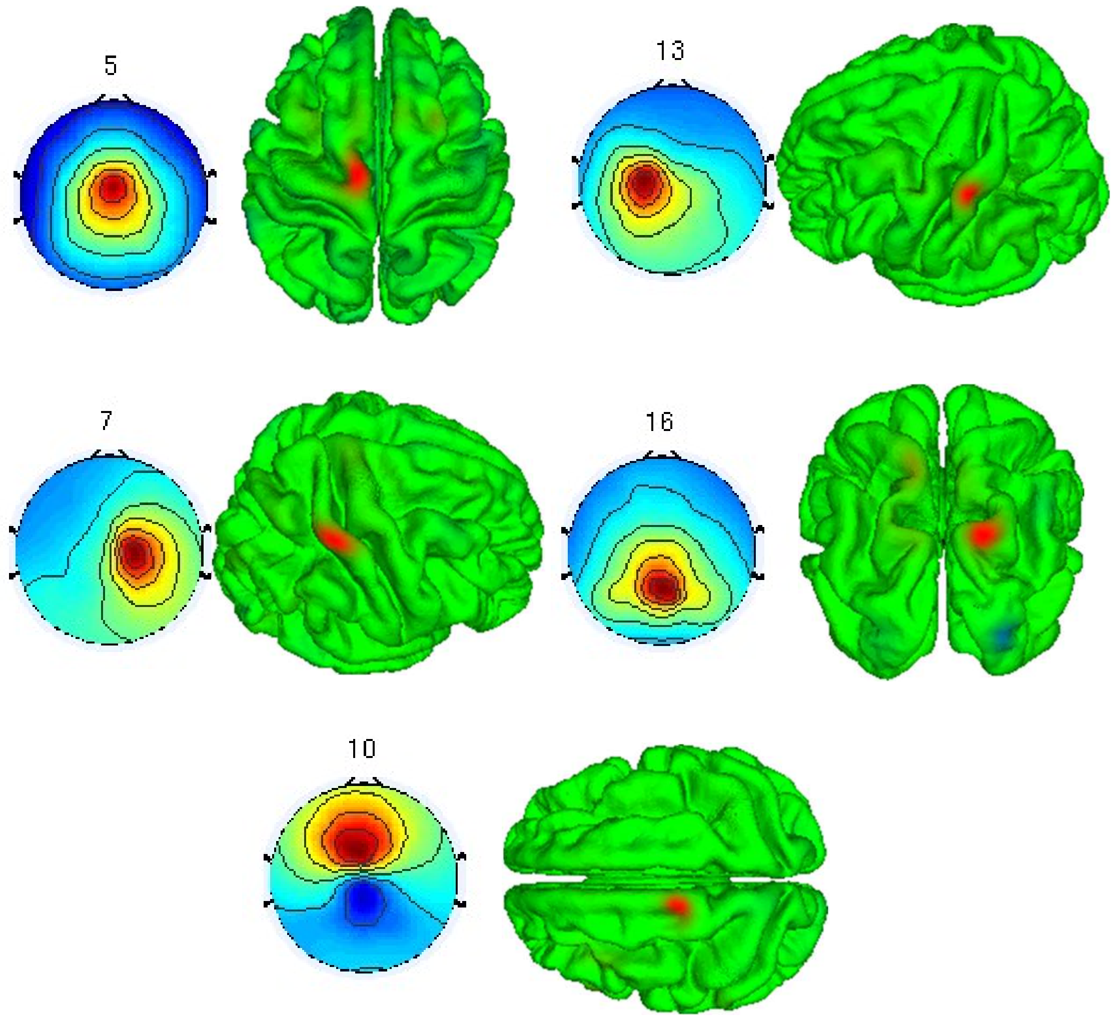
sLORETA



c/o A. Ojeda

EEG source localization





Swartz Center for Computational Neuroscience

Generation of individual head models

NFT

NFT (sccn.ucsd.edu/nft/) was used to generate four-layer Finite Element (FEM) head models.

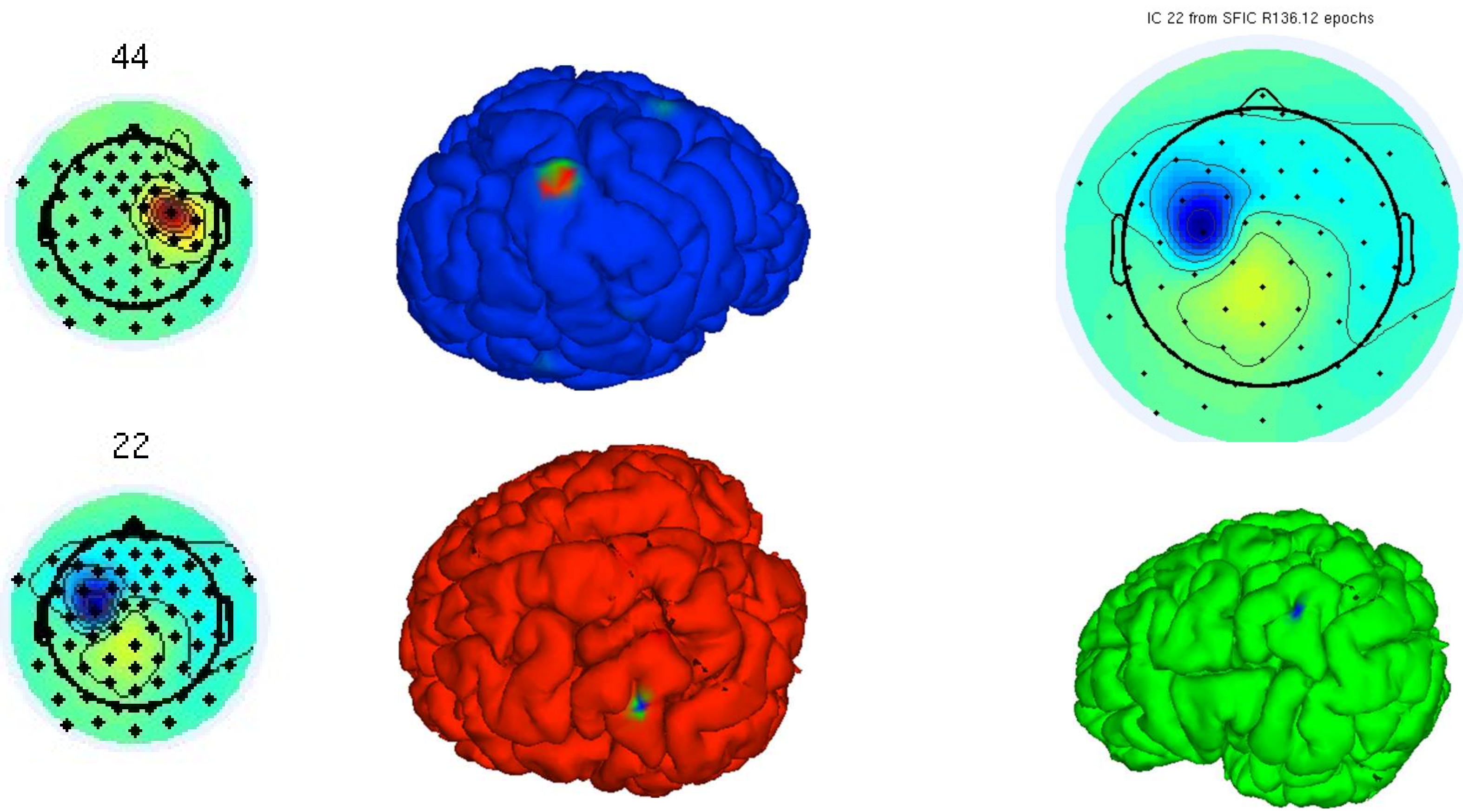
6 months

12 months

1,441,777 tetrahedral volume elements

1,025,643 tetrahedral volume elements

Source localization for 1-year old



Future work

- Neuroscience Gateway Portal (NSG)
  - Dipole source localization
  - Distributed source localization
- SCALE
  - Simultaneous skull conductivity and source localization estimation.
  - Inputs: EEG and MRI
  - Outputs: Source estimates and skull conductivity estimate.

NFT download

NFT wiki

- https://github.com/sccn/NFT/wiki

NFT code

- https://github.com/sccn/NFT

NFT test data

- https://github.com/sccn/NFT_test

NFT reference

- Akalin Acar Z, Makeig S, Neuroelectromagnetic Forward Head Modeling Toolbox, J. of Neuroscience Methods, vol 190(2), 258-270, 2010.